Designing Leadership under Social Conformity

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Abstract

This paper studies the optimal design of leadership in a local-average network game, in which players differ in individual productivity and incur a conformity cost for deviating from their neighbors' average effort—the social norm. An organizational planner divides players into leaders and followers across two sequential stages to maximize aggregate effort. We first show that despite the presence of strategic complementarities, leadership can either increase or decrease equilibrium effort relative to simultaneous play, depending on whether a leader's productivity exceeds the social norm. Next, we derive an exact index to identify the optimal single leader, and demonstrate that the selected player can differ substantially from the key leader in the local-aggregate model. We then characterize the limiting structure of optimal leader groups as the taste for conformity approaches zero or infinity. An optimal group favors locally productive players under weak conformity, but collapses to the most productive player in the network under strong conformity. We also analyze multi-stage hierarchies and establish that finer sequencing can be suboptimal. Finally, we extend the model to incorporate incomplete information, anti-conformity, and more general preferences and interactions.

Keywords: Leadership; Social conformity; Local-average model; Network games; Organization design.

JEL classification: C72, D23, D85, D91, Z13.

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1 Introduction

Mobilizing individual effort is central to collective performance, yet private incentives need not align with broader societal interests.[1] Leading by example offers an important mechanism for bridging this gap, even in the absence of contractual relationships or hierarchical authority (Hermalin, 1998; Güth et al., 2007). For instance, in Bolivian communities, asking elected leaders to donate publicly before other residents substantially increased total giving for schoolbooks (Jack and Recalde, 2015). During the American Civil War, military units whose officers led from the front and shared battlefield risks experienced lower desertion rates (Costa, 2014). A key channel behind the effectiveness of leadership is that the actions of early movers can establish a behavioral reference point, thereby shaping the incentives of those who follow (Acemoglu and Jackson, 2015; Tankard and Paluck, 2016). Although these observations underscore the economic significance of leadership, fundamental questions remain: how should leadership be organized, what determines the selection of key leaders, and to what extent does the surrounding social environment shape the outcomes of such leadership arrangements?

Social conformity—the tendency for individuals to align their behavior with that of their peers—is a prominent feature of many social interactions. In a variety of settings, prevailing practices and shared expectations can serve as social reference points that guide individual choices (Andreoni and Bernheim, 2009; Bursztyn et al., 2020). Existing research provides several behavioral foundations for this reliance, including social comparison (Festinger, 1954), social-image concerns (Bernheim, 1994; Bursztyn and Jensen, 2017), and identity-based preferences (Akerlof and Kranton, 2000).[2] In the context of leadership, the desire to conform has two contrasting implications. On the one hand, it makes followers more receptive to a leader's example; on the other hand, it discourages leaders from deviating from their peers. Effective leadership thus hinges on both the ability to influence others and the incentive to take the lead. Despite this

[1] This discrepancy arises in settings such as charitable fundraising (Frey and Meier, 2004; Shang and Croson, 2009), workplace production (Mas and Moretti, 2009; Bandiera et al., 2010), and school initiatives (Paluck and Shepherd, 2012; Paluck et al., 2016).

[2] Social comparison provides a basis for self-evaluation: individuals assess their opinions and abilities relative to those of others, particularly when objective standards are unavailable. Social-image concerns arise when individuals care about how others perceive them. Because observable actions convey information about underlying characteristics, individuals may conform to gain social approval or avoid disapproval. Identity-based preferences link individuals' utility to their social identities and associated behavioral prescriptions, making conformity to these prescriptions intrinsically rewarding and deviations psychologically costly.

tension, theory offers limited guidance on how to organize leadership while taking into account the dual role of conformity.

This paper studies the optimal design of leading by example in the presence of social conformity. Organizational members (hereafter players) interact on a social network and differ in individual productivity, which governs the intrinsic incentive to exert effort. Following Ushchev and Zenou (2020), we introduce a preference for conformity through local-average interactions: each player incurs a cost for deviating from the average effort of her neighbors, which serves as a *social norm*. In this setting, the game features both *payoff externalities*, as neighbors' efforts shift the social norm, and *strategic complementarities*, as higher peer effort weakly increases a player's marginal return to effort. We model leadership through the timing of effort choices. Our main analysis considers a two-stage sequential game in which an organizational planner selects a subset of players to move first as *leaders*, while the remaining players observe these choices and then respond as *followers*. The planner's objective is to maximize aggregate effort.

1.1 Main Results

Our main findings are summarized as follows. First, we characterize the subgame-perfect Nash equilibrium (SPNE) for a given leader group. At any level of social conformity, each player's effort is a weighted average of individual productivities in the network, with weights jointly determined by the strength of conformity and the allocation of leadership. In contrast to the local-aggregate model of Zhou and Chen (2015),[3] we show that introducing leadership under social conformity can either increase or decrease individual and aggregate effort relative to the simultaneous-move Nash equilibrium.

The central mechanism stems from the dual role of social conformity. To see this, consider a single leader whose productivity exceeds the social norm, so that conformity initially depresses her Nash effort. Moving first creates a *strategic social-norm effect*: anticipating that higher effort will induce followers to adjust upward, the leader can shift her own social norm and thereby ease the pressure to conform. These responses propagate through the network and increase aggregate effort. Conversely, when the leader's productivity falls below the social norm, the same mechanism works in reverse and reduces aggregate effort.

[3] Zhou and Chen (2015) establish that each player's equilibrium contribution is weakly higher under sequential moves than under simultaneous moves. The result is driven by positive network spillovers and strategic complementarities, whereby leaders anticipate that greater contributions induce followers to contribute more.

Second, we address the key-leader problem under social conformity: if the planner chooses a single player to move first, who should it be? We construct an exact leading index to identify the key leader—the single leader who maximizes aggregate effort. Because effective leadership requires both a strong incentive to increase effort and the ability to propagate that increase through the network, the most productive player need not be the key leader. Moreover, the key leader in the presence of social norms can differ substantially from the key leader of Zhou and Chen (2015), even when simultaneous-move best responses coincide. This divergence arises because the marginal return to leading is proportional to Nash effort in the local-aggregate model, but to the difference between Nash effort and the social norm in the local-average model. Consequently, more productive peers raise a candidate's Nash effort but reduce her leadership gain by elevating the social norm she already faces.

We further show that the taste for conformity systematically reshapes the selection of the key leader. To understand this result, consider the extreme cases approaching *pure individualism* and *pure conformism*.[4] When the taste for conformity is sufficiently small, the effect of local connections dominates: the leadership gain is proportional to a player's *local productivity advantage* over the average productivity of her neighbors, weighted by *local reciprocity*, defined as the average weight those neighbors assign to the leader. Because neighbors with fewer links dilute this influence less, a player who stands out locally can outperform a globally more productive peer. When the taste for conformity is sufficiently large, by contrast, indirect connections throughout the network come into play. Although Nash efforts approach a consensus given by the degree-weighted average of individual productivities, a single leader anchors the entire population at her own productivity. As a result, the most productive player must be the key leader. Between these two limits, leadership gains can vary nonmonotonically with the taste for conformity, and stronger conformity can reverse the ranking of candidates.

Third, we extend the analysis to multiple leaders and characterize optimal group selection. The key trade-off is that appointing more leaders provides more examples but leaves fewer followers to respond. Due to the combinatorial nature of group selection, simply ranking candidates by their individual leading indices is not sufficient to ensure a global optimum. Nevertheless, for any level of social conformity, we establish two

[4]Under pure individualism, each player's equilibrium effort equals her individual productivity and is independent of the social norm, so leadership has no effect. However, under pure conformism, all players in the simultaneous-move benchmark choose the same effort level, equal to the degree-weighted average of individual productivities.

structural properties of an optimal leader group. First, the group consists only of players whose productivity exceeds the social norm, or equivalently, who would increase aggregate effort if selected as a single leader. Second, the remaining followers must form a *dominating set* in the network, meaning that every leader has at least one follower as a neighbor. This result reveals a substitutability between connected leaders, who cannot respond to one another as followers and therefore make tightly clustered leadership less attractive.

Beyond these general characterizations, more explicit results obtain in the limiting cases of weak and strong conformity. For sufficiently small conformity, only players with a positive local productivity advantage qualify as leaders, and group selection among these candidates reduces to a maximum directed cut problem (Ghuge et al., 2022). The planner seeks a partition that maximizes the total weight of links directed from leaders to followers, where each link is weighted by the leader's local productivity advantage and reciprocal influence. Because this program contains the classical maximum cut problem as a special case, optimal group selection under weak conformity is NP-hard in general. For sufficiently large conformity, by contrast, a "*less is more*" principle emerges: every optimal leader group collapses to a single player with the highest individual productivity. Appointing additional leaders leaves fewer followers to respond and retains mutual norm constraints among leaders, which reduces aggregate effort even when all selected leaders are equally productive.

Moreover, we investigate the leadership design problem for specific network architectures. In the case of complete networks, we show that for any given group size, assigning players with higher productivity to the leader group is always beneficial. In addition, an optimal leader group shrinks monotonically as the taste for conformity strengthens. In complete bipartite networks, key-leader selection favors the part with higher maximum and average productivities, or the part with smaller size when these moments coincide. Furthermore, when one part is uniformly more productive, optimal group selection exhibits a bang-bang pattern: at a conformity threshold, the optimal leader group drops abruptly from that entire part to a singleton.

Fourth, we relax the two-stage restriction and allow the planner to organize leadership across an arbitrary sequence of stages. A natural question in organizational design is what constitutes the optimal hierarchy of leadership. Existing theories suggest that a more

granular hierarchy—ultimately a sequential chain—always benefits the organization.[5] Instead, we show that in the presence of social conformity, finer sequencing need not increase aggregate effort. The main reason is that moving earlier allows a player to reshape the social norm through subsequent responses, raising effort only when her productivity exceeds that norm. Consequently, introducing intermediate tiers into the leadership hierarchy is valuable when a player should first follow a more productive neighbor and then lead a less productive one. To obtain more intuition, we consider the limiting cases of weak and strong conformity. If conformity is sufficiently weak, an optimal sequence assigns players with positive local productivity advantages to move one by one in decreasing order of those advantages. All remaining players then act simultaneously in the final stage. If conformity is sufficiently strong, by contrast, any multi-stage arrangement beyond a direct leader–follower hierarchy is suboptimal. As a result, a single player with the highest productivity moves first, and the rest of the players follow together.

Finally, we consider several extensions of our baseline model. In many organizational settings, leaders not only move first to set an example, but also hold superior information about an unknown state that affects the returns to effort (Hermalin, 1998; Zhou and Chen, 2016). We therefore allow leaders to privately observe the state before choosing effort, while followers infer the state from the leaders' actions. In a separating equilibrium, we show that signaling raises equilibrium effort above the complete-information benchmark, as revealing a favorable state induces followers to exert higher effort, which shifts the social norm upward and eases peer pressure for leaders whose effort exceeds that norm. At the same time, incomplete information introduces a novel trade-off in leadership design. Because more productive followers respond more strongly to the signaled state, appointing a productive player to lead sacrifices her strong response as a follower. Under weak conformity, the leadership gain is proportional to a leader's local productivity advantage, weighted by the *follower-to-leader productivity ratio*. Consequently, in a complete network the most productive player need not be selected to lead, and in a complete bipartite network an intermediate group size can become optimal; under strong

[5] In the local-aggregate network model, Zhou and Chen (2015) show that, starting from any fixed sequence, aggregate effort always increases when a set of simultaneously moving players instead chooses effort sequentially, suggesting that any local modification toward the sequential-move game is beneficial. Similarly, in a team-production model with hierarchical signaling, Zhou (2016) shows that a full chain is the optimal hierarchy because it maximizes the incentive to lead by example across multiple stages.

conformity, by contrast, the informational gain vanishes and optimal leadership again collapses to a single player with the highest productivity.

We also examine how variations in the baseline specification of conformity preferences and network interactions affect our results. Specifically, we show that our main insights continue to hold when players have heterogeneous tastes for conformity and interact through weighted and directed networks. In addition, we study *anti-conformity*, whereby players benefit from differentiating their efforts from the social norm. Although efforts become strategic substitutes, a unique equilibrium persists under a moderate level of anti-conformity and preserves our characterization of leader selection. Finally, we consider *shifted social norms* that set a player's reference level above or below the average effort of her neighbors. In this case, even under homogeneous productivities, a downward shift in the social norm enables leaders to raise aggregate effort, whereas an upward shift induces leaders to reduce aggregate effort.

1.2 Related Literature

Our study contributes to the economic theory of leadership and organizational hierarchy (for reviews, see Bolton et al., 2010; Hermalin, 2012). In existing models, leaders improve organizational performance by signaling private information (Hermalin, 1998; Komai et al., 2007; Zhou and Chen, 2016), by guiding expectations and actions in coordination games (Dewan and Myatt, 2008; Bolton et al., 2013; Landa and Tyson, 2017), or by leveraging local-aggregate complementarities in networks (Zhou and Chen, 2015). Huck and Rey-Biel (2006) and Matsuno (2026) introduce conformity into this literature and show that it makes leading by example more effective, either by raising team output or by improving coordination among followers. To the best of our knowledge, ours is the first study to analyze the selection of leaders and the design of hierarchies in networks under social conformity. Our findings challenge a common implication of this literature, namely that the responses of followers make leading by example more valuable, so that leadership never lowers performance. In our setting, conformity cuts both ways: it makes followers receptive to an early example but also exposes the leader to peer pressure. As a result, whether leadership raises or lowers aggregate effort hinges on the relationship between the leader's productivity and the social norm.

Our analysis of the optimal hierarchy of leadership contributes to the design of organizational architectures. Varian (1994) shows that sequential moves can reduce voluntary public-good provision relative to simultaneous play. When efforts are strategic

complements or early moves convey information, however, finer sequencing strengthens incentives: a chain of single movers is optimal in local-aggregate network games (Zhou and Chen, 2015), in hierarchical team production (Zhou, 2016), and in sequential selling (Zhou et al., 2023). Sequential structures also shape incentive provision and transparency among peers (Winter, 2006, 2010) as well as dynamic link formation (Sun et al., 2026). From these previous studies, two natural questions arise: (i) does a more granular hierarchy always improve aggregate performance, and (ii) which members should be separated into different tiers? While strategic complementarity in efforts favors finer sequencing in existing models regardless of member characteristics, we show that this property alone does not determine the optimal hierarchy. In the presence of social conformity, however, refining the sequence of moves can strictly reduce aggregate effort, as only members who are more productive than their peers should move in earlier stages. The key implication is that a deep hierarchy in organizations can be suboptimal, providing a rationale for the flat organizational structures documented in theory and practice (Rajan and Zingales, 2001; Rajan and Wulf, 2006).

Our work also contributes to the analysis of social norms in networks. A large literature characterizes equilibrium in network games with local-aggregate spillovers (Ballester et al., 2006; Galeotti et al., 2010; Bramoullé et al., 2014). Ushchev and Zenou (2020) provide a comprehensive investigation of the local-average model, including its comparative statics, welfare properties, and policy implications. Boucher et al. (2024) nest spillovers and conformity in a general theory of peer effects and show that the two mechanisms, observationally equivalent in simultaneous play, carry different policy implications. Building on the local-average model, Meng (2025) studies budgeted targeting interventions in the spirit of Galeotti et al. (2020), and Meng et al. (2026) study revenue-maximizing mechanisms when agents' conformity types are private information. These analyses consider simultaneous play, in which each player's social norm forms at once from the actions of neighbors. Under sequential moves, by contrast, the norm becomes stage-dependent: a leader's effort shapes the social norm through the subsequent responses of followers. This stage-dependent feedback breaks the simultaneous-move equivalence with local-aggregate games and yields distinct optimal leaders and hierarchies.

Finally, our paper relates to studies of how social norms and leadership shape behavior in groups and organizations. Early work shows that peer pressure can mitigate free riding in partnerships (Kandel and Lazear, 1992) and that endogenous norms anchored to average peer actions alter optimal incentive pay (Fischer and Huddart, 2008). Subsequent studies

examine prominent agents whose visible actions overturn low-cooperation conventions (Acemoglu and Jackson, 2015), leadership structures that emerge from rewards for conformity and influence (Goldbaum, 2019), leaders who sustain conventions through fixed reference points or norm recommendations (Cabrales and Hauk, 2024; Block et al., 2025), and the adaptation of conformist preferences over time (Olcina et al., 2024). In our model, an organizational designer influences the formation of the social norm through the timing of effort choices. Moving first shifts the social norm only by inducing later movers to adjust their efforts, while the early mover remains subject to the resulting norm. Appointing additional leaders therefore leaves fewer followers to adjust the norm and preserves mutual peer pressure among leaders, which explains why concentrating leadership in fewer hands can raise effort across the network.

The remainder of the paper is organized as follows. Section 2 presents the model, the simultaneous-move Nash benchmark, and the leadership design problem. Section 3 establishes the two-stage equilibrium characterization, analyzes key-leader and key-group selection, and derives explicit results for special network architectures. Section 4 extends the analysis to general sequences and characterizes the optimal hierarchy of leadership. Section 6 concludes. All proofs are relegated to the Appendix.

2 Model

Our main departure from standard models of leading by example is that social conformity shapes the incentives of both leaders and followers. While followers adjust their efforts toward a leader's example, leaders themselves face pressure to conform to a social norm that shifts with followers' responses.

2.1 Preferences and the Environment

Consider an organization of $n \geq 2$ members (or players), indexed by $i \in N := \{1, \ldots, n\}$, who interact in a social network g. The adjacency matrix $\mathbf{G} = [g_{ij}]$ is an $(n \times n)$ matrix with $\{0, 1\}$ entries that keeps track of direct connections in the network: players i and j are directly connected if and only if $g_{ij} = 1$, and $g_{ij} = 0$ otherwise. We assume that g is connected, undirected ($g_{ij} = g_{ji}$), and has no self-loops ($g_{ii} = 0$). For each player i, define $N(i) := \{j \in N : g_{ij} = 1\}$ as the set of direct neighbors and $d_i := \sum_{j=1}^{n} g_{ij} > 0$ as player i's *degree*. For any nonempty subset $S \subseteq N$, the *subnetwork* induced by S consists

of the players in S and all links in g with both endpoints in S; a *connected component* of this subnetwork is a maximal nonempty subset of S that forms a connected subnetwork.

Let $\widehat{\mathbf{G}} = [\widehat{g}_{ij}]$ denote the $(n \times n)$ row-normalized adjacency matrix with entries $\widehat{g}_{ij} := g_{ij}/d_i$, where $\widehat{g}_{ij}$ is the weight that player i assigns to player j in forming a local reference group; every direct neighbor receives equal weight $1/d_i$, so each row of $\widehat{\mathbf{G}}$ sums to one.

Each player $i \in N$ is characterized by an exogenous *individual productivity* $\alpha_i > 0$ and chooses an effort level $x_i \geq 0$. Let $\boldsymbol{\alpha} := (\alpha_1, \ldots, \alpha_n)^\top$ and $\mathbf{x} := (x_1, \ldots, x_n)^\top$ denote the productivity and effort profiles. Player i's *social norm* is the average effort of her direct neighbors, $\bar{x}_i := \sum_{j=1}^n \widehat{g}_{ij} x_j$. Preferences take the linear-quadratic form of Ushchev and Zenou (2020):

$$u_i(x_i, \mathbf{x}_{-i}; g) := \alpha_i x_i - \frac{1}{2}x_i^2 - \frac{\gamma}{2}(x_i - \bar{x}_i)^2, \qquad \gamma \geq 0, \tag{1}$$

where $\gamma \geq 0$ measures the taste for conformity, and the network, productivities, and payoff parameters are common knowledge. In (1), $\alpha_i x_i - \frac{1}{2}x_i^2$ is the intrinsic (net) return from effort in the absence of peer effects. The *conformity cost* $-\frac{\gamma}{2}(x_i - \bar{x}_i)^2$ captures the cost of deviating from the social norm $\bar{x}_i$, which penalizes player i in proportion to the squared distance between own effort and the average effort of direct neighbors.[6] Throughout the analysis, we use $\lambda := \gamma/(1+\gamma) \in [0, 1)$ to parameterize the relative strength of conformity.

Because the social norm $\bar{x}_i$ is formed from neighbors' efforts, each player's effort generates payoff externalities across links: for any $j \neq i$,

$$\frac{\partial u_i(x_i, \mathbf{x}_{-i}; g)}{\partial x_j} = \gamma \widehat{g}_{ij}(x_i - \bar{x}_i).$$

When $\gamma > 0$ and $\widehat{g}_{ij} > 0$, higher effort by neighbor j raises player i's social norm $\bar{x}_i$. Whether this norm shift benefits or hurts player i depends on how x_i compares with $\bar{x}_i$: when $x_i > \bar{x}_i$, a higher $\bar{x}_i$ brings the social norm closer to player i's effort and reduces the conformity cost, whereas when $x_i < \bar{x}_i$, a higher $\bar{x}_i$ pushes the norm further above x_i and increases that cost. At the same time, efforts are strategic complements, since $\partial^2 u_i(x_i, \mathbf{x}_{-i}; g)/(\partial x_i \partial x_j) = \gamma \widehat{g}_{ij} \geq 0$ for any $j \neq i$: higher effort by a neighbor always raises player i's marginal return to effort. Together, these two features govern the incentives for leadership by example: when a leader chooses higher effort before followers

[6] The qualitative trade-offs in our analysis extend to a general convex conformity cost $\gamma\phi(x_i - \bar{x}_i)$, where $\phi(\cdot)$ is twice continuously differentiable and satisfies $\phi(0) = \phi'(0) = 0$ and $\phi''(z) > 0$ for all $z \in \mathbb{R}$.

act, strategic complementarity induces followers to raise their subsequent efforts, and the resulting upward shift in the leader's social norm rewards a leader whose effort exceeds the norm ($x_i > \bar{x}_i$) but penalizes a leader whose effort falls short of it ($x_i < \bar{x}_i$).[7]

We use the following notation throughout the paper.

- Vectors are columns, and $\top$ denotes transposition. For a vector $\mathbf{z}$ and a set $A \subseteq N$, $\mathbf{z}_A$ is the subvector indexed by A, and $\mathbf{z}_{-i} := \mathbf{z}_{N\setminus\{i\}}$ omits coordinate i. For an $n \times n$ matrix $\mathbf{T} = [t_{ij}]$, $\mathbf{T}_{i\cdot}$ and $\mathbf{T}_{\cdot i}$ denote row i and column i, respectively; for sets $A, B \subseteq N$, $\mathbf{T}_{AB}$ retains the rows in A and columns in B, where a single index i and $-i$ in subscripts stand for $\{i\}$ and $N \setminus \{i\}$: in particular, $\mathbf{T}_{-i} := \mathbf{T}_{-i,-i}$ deletes row and column i, while $\mathbf{T}_{i,-i}$ and $\mathbf{T}_{-i,i}$ retain row i across columns $N \setminus \{i\}$ and rows $N \setminus \{i\}$ in column i, respectively. Moreover, $\mathrm{diag}(\cdot)$ forms the diagonal matrix with the specified diagonal entries (or blocks), whereas for a square matrix $\mathbf{T}$, $\mathbf{T}^{\mathrm{D}}$ retains only the diagonal entries and $\rho(\mathbf{T})$ denotes the spectral radius.
- For $A \subseteq N$, $\mathbf{1}_A$ is the vector of ones on coordinates in A, $\mathbf{I}_A$ is the identity matrix on A (subscripts on $\mathbf{1}$ and $\mathbf{I}$ are omitted when clear from context), and $\mathbf{0}$ denotes a zero vector or zero matrix of conformable dimension. For sets $A, B \subseteq N$, $|A|$ is the cardinality of A, and $A \setminus B$ is the set of elements in A outside B.
- Vector and matrix inequalities are componentwise: $\mathbf{A} \geq \mathbf{B}$ (equivalently, $\mathbf{B} \leq \mathbf{A}$) means that every entry of $\mathbf{A} - \mathbf{B}$ is nonnegative, and $\mathbf{A} > \mathbf{B}$ (equivalently, $\mathbf{B} < \mathbf{A}$) means that every entry is strictly positive; in particular, comparisons with $\mathbf{0}$ apply entry by entry.

2.2 The Organizational Problem

We model leading by example through the timing of effort choices. As in Zhou and Chen (2015), an organizational planner who knows the network and all individual productivities selects a group of early movers before effort decisions take place in two stages; Figure 1 summarizes the timeline of the game.

In Stage 0, the planner publicly partitions the players into a leader group $L \subseteq N$ and a follower group $F := N \setminus L$. When players are ordered so that leaders precede

[7] In the standard local-aggregate model (see, for example, Ballester et al., 2006; Zhou and Chen, 2015), payoffs take the form $u_i^{\mathrm{agg}}(x_i, \mathbf{x}_{-i}; g) = \alpha_i x_i - \frac{1}{2}x_i^2 + \delta x_i \sum_{j=1}^{n} g_{ij}x_j$, where $\delta \geq 0$ measures the strength of network interaction. Higher effort by a neighbor raises both the marginal return to own effort and the payoff level ($\partial u_i^{\mathrm{agg}}(x_i, \mathbf{x}_{-i}; g)/\partial x_j = \delta g_{ij} x_i \geq 0$) without creating a norm-deviation penalty. In Section 3, we compare leadership in the two models.

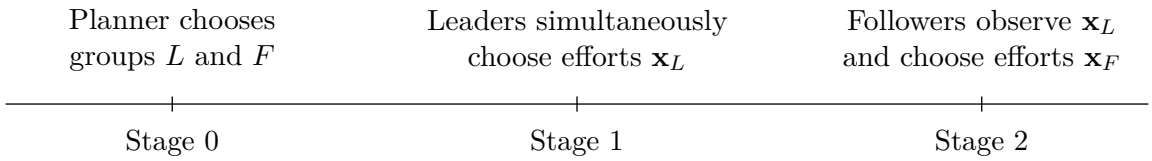


Figure 1: Timeline of the organizational problem.

followers, the adjacency matrix $\mathbf{G}$ and the row-normalized matrix $\widehat{\mathbf{G}}$ admit the block decompositions

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_{LL} & \mathbf{G}_{LF} \\ \mathbf{G}_{FL} & \mathbf{G}_{FF} \end{pmatrix}, \qquad \text{and} \qquad \widehat{\mathbf{G}} = \begin{pmatrix} \widehat{\mathbf{G}}_{LL} & \widehat{\mathbf{G}}_{LF} \\ \widehat{\mathbf{G}}_{FL} & \widehat{\mathbf{G}}_{FF} \end{pmatrix}. \tag{2}$$

In Stage 1, leaders in L simultaneously choose their efforts $\mathbf{x}_L$, taking into account the induced Stage 2 response of followers. In Stage 2, followers in F observe $\mathbf{x}_L$ and simultaneously choose their efforts $\mathbf{x}_F$. Each player acts once. Payoffs are realized according to (1) at the final effort profile.

For a leader group $L \subseteq N$, let $\mathbf{x}^S(L)$ denote the subgame-perfect equilibrium effort profile in the two-stage game and $X^S(L) := \sum_{i=1}^n x_i^S(L)$ denote the corresponding aggregate effort. The planner selects a leader group to maximize aggregate equilibrium effort:

$$\max_{L \subseteq N} X^S(L) = \sum_{i=1}^n x_i^S(L). \tag{3}$$

We refer to a leader group L^* that solves (3) as an *optimal leader group* (or a *key group*). To measure the effect of leadership relative to the simultaneous-move benchmark, let $\mathbf{x}^N$ denote the simultaneous-move Nash equilibrium (hereafter the *Nash equilibrium*) effort profile and $X^N := \sum_{i=1}^n x_i^N$ the Nash-benchmark aggregate effort. The *leadership gain* of group L is defined as $\ell(L) := X^S(L) - X^N$. Because X^N does not depend on L, solving the organizational problem in (3) is equivalent to maximizing the leadership gain $\ell(L)$.

2.3 Simultaneous-move Benchmark

In the simultaneous-move benchmark (which arises when $L = \varnothing$ or $L = N$), all players move at the same time. Each player $i \in N$ chooses $x_i \geq 0$ to maximize (1), taking the efforts $\mathbf{x}_{-i}$ of all other players—and hence the social norm $\bar{x}_i$—as given. Because (1) is strictly concave in x_i, the first-order condition $(\alpha_i - x_i) - \gamma(x_i - \bar{x}_i) = 0$, or equivalently

$x_i + \gamma(x_i - \bar{x}_i) = \alpha_i$, characterizes player i's unique best response. With $\lambda = \gamma/(1+\gamma)$, this condition yields the linear best-response function

$$x_i = (1-\lambda)\alpha_i + \lambda \bar{x}_i. \tag{4}$$

Each player's best response is a convex combination of own productivity α_i and the social norm $\bar{x}_i$, with weights $1-\lambda$ and λ that balance intrinsic effort incentives against the desire to conform. A stronger taste for conformity shifts weight toward the social norm and increases responsiveness to neighbors' efforts; because $\alpha_i > 0$, the best response is strictly positive for every nonnegative peer-effort profile and every $\lambda \in [0, 1)$. In the limiting case $\lambda = 0$ (*pure individualism*), each player ignores peer behavior and chooses $x_i = \alpha_i$; as $\lambda \to 1$ (*pure conformism*), each player seeks solely to match the social norm.

In vector form, the social norms are $\bar{\mathbf{x}} := \widehat{\mathbf{G}}\mathbf{x}$, so the best-response system (4) writes as $\mathbf{x} = (1-\lambda)\boldsymbol{\alpha} + \lambda\widehat{\mathbf{G}}\mathbf{x}$. Proposition 1 of Ushchev and Zenou (2020) characterizes the resulting Nash equilibrium.

Lemma 1 (Simultaneous-move equilibrium)**.** *The simultaneous-move game admits a unique Nash equilibrium, which is given by*

$$\mathbf{x}^N = \mathbf{M}^N \boldsymbol{\alpha}, \tag{5}$$

where $\mathbf{M}^N = [m_{ij}^N]$ *is an* $(n \times n)$ *simultaneous-move influence matrix defined as*[8]

$$\mathbf{M}^N = (1-\lambda)(\mathbf{I} - \lambda\widehat{\mathbf{G}})^{-1} = (1-\lambda)\sum_{t=0}^{\infty} \lambda^t \widehat{\mathbf{G}}^t. \tag{6}$$

Proof: See Appendix A.1.

By (5), player i's Nash effort can be written as $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$, where $m_{ij}^N = \partial x_i^N / \partial \alpha_j$ measures the marginal effect of player j's productivity on player i's equilibrium effort. As in Ushchev and Zenou (2020), the Neumann series in (6) expands this cross-effect across walk lengths:

$$m_{ij}^N = \frac{\partial x_i^N}{\partial \alpha_j} = \sum_{t=0}^{\infty} \underbrace{(1-\lambda)\lambda^t}_{\text{geometric weight}} (\widehat{\mathbf{G}}^t)_{ij},$$

where $(\widehat{\mathbf{G}}^t)_{ij}$ records the normalized weight of length-t walks from player i to player j, and the coefficients $(1-\lambda)\lambda^t$ form a geometric distribution over walk lengths $t \geq 0$.

[8]The adjacency matrix $\widehat{\mathbf{G}}$ is row-normalized, so $\rho(\widehat{\mathbf{G}}) = 1$. For $0 \leq \lambda < 1$, $\rho(\lambda\widehat{\mathbf{G}}) = \lambda < 1$, ensuring that the inverse exists and admits the Neumann series in (6).

Because $\widehat{\mathbf{G}}$ is row-stochastic, the influence matrix $\mathbf{M}^N$ is also row-stochastic, and connectedness ensures $m_{ij}^N > 0$ for all $i, j \in N$ whenever $0 < \lambda < 1$. Therefore, each player's Nash effort is a strictly positive convex combination of all individual productivities. When productivities are homogeneous ($\alpha_i = \alpha$ for all $i \in N$), row-stochasticity implies $x_i^N = \alpha$ for every player regardless of network position; unlike the local-aggregate model, network structure generates effort dispersion in Nash equilibrium only when productivities are heterogeneous.

In the absence of conformity ($\gamma = 0$), each player chooses her standalone effort, so $x_i^N = \alpha_i$ for all $i \in N$. The presence of conformity ($\gamma > 0$), however, creates a gap between player i's productivity α_i and Nash effort x_i^N, proportional to her deviation from the equilibrium social norm $\bar{x}_i^N := \sum_{j=1}^n \widehat{g}_{ij} x_j^N$:

$$\alpha_i - x_i^N = \gamma(x_i^N - \bar{x}_i^N) = \lambda(\alpha_i - \bar{x}_i^N). \tag{7}$$

Under heterogeneous productivities, the three differences in (7) share the same sign and can be either positive or negative depending on how player i's productivity compares with the social norm. When $\alpha_i > \bar{x}_i^N$, peer pressure constrains player i and depresses her effort below α_i; when $\alpha_i < \bar{x}_i^N$, peer pressure instead disciplines player i and raises her effort above α_i. In addition, because $\min_{j \in N} \alpha_j < \bar{x}_i^N < \max_{j \in N} \alpha_j$ for all $i \in N$ when $\gamma > 0$, a player with the highest productivity always exceeds the social norm, whereas a player with the lowest productivity always falls below it.

3 Leadership and Conformity

3.1 Equilibrium Characterization

For a leader group $L \subseteq N$, we solve for equilibrium efforts by backward induction, starting with followers' responses to leading efforts. The block expressions in this subsection apply to $0 < |L| < n$; the empty and full groups reproduce the Nash equilibrium.

Followers. In Stage 2, each follower $k \in F$ observes the leading effort vector $\mathbf{x}_L$ and chooses $x_k \geq 0$ to maximize (1), taking the efforts of all other players as given. By (4), follower k's best response is $x_k = (1 - \lambda)\alpha_k + \lambda \bar{x}_k$, where the social norm is a weighted average of the efforts of neighboring leaders and followers: $\bar{x}_k = \sum_{i \in L} \widehat{g}_{ki} x_i + \sum_{j \in F} \widehat{g}_{kj} x_j$. In vector form, under the block partition in (2), the follower best responses satisfy $\mathbf{x}_F = (1 - \lambda)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL} \mathbf{x}_L + \lambda \widehat{\mathbf{G}}_{FF} \mathbf{x}_F$. Define $\mathbf{U}_F := (\mathbf{I}_F - \lambda \widehat{\mathbf{G}}_{FF})^{-1}$. For a nonnegative

effort vector $\mathbf{x}_L$, the follower subgame admits a unique equilibrium with strictly positive efforts,

$$\mathbf{x}_F(\mathbf{x}_L) = \mathbf{U}_F\Big[(1-\lambda)\boldsymbol{\alpha}_F + \lambda\widehat{\mathbf{G}}_{FL}\mathbf{x}_L\Big]. \tag{8}$$

Leaders. In Stage 1, each leader $k \in L$ anticipates how $\mathbf{x}_L$ shapes followers' responses $\mathbf{x}_F(\mathbf{x}_L) = \big(x_j(\mathbf{x}_L)\big)_{j\in F}$ in (8) and hence the social norm against which leader k's own effort is evaluated:

$$\bar{x}_k(\mathbf{x}_L) = \sum_{i\in L}\widehat{g}_{ki}x_i + \sum_{j\in F}\widehat{g}_{kj}x_j(\mathbf{x}_L).$$

Because $g_{kk} = 0$ and $\mathbf{x}_F(\mathbf{x}_L)$ is linear in $\mathbf{x}_L$, leader k's effort x_k affects $\bar{x}_k(\mathbf{x}_L)$ entirely through followers' responses at the constant rate $c_k := \partial\bar{x}_k(\mathbf{x}_L)/\partial x_k = \sum_{j\in F}\widehat{g}_{kj}\,\partial x_j(\mathbf{x}_L)/\partial x_k$. A unit increase in x_k therefore raises leader k's norm by c_k and increases the deviation $x_k - \bar{x}_k(\mathbf{x}_L)$ by $1 - c_k$. Let $\mathbf{x}_{-k}(\mathbf{x}_L) := (\mathbf{x}_{L\setminus\{k\}}, \mathbf{x}_F(\mathbf{x}_L))$. With followers' responses given by (8), leader k's payoff in (1) is strictly concave in x_k, and the first-order condition decomposes the marginal payoff into three terms:

$$\frac{du_k\left(x_k, \mathbf{x}_{-k}(\mathbf{x}_L); g\right)}{dx_k} = \underbrace{\alpha_k - x_k}_{\substack{\text{standalone}\\ \text{productivity effect}}} - \underbrace{\gamma(x_k - \bar{x}_k)}_{\substack{\text{direct}\\ \text{social-norm effect}}} + \underbrace{\gamma c_k(x_k - \bar{x}_k)}_{\substack{\text{indirect}\\ \text{social-norm effect}}} = 0. \tag{9}$$

When leader k increases effort x_k, her marginal payoff is affected in three ways. First, there is a *standalone productivity effect*, $\alpha_k - x_k$, which is the marginal net return to effort in the absence of peer interactions. Second, there is a *direct social-norm effect*, $-\gamma(x_k - \bar{x}_k)$, which captures the marginal cost of departing from the social norm when $\bar{x}_k$ is held fixed. Whereas a follower in Stage 2 takes $\bar{x}_k$ as given and faces only these first two effects, a leader in Stage 1 also internalizes a third, strategic channel: the *indirect social-norm effect*, $\gamma c_k(x_k - \bar{x}_k)$, which arises because followers' responses shift $\bar{x}_k$ in the direction of x_k at rate $c_k \geq 0$. For a leader, the indirect social-norm effect always acts in the *opposite* direction to the direct social-norm effect, reflecting the dual role of social conformity.

Let $\bar{x}_k^S$ and $\bar{x}_k^N$ denote player k's equilibrium social norm in the two-stage and simultaneous-move games (we omit the argument L when the leader group is fixed). By (9), each leader $k \in L$ chooses an equilibrium effort that is a weighted average of individual productivity and the final social norm:

$$x_k^S = \frac{1-\lambda}{1-\lambda c_k}\,\alpha_k + \frac{\lambda(1-c_k)}{1-\lambda c_k}\,\bar{x}_k^S. \tag{10}$$

Under social conformity, compared with the best response in the Nash equilibrium (4), moving first allows the leader to shift weight from the social norm toward own productivity. Individual productivity exceeds (falls below) the social norm if and only if the leader's effort lies below (above) individual productivity, that is, $\bar{x}_k^S < x_k^S < \alpha_k$ ($\alpha_k < x_k^S < \bar{x}_k^S$).[9]

In vector form, the leaders' social norms are $\bar{\mathbf{x}}_L(\mathbf{x}_L) := \big(\bar{x}_k(\mathbf{x}_L)\big)_{k\in L} = \widehat{\mathbf{G}}_{LL}\mathbf{x}_L + \widehat{\mathbf{G}}_{LF}\mathbf{x}_F(\mathbf{x}_L)$. By (8), we obtain

$$\bar{\mathbf{x}}_L(\mathbf{x}_L) = (1-\lambda)\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F + \mathbf{C}_L\mathbf{x}_L, \tag{11}$$

with the *norm-response matrix*

$$\mathbf{C}_L := \frac{\partial \bar{\mathbf{x}}_L(\mathbf{x}_L)}{\partial \mathbf{x}_L} = \widehat{\mathbf{G}}_{LL} + \lambda\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}, \tag{12}$$

where $\widehat{\mathbf{G}}_{LL}$ represents the direct influence of other leaders' efforts on the social norm and $\lambda\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}$ represents the indirect response through followers. Because the network has no self-loops, each diagonal entry $c_k := (\mathbf{C}_L)_{kk}$ arises entirely through followers:

$$c_k = \frac{\partial \bar{x}_k(\mathbf{x}_L)}{\partial x_k} = \lambda(\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL})_{kk} = \sum_{t=0}^{\infty}\lambda^{t+1}\Big(\widehat{\mathbf{G}}_{LF}\widehat{\mathbf{G}}_{FF}^{\,t}\widehat{\mathbf{G}}_{FL}\Big)_{kk}. \tag{13}$$

Each term records return walks that leave leader k for a neighboring follower, take t steps within the subnetwork induced by the followers, and return to k. The response matrices $\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}$ and $\mathbf{C}_L$ are nonnegative, with $\mathbf{C}_L\mathbf{1}_L \le \mathbf{1}_L$ and $0 \le c_k \le \lambda < 1$. Rewriting (10) in vector form gives

$$(1-\lambda)\mathbf{x}_L^S + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{x}_L^S - \bar{\mathbf{x}}_L^S) = (1-\lambda)\boldsymbol{\alpha}_L.$$

Together with $\bar{\mathbf{x}}_L^S = \bar{\mathbf{x}}_L(\mathbf{x}_L^S)$ from (11), define

$$\mathbf{V}_L := \Big[(1-\lambda)\mathbf{I}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L)\Big]^{-1}.$$

Then the equilibrium leading efforts are

$$\mathbf{x}_L^S = (1-\lambda)\mathbf{V}_L\Big[\boldsymbol{\alpha}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F\Big].$$

[9] Analogously to (7), rearranging (10) yields $\alpha_k - x_k^S = \gamma(1-c_k)(x_k^S - \bar{x}_k^S) = \frac{\lambda(1-c_k)}{1-\lambda c_k}(\alpha_k - \bar{x}_k^S)$. Because $0 \le c_k < 1$, the three differences share the same sign whenever $\gamma > 0$.

Theorem 1 (Subgame-perfect equilibrium)**.** *For any leader group $L \subseteq N$, the two-stage game admits a unique subgame-perfect Nash equilibrium. With leaders ordered first, equilibrium efforts are given by*

$$\begin{pmatrix} \mathbf{x}_L^S \\ \mathbf{x}_F^S \end{pmatrix} = \mathbf{M}^S(L) \begin{pmatrix} \boldsymbol{\alpha}_L \\ \boldsymbol{\alpha}_F \end{pmatrix},$$

where $\mathbf{M}^S(L) = [m_{ij}^S(L)]$ is an $(n \times n)$ two-stage influence matrix defined as

$$\mathbf{M}^S(L) := (1-\lambda) \begin{pmatrix} \mathbf{V}_L & \lambda \mathbf{V}_L (\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}}) \widehat{\mathbf{G}}_{LF} \mathbf{U}_F \\ \lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL} \mathbf{V}_L & \mathbf{U}_F + \lambda^2 \mathbf{U}_F \widehat{\mathbf{G}}_{FL} \mathbf{V}_L (\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}}) \widehat{\mathbf{G}}_{LF} \mathbf{U}_F \end{pmatrix}. \tag{14}$$

Proof: See Appendix A.2.

The entry $m_{ij}^S(L) = \partial x_i^S(L)/\partial \alpha_j$ measures the marginal effect of player j's productivity on player i's equilibrium effort (with the argument L omitted when the leader group is fixed). For $0 < \lambda < 1$, all entries of $\mathbf{M}^S(L)$ are strictly positive, so a higher productivity for any player raises effort throughout the network. As in the simultaneous-move model of Ushchev and Zenou (2020), $\mathbf{M}^S(L)$ is row-stochastic: each equilibrium effort $x_i^S(L) = \sum_{j=1}^n m_{ij}^S(L)\alpha_j$ is a convex combination of productivities bounded between $\min_{j \in N} \alpha_j$ and $\max_{j \in N} \alpha_j$. If players are homogeneous ($\alpha_i = \alpha$ for all $i \in N$), then $x_i^S(L) = x_i^N = \alpha$ for all $i \in N$ and $L \subseteq N$, so $\ell(L) = 0$: leadership alters effort only when productivities are heterogeneous.

When productivities are heterogeneous, the block structure of $\mathbf{M}^S(L)$ in (14) reflects how leadership interacts with the dual role of social conformity. On the one hand, followers adjust their efforts in response to changes in leaders' efforts, as described by $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}$. By changing the leaders' social norms, these responses reduce the effective pressure to conform and shift equilibrium weight toward the leaders' own productivities. On the other hand, leaders continue to face peer pressure from both other leaders and followers, so followers' productivities $\boldsymbol{\alpha}_F$ shape leaders' equilibrium efforts through $(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}}) \widehat{\mathbf{G}}_{LF} \mathbf{U}_F$ and spill back over to the follower group. Unlike the local-aggregate model of Zhou and Chen (2015), where moving first weakly raises all efforts under sufficiently weak interactions, row-stochasticity of $\mathbf{M}^S(L)$ and $\mathbf{M}^N$ implies that leadership under social conformity redistributes productivity weights and can either increase or decrease equilibrium efforts relative to the Nash benchmark.

Proposition 1 (Equilibrium effort comparison)**.** *For a leader group $L \subseteq N$, leadership increases the equilibrium effort of player $j \in N$ relative to the Nash benchmark ($x_j^S > x_j^N$) if and only if*

$$\sum_{i \in L} m_{ji}^S c_i (\alpha_i - x_i^N) > 0. \tag{15}$$

Proof: See Appendix A.3.

To see why $c_i(\alpha_i - x_i^N)$ governs leader i's impact in (15), evaluate leader i's Stage 1 marginal payoff in (9) at the Nash equilibrium $\mathbf{x}^N$, where the standalone productivity effect and the direct social-norm effect cancel by (7), leaving only the indirect social-norm effect:

$$\left.\frac{du_i\left(x_i, \mathbf{x}_{-i}(\mathbf{x}_L); g\right)}{dx_i}\right|_{\mathbf{x}=\mathbf{x}^N} = \underbrace{(\alpha_i - x_i^N) - \gamma(x_i^N - \bar{x}_i^N)}_{=0} + \gamma c_i (x_i^N - \bar{x}_i^N) = c_i(\alpha_i - x_i^N).$$

The difference $\alpha_i - x_i^N$ between productivity and Nash effort therefore determines the direction of leader i's first-mover incentive, the induced norm shift c_i scales its magnitude, and the equilibrium weight m_{ji}^S transmits the resulting adjustment to player j (with the reverse strict inequality in (15) characterizing $x_j^S < x_j^N$). With multiple leaders, opposing incentives can offset one another and generate different net effects across players. However, in the case of a single leader, only one term enters (15) and all equilibrium efforts move in the same direction.

Remark 1. For a single leader $L = \{i\}$, equilibrium efforts move in the same direction across the network, increasing relative to the Nash benchmark if and only if $\alpha_i > x_i^N > \bar{x}_i^N$, or equivalently,

$$\alpha_i > \sum_{j \neq i} \frac{m_{ij}^N}{1 - m_{ii}^N} \alpha_j. \tag{16}$$

Proof: See Appendix A.4.

Since $\sum_{j \neq i} m_{ij}^N / (1 - m_{ii}^N) = 1$, the right-hand side of (16) is a weighted average of all other players' productivities, with weights determined by their simultaneous-move influence on player i. A single leader whose productivity exceeds this average productivity is constrained by a lower social norm in the Nash equilibrium ($\alpha_i > x_i^N > \bar{x}_i^N$), so moving first increases her effort and induces higher efforts from all followers; conversely, a leader whose productivity falls below this average productivity is disciplined by a higher social

norm in the Nash equilibrium ($\alpha_i < x_i^N < \bar{x}_i^N$), so moving first reduces her effort and lowers all others' efforts.

3.2 Single Leader

We begin with the case where the organizational planner selects a single leader $L = \{i\}$. The planner then solves the *key leader problem*, defined as $\max_{i \in N} X^S(\{i\})$, and a player who solves this problem is referred to as a *key leader*. For notational simplicity, we write $\ell_i := \ell(\{i\}) = X^S(\{i\}) - X^N$ for the leadership gain of a single leader i.

The choice of a single leader trades off the candidate's incentive to adjust effort against the network propagation of that adjustment. When $L = \{i\}$, let $x_j(x_i)$ denote follower j's response in (8). Each follower $j \neq i$ responds at the constant rate $\partial x_j(x_i)/\partial x_i = m_{ji}^N / m_{ii}^N$, and leader i's norm response is

$$c_i = \frac{m_{ii}^N + \lambda - 1}{\lambda m_{ii}^N} \in (0, \lambda]. \tag{17}$$

Together with $x_i^N = \sum_{k=1}^n m_{ik}^N \alpha_k$, (17) implies that the simultaneous-move influence matrix $\mathbf{M}^N = [m_{ij}^N]$ pins down followers' responsiveness, leader i's norm response c_i (which increases in m_{ii}^N), and the gap between productivity and Nash effort.

Proposition 2 (Key leader under conformity)**.** *A key leader i^* maximizes the leadership gain*

$$\ell_i = \frac{(1-\lambda)c_i}{1 - \lambda c_i(2 - c_i)} \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N} (\alpha_i - x_i^N). \tag{18}$$

We refer to ℓ_i as the local-average leading index *of player i.*

Proof: See Appendix A.5.

The leading index ℓ_i decomposes into the leader's own effort adjustment $x_i^S - x_i^N$ and a network propagation multiplier. In the first factor, $(1-\lambda)c_i$ captures the marginal incentive to adjust effort by shifting the social norm, while the denominator $1-\lambda c_i(2-c_i) = (1-\lambda)+\lambda(1-c_i)^2$ reflects the curvature of the leader's objective, combining intrinsic effort-cost curvature $(1-\lambda)$ with the residual norm-deviation curvature $\lambda(1-c_i)^2$. The second factor, $\sum_{j=1}^n m_{ji}^N / m_{ii}^N = 1 + \sum_{j \neq i} \partial x_j(x_i)/\partial x_i$, aggregates followers' resulting responses across the network. Because both network factors are strictly positive, $\text{sign}(\ell_i) = \text{sign}(\alpha_i - x_i^N)$: under heterogeneous productivities, a player with the highest productivity

always raises aggregate effort ($\ell_i > 0$), whereas a player with the lowest productivity lowers it ($\ell_i < 0$). Among candidates with $\alpha_i > x_i^N$, however, a more productive player can be a weaker leader than a candidate whose network position induces a larger shift in her social norm and a larger aggregate response among followers.

The role of social conformity. How a candidate's leading index $\ell_i(\lambda)$ varies with the taste for conformity depends on the peer benchmark embedded in $x_i^N(\lambda)$. As conformity strengthens, that benchmark shifts from the local neighbor average $\bar{\alpha}_i := \sum_{j=1}^n \widehat{g}_{ij}\alpha_j$ to the degree-weighted network average $\bar{\alpha}_g := \sum_{i=1}^n \pi_i \alpha_i$, where $\pi_i := d_i / \sum_{j=1}^n d_j$.

Corollary 1 (Key leader under weak and strong conformity)**.**

(i) *For sufficiently small conformity, a key leader maximizes $r_i(\alpha_i - \bar{\alpha}_i)$, where $r_i := (\widehat{\mathbf{G}}^2)_{ii} = \frac{1}{d_i}\sum_{j\in N(i)} \frac{1}{d_j}$.*

(ii) *For sufficiently large conformity, a key leader is a player with the highest productivity.*

Proof: See Appendix A.6.

In part (i), when the taste for conformity is sufficiently small ($\lambda \to 0$), the gap between productivity and Nash effort satisfies $\alpha_i - x_i^N = \lambda(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$. Meanwhile, expanding (13) for $L = \{i\}$ gives

$$c_i = \lambda(\widehat{\mathbf{G}}^2)_{ii} + \sum_{t=1}^{\infty} \lambda^{t+1} \widehat{\mathbf{G}}_{i,-i} \widehat{\mathbf{G}}_{-i}^t \widehat{\mathbf{G}}_{-i,i} = \lambda \sum_{j\in N(i)} \widehat{g}_{ij}\widehat{g}_{ji} + \mathcal{O}(\lambda^2) = \lambda r_i + \mathcal{O}(\lambda^2).$$

Substituting these expansions into (18) yields $\ell_i(\lambda) = \lambda^2 r_i(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^3)$. Define $r_{ij} := \widehat{g}_{ij}\widehat{g}_{ji} = g_{ij}/(d_i d_j)$, so that $r_i = \sum_{j\in N(i)} r_{ij}$; we refer to r_{ij} as the *link reciprocity* between players i and j, and to r_i as the *local reciprocity* of player i.[10] For leader i's neighbor $j \in N(i)$, $\widehat{g}_{ji} = 1/d_j$ represents the weight that neighbor j places on player i's effort, so $r_i = d_i^{-1} \sum_{j\in N(i)} \widehat{g}_{ji}$ equals the average weight that i's neighbors assign to i in their own social norms. In a largely individualist organization, the planner therefore favors a candidate with a large local productivity advantage $\alpha_i - \bar{\alpha}_i$ and high local reciprocity r_i: when a candidate's neighbors have fewer links, those neighbors do not dilute i's influence across many peers and thus induce a larger shift in i's social norm.

[10] In the probabilistic interpretation of the local-average model (Ushchev and Zenou, 2020), where $\widehat{\mathbf{G}} = [\widehat{g}_{ij}]$ is the transition probability matrix of a Markov chain on N, $r_{ij} = \widehat{g}_{ij}\widehat{g}_{ji}$ is the probability of moving from state i to state j and returning to i in the next step, and $r_i = \sum_{j\in N(i)} r_{ij} = (\widehat{\mathbf{G}}^2)_{ii}$ is the two-step return probability of state i.

In part (ii), when the taste for conformity is sufficiently large ($\lambda \to 1$), leader selection depends only on individual productivity. Proposition 3 of Ushchev and Zenou (2020) shows that Nash efforts converge to the degree-weighted average productivity, $\bar{\alpha}_g$.[11] When player i moves first, however, followers' efforts respond increasingly one for one to the leader's effort. The resulting adjustment of the social norm allows the leader to choose effort closer to own productivity; in the limit, all players exert effort α_i. The limiting leadership gain is therefore $\lim_{\lambda\to 1} \ell_i(\lambda) = n(\alpha_i - \bar{\alpha}_g)$. Since all candidates face the same Nash benchmark, a player with the highest productivity is a key leader.

For intermediate levels of conformity, leader selection also depends on network position and the productivity of other players. Corollary 2 characterizes how the leading index responds to individual productivities, the taste for conformity, and the addition of links between followers.

Corollary 2 (Comparative statics)**.** *The local-average leading index ℓ_i of a player $i \in N$ satisfies the following properties:*

(i) Productivities. *It strictly increases in player i's own productivity α_i but strictly decreases in any follower's productivity α_j $(j \neq i)$.*

(ii) Conformity. *If $\bar{\alpha}_i < \alpha_i \leq \bar{\alpha}_g$, ℓ_i is nonmonotone in λ and attains an interior global maximum; if $\bar{\alpha}_g \leq \alpha_i < \bar{\alpha}_i$, it attains an interior global minimum.*

(iii) Link dilution. *If $\alpha_i > \bar{\alpha}_i$, adding a new link between a neighbor $j \in N(i)$ and another follower strictly decreases ℓ_i for all sufficiently small λ.*

Proof: See Appendix A.7.

In part (i), higher own productivity widens the gap between productivity and Nash effort because Nash effort rises less than one for one. An increase in a follower's productivity instead raises the candidate's Nash effort without changing own productivity, which narrows that gap and reduces the gain from moving first. Thus, more productive followers raise aggregate effort but reduce the additional effort generated by appointing player i as the leader.

In part (ii), the candidate's productivity exceeds the local neighbor average but does not exceed the degree-weighted network average. Leadership therefore raises aggregate effort under weak conformity, whereas its gain converges to a nonpositive value as

[11] Because g is connected, all degree weights $\pi_i = d_i / \sum_{j=1}^{n} d_j$ lie in $(0, 1)$ and sum to one. Under heterogeneous productivities, $\bar{\alpha}_g$ is therefore a strict convex combination satisfying $\min_{i\in N} \alpha_i < \bar{\alpha}_g < \max_{i\in N} \alpha_i$.

$\lambda \to 1$. Since the gain also vanishes at $\lambda = 0$, it must attain a global maximum at an intermediate level of conformity. The reversed productivity comparisons yield an interior global minimum.

In part (iii), an additional link between a neighbor and another follower reduces the weight that the neighbor places on the leader's effort. Local reciprocity therefore falls, while the candidate's productivity advantage over the local neighbor average remains unchanged. Under weak conformity, the weaker response of the leader's social norm reduces the gain from moving first.

Comparison with the local-aggregate model. To isolate how social conformity affects leader selection, we compare the two games while holding simultaneous-move behavior and followers' responses fixed. The payoff in (1) can be written as

$$u_i(x_i, \mathbf{x}_{-i}; g) = \alpha_i x_i - \frac{1}{2}x_i^2 - \frac{\gamma}{2}(x_i - \bar{x}_i)^2 = \underbrace{\alpha_i x_i - \frac{1+\gamma}{2}x_i^2 + \gamma x_i \sum_{j=1}^{n} \widehat{g}_{ij} x_j}_{=:u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)} - \frac{\gamma}{2}\bar{x}_i^2, \quad (19)$$

where $u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)$ is a local-aggregate payoff.[12] For given actions of other players, the additional term $-\gamma \bar{x}_i^2/2$ is independent of own effort. The two games therefore have the same simultaneous-move best responses, Nash equilibrium $\mathbf{x}^N$, and followers' responses in Stage 2 (Ushchev and Zenou, 2020).

The difference arises in Stage 1. A leader anticipates how an early action changes followers' efforts and, through those responses, the leader's own social norm. The additional payoff term therefore affects the leader's effort choice even though it leaves simultaneous-move behavior unchanged.

In the local-aggregate game, the gain in aggregate effort from appointing player i as the single leader is measured by the *local-aggregate leading index* (Zhou and Chen, 2015):

$$\ell_i^{\text{agg}} = \frac{\lambda c_i}{1 - 2\lambda c_i} \frac{\sum_{j=1}^{n} m_{ji}^N}{m_{ii}^N} x_i^N, \quad (20)$$

[12] Dividing $u_i^{\text{agg}}(x_i, \mathbf{x}_{-i}; g)$ by $1+\gamma$ gives the standard local-aggregate payoff $(1-\lambda)\alpha_i x_i - \frac{1}{2}x_i^2 + \lambda x_i \sum_{j=1}^{n} \widehat{g}_{ij} x_j$ with row-normalized adjacency matrix $\widehat{\mathbf{G}}$ (Ushchev and Zenou, 2020). Throughout our comparisons, the local-aggregate model (or local-aggregate benchmark) refers to u_i^{agg} in (19).

which is well defined when $1 - 2\lambda c_i > 0$.[13] By the Nash equilibrium relation in (7), the local-average leading index in (18) can be written in a comparable form:

$$\ell_i = \frac{\lambda c_i}{1 - 2\lambda c_i + \lambda c_i^2} \frac{\sum_{j=1}^{n} m_{ji}^N}{m_{ii}^N} (x_i^N - \bar{x}_i^N). \tag{21}$$

Comparing (20) and (21) highlights two main differences. First, the local-aggregate leading index depends on the candidate's Nash effort level, whereas the local-average leading index depends on the difference between Nash effort and the social norm. Consequently, an increase in a neighbor's productivity raises the local-aggregate leading index but lowers the local-average leading index. This is because more productive neighbors increase the candidate's Nash effort in both games; under conformity, however, the social norm rises by more, which reduces the gain from moving first.

Second, the denominator of the local-average leading index contains an additional curvature term (λc_i^2). Because moving first changes the leader's own social norm, the additional payoff term in (19) makes the leader's objective more concave. This limits the effort adjustment and ensures a finite equilibrium effort for all $\lambda \in (0, 1)$.

Example 1. Consider the bridge network in Figure 2, where the triangles formed by players $\{1, 2, 3\}$ and $\{4, 5, 6\}$ are joined by the bridge link $\{3, 4\}$. Players 1 and 6 have the lowest productivity, α_B, while the remaining players have the highest productivity, $\alpha_A > \alpha_B > 0$. Designating either player 1 or player 6 as the leader lowers aggregate effort. The planner therefore compares the equally productive bridge players 3 and 4 with the non-bridge players 2 and 5. By symmetry, it suffices to compare players 2 and 3.

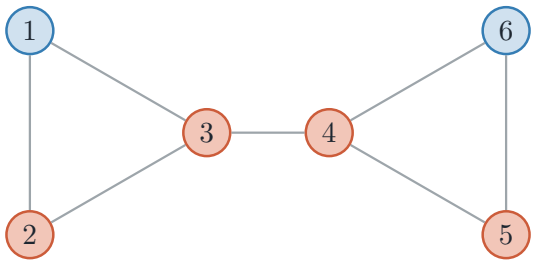


Figure 2: A bridge network

Note: Red and blue nodes indicate high and low individual productivities, respectively.

For any admissible strength of network effects, player 3 has a higher Nash effort than player 2 and is a key leader in the local-aggregate benchmark. Under social conformity,

[13] See Appendix A.8 for the derivation of (20).

however, the relevant comparison concerns the difference between Nash effort and the social norm. Although player 3 exerts more effort, player 3 also faces a higher social norm. In fact, player 2's effort exceeds the social norm by a larger margin:

$$\frac{x_2^N - \bar{x}_2^N}{x_3^N - \bar{x}_3^N} = \frac{3}{\lambda + 2} > 1.$$

This larger margin gives player 2 a stronger incentive to increase effort by moving first, which can outweigh player 3's advantage in network position.

For illustration, set $\alpha_A = 2$ and $\alpha_B = 1$. The calculations in Panel A of Table 1 show that at $\lambda = 0.20$, player 3 is a key leader in the local-aggregate model, whereas player 2 is a key leader in the local-average model. As the taste for social conformity increases, player 2's relative advantage in effort above the social norm diminishes. At $\lambda = 0.60$, player 3's advantage in network position outweighs that difference, and both models select a bridge player, as shown in Panel B.

Table 1: Local-aggregate and local-average leading indices in the bridge network

Player	c_i	$\sum_j m_{ji}^N / m_{ii}^N$	x_i^N	$x_i^N - \bar{x}_i^N$	ℓ_i^{agg}	ℓ_i
Panel A. $\lambda = 0.20$						
1	0.0908	1.1877	1.1848	−0.7390	0.0265	−0.0165
2	0.0908	1.1877	1.9120	0.3519	0.0428	**0.0079**
3	0.0966	1.3049	1.9355	0.2581	**0.0508**	0.0068
Panel B. $\lambda = 0.60$						
1	0.3395	1.8097	1.4825	−0.3217	0.9223	−0.1792
2	0.3395	1.8097	1.7902	0.1399	1.1138	0.0779
3	0.3662	2.3054	1.8182	0.1212	**1.6428**	**0.0958**

Notes: $\alpha_A = 2$ and $\alpha_B = 1$. Players 1, 2, and 3 represent the symmetric pairs $\{1, 6\}$, $\{2, 5\}$, and $\{3, 4\}$, respectively. Both indices measure the increase in aggregate effort relative to the common simultaneous-move benchmark. Bold entries identify the largest index in each panel.

3.3 Multiple Leaders

We now allow the planner to select multiple players into the leader group to solve (3), which we refer to as the *key group problem*; we call a group L^* that solves (3) an *optimal leader group* (or a *key group*). Under social conformity, group selection entails two trade-offs that do not arise in the single-leader problem. First, whereas all players have a positive first-mover incentive in the local-aggregate model, under conformity a leader

with productivity below the social norm adjusts effort downward and can offset the effort increases induced by other leaders. Second, because leaders move simultaneously in Stage 1, promoting an additional player into L allows one more player to move first but removes a follower from F, which dampens the induced shift in neighboring leaders' social norms (c_i) and leaves those leaders more constrained by the social norm. Selecting a key group therefore requires screening out low-productivity candidates while balancing the gains from additional leaders against the preservation of followers' responses.

Definition 1 (Dominating set)**.** A set $S \subseteq N$ is a *dominating set* if each player outside S has at least one neighbor in S, that is, $N(i) \cap S \neq \varnothing$ for $i \in N \setminus S$.

Proposition 3 (Leader eligibility and follower domination)**.** *Suppose that productivities are heterogeneous. For any optimal leader–follower partition* (L^*, F^*) *with* $F^* := N \setminus L^*$*:*

(i) *The leader group* L^* *consists only of players with a positive local-average leading index* ($\ell_i > 0$)*.*

(ii) *The follower group* F^* *forms a* dominating set*.*

Proof: See Appendix A.10.

In part (i), by Remark 1 and Proposition 2, $\ell_i > 0$ is equivalent to $\alpha_i > x_i^N > \bar{x}_i^N$. To understand this result, consider a leader in L whose productivity lies at or below the social norm, i.e., $\alpha_i \leq \bar{x}_i^S(L)$, so that $\alpha_i \leq x_i^S(L)$. Moving first allows such a player to shift her social norm downward and reduce her effort. Reassigning any such leaders to the follower group eliminates their downward adjustments, amplifies the induced shift in the remaining leaders' social norms, and raises effort across the network. At an optimum, every leader therefore remains below her own productivity while raising efforts above the Nash benchmark, which yields $\bar{x}_i^N < x_i^N < x_i^S(L^*) < \alpha_i$ for all $i \in L^*$. In particular, a direct implication of part (i) is that assigning a least productive player to the leader group is never optimal.

Whereas part (i) restricts the leader group to candidates with $\ell_i > 0$, part (ii) implies that not all such candidates can lead together: even a high-productivity player may need to remain in the follower group so that F^* forms a dominating set. To see why, suppose that a leader has all of her neighbors in the leader group and none in the follower group. Because all of her neighbors act simultaneously with her, none of them can adjust to her effort subsequently; moving first therefore fails to shift her social norm and leaves

her effort rule identical to that of a follower. Reassigning that player to the follower group does not alter her own response rule, but allows her to respond to her neighboring leaders, which allows the remaining leaders to induce larger shifts in their social norms and raises effort across the network.

Follower domination consequently precludes appointing a leader together with all of her neighbors. Because both parts of Proposition 3 rely only on reassigning a leader to the follower group, the same conditions continue to hold under any upper bound on the leader group size.

Figure 3 provides an example to illustrate this tension. In the 3-tier tree shown in the figure, individual productivities decrease from the top player 1 (α_1) through the middle-tier players 2 and 3 ($\alpha_2 = \alpha_3$) to the bottom players 4 and 5 ($\alpha_4 = \alpha_5$), and it can be readily shown that the leading indices satisfy $\ell_1 > \ell_2 = \ell_3 > 0 > \ell_4 = \ell_5$.

Consider first the key leader problem in panel (a): because player 1 has the highest leading index, player 1 is selected as the unique key leader. Consider next the key group problem in panel (b). Although players 1, 2, and 3 all have positive leading indices, appointing all three would leave player 1 connected only to co-leaders, violating the dominating structure of part (ii). Instead, the optimal leader group consists of the middle-tier players $L^* = \{2, 3\}$, while the most productive player 1 joins the bottom players in the follower group $F^* = \{1, 4, 5\}$ to form a dominating set.

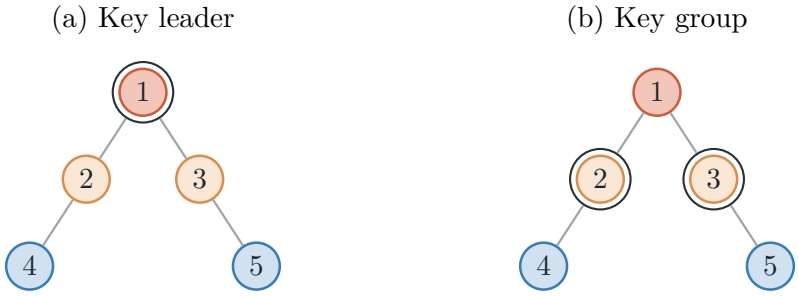


Figure 3: Key leader and key group in a 3-tier tree

Note: Red, orange, and blue nodes indicate high, medium, and low individual productivities, respectively; an outer ring identifies a leader. Panel (a) shows the unique key leader $L^* = \{1\}$, and panel (b) shows the unique key group $L^* = \{2, 3\}$.

Proposition 4 (Key groups under weak and strong conformity)**.** *Suppose that productivities are heterogeneous.*

(i) *For sufficiently small conformity, an optimal leader group maximizes*

$$\Lambda(L) := \sum_{i \in L} \sum_{j \in F} r_{ij}(\alpha_i - \bar{\alpha}_i), \tag{22}$$

where $r_{ij} := \widehat{g}_{ij}\widehat{g}_{ji}$.

(ii) *For sufficiently large conformity, an optimal leader group is a singleton consisting of a player with the highest productivity.*

Proof: See Appendix A.11.

When the taste for conformity is sufficiently small ($\lambda \to 0$), the leadership gain satisfies $\ell(L) = \lambda^2 \Lambda(L) + \mathcal{O}(\lambda^3)$. A leader–follower link allows a change in the leader's effort to feed back into the leader's own social norm. The weight $\widehat{g}_{ji}$ governs follower j's response to leader i, while $\widehat{g}_{ij}$ determines how that response affects leader i's social norm. The product $r_{ij} = \widehat{g}_{ij}\widehat{g}_{ji}$ therefore enters the link's contribution $r_{ij}(\alpha_i - \bar{\alpha}_i)$ to $\Lambda(L)$. By contrast, players within either group choose efforts simultaneously, taking one another's actions as given. A change in one player's effort therefore induces no subsequent response from the other along their shared link, so the link does not generate this feedback.

Definition 2 (Maximum directed cut)**.** Given a directed graph (V, E) with edge weights $w : E \to \mathbb{R}$, the *maximum directed cut* (MAX-DiCUT) problem selects a vertex subset $S \subseteq V$ that maximizes the total weight of outgoing edges, that is,

$$\max_{S \subseteq V} \sum_{\substack{(i,j) \in E \\ i \in S,\, j \notin S}} w_{ij}.$$

Definition 2 follows Ghuge et al. (2022). When the taste for conformity is sufficiently small, the group selection problem admits this representation: each social link gives rise to two directed edges, whose weights reflect the gains from the two possible orders of moves. Let $V = N$ and assign

$$w_{ij} := r_{ij} \max\{\alpha_i - \bar{\alpha}_i, 0\}.$$

For a player whose productivity exceeds the neighborhood average, this weight captures the contribution to aggregate effort from moving before a neighbor, as the neighbor's response shifts the leader's social norm. Otherwise, the weight is zero. The selected group

and its complement correspond to leaders and followers, respectively, so the planner chooses the partition that generates the largest total contribution across groups.[14]

Unlike the maximum cut (MAX-CUT) problem in the local-aggregate model of Zhou and Chen (2015), link weights in the local-average setup depend on individual productivity relative to the neighborhood average. Reversing two adjacent players' roles therefore generally changes the gain. The partition determines both which links support sequential responses and whose productivity drives those responses.

Let $N_+ := \{i \in N : \alpha_i > \bar{\alpha}_i\}$ denote the set of candidates with positive local productivity gaps. Nevertheless, when candidates in N_+ share the same local productivity advantage and link reciprocities coincide, maximizing $\Lambda(L)$ contains the classical maximum cut problem as a special case.

Claim 1. *Solving the key group problem in (22) is NP-hard in general. On the candidate domain 2^{N_+}, the second-order objective $\Lambda(L)$ is submodular:*[15] *the marginal gain from appointing a candidate falls strictly whenever an adjacent candidate also leads.*

Proof: See Appendix A.12.

To see the source of submodularity, rewrite $\Lambda(L)$ for any candidate group $L \subseteq N_+$ using local reciprocity r_i:

$$\Lambda(L) = \sum_{i \in L} r_i(\alpha_i - \bar{\alpha}_i) - \sum_{\substack{i,j \in L \\ i<j}} r_{ij}[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)]. \tag{23}$$

The first sum adds the candidates' standalone second-order gains $\Lambda(\{i\}) = r_i(\alpha_i - \bar{\alpha}_i)$, while the second sum deducts the two responses lost whenever adjacent candidates move together in Stage 1 instead of responding to each other. Because this pairwise loss can outweigh a candidate's standalone gain, an optimal leader group may keep a high-index candidate as a follower rather than ranking leaders by individual indices ℓ_i alone. Moreover, since submodularity makes each candidate's marginal contribution

[14] Since $r_{ij} \geq 0$, taking the positive part of the local productivity gap gives nonnegative edge weights for MAX-DiCUT problem. In addition, a maximizing set yields a maximizer of Λ after removing any selected players with zero outgoing weight. This removal cannot reduce the cut value. Conversely, removing all players whose productivity does not exceed their neighbors' average from any group cannot reduce Λ, so the two problems have the same optimal value.

[15] A set function $f : 2^V \to \mathbb{R}$ is *submodular* if $f(S) + f(T) \geq f(S \cup T) + f(S \cap T)$ for all $S, T \subseteq V$, or equivalently if the marginal contribution $f(S \cup \{i\}) - f(S)$ decreases with S.

lowest when all other candidates lead, appointing the entire candidate set N_+ maximizes Λ if and only if, for each $i \in N_+$,

$$(\alpha_i - \bar{\alpha}_i) \sum_{j \notin N_+} r_{ij} \geq \sum_{j \in N_+} r_{ij}(\alpha_j - \bar{\alpha}_j). \tag{24}$$

Condition (24) requires candidate i's leading contribution through noncandidate followers (left-hand side) to exceed the responses from followers that adjacent candidates who also lead lose when i leads rather than follows (right-hand side). In particular, if N_+ is an *independent set* (so candidates share no links), the right-hand side vanishes and all candidates in N_+ are appointed; conversely, if some candidate $i \in N_+$ has only candidates as neighbors, that is, $N(i) \subseteq N_+$, the left-hand side vanishes and a key group must leave at least one member of N_+ as a follower.

Part (ii) shows that when the taste for conformity is sufficiently large, the planner maximizes aggregate effort by appointing a single player with the highest productivity. Denote $c_i(1) := \lim_{\lambda \to 1} c_i(\lambda)$. Consider a leader group with multiple leaders, that is, $|L| \geq 2$. As $\lambda \to 1$, the equilibrium conditions imply that the effort of every player $k \in N$ converges to the common limit (see Appendix A.11)

$$\lim_{\lambda \to 1} x_k^S(L;\lambda) = \frac{\displaystyle\sum_{i \in L} \frac{\pi_i}{1 - c_i(1)} \alpha_i + \sum_{j \in F} \pi_j \alpha_j}{\displaystyle\sum_{i \in L} \frac{\pi_i}{1 - c_i(1)} + \sum_{j \in F} \pi_j} = \frac{\bar{\alpha}_g + \displaystyle\sum_{i \in L} \frac{\pi_i c_i(1)}{1 - c_i(1)} \alpha_i}{1 + \displaystyle\sum_{i \in L} \frac{\pi_i c_i(1)}{1 - c_i(1)}}, \tag{25}$$

where $\bar{\alpha}_g = \sum_{i \in N} \pi_i \alpha_i$ represents the common Nash effort in the limiting case of strong conformity.

Equation (25) expresses the common effort level as a convex combination of the degree-weighted average productivity $\bar{\alpha}_g$, which is the Nash benchmark under strong conformity, and the leaders' individual productivities. Suppose that individual productivities are heterogeneous; then the degree-weighted average productivity is strictly below the maximum productivity. Since all weights in this convex combination are nonnegative, the common effort level is therefore strictly below the maximum productivity, even when all leaders have that productivity.

With a single leader, all other players respond to that leader's effort, so the leader can raise the social norm throughout the network toward own productivity. Appointing another leader removes that player's response to the original leader: the two now choose their efforts simultaneously, and lower-productivity followers continue to affect the effort chosen by both. Appointing additional leaders therefore lowers effort throughout the

network when conformity is sufficiently strong, even when the additional leaders have the same highest productivity.

Example 2. Consider the 3-regular network in Figure 4, where the inner triangle $A = \{1, 2, 3\}$ and the outer triangle $B = \{4, 5, 6\}$ are joined by matching links $\{1, 4\}, \{2, 5\}, \{3, 6\}$. Players in A have productivity α_A, and players in B have productivity α_B, with $\alpha_A > \alpha_B > 0$. Each player has two neighbors in the same triangle and one in the other. The social norm therefore depresses Nash effort below individual productivity in A and raises it above individual productivity in B. By Proposition 3(i), an optimal leader group under social conformity contains only players in A.

When the taste for conformity is sufficiently small, the planner assigns two players from A to the leader group and places the third in the follower group. For $L \subseteq A$ with $l = |L| \in \{1, 2, 3\}$, equation (23) gives

$$\Lambda(L) = \frac{\alpha_A - \alpha_B}{27}\, l(4 - l).$$

The leadership gain is largest at $l = 2$. Starting from $L = \{1, 2\}$, assigning player 3 to the leader group prevents player 3 from responding to the efforts of players 1 and 2. Each existing leader consequently has less influence on that leader's own social norm. The loss of these responses outweighs the gain from allowing player 3 to move first.

The local-aggregate benchmark instead benefits from adding player 6 to $\{1, 2\}$. Player 6 is adjacent to neither existing leader, so assigning player 6 to the leader group preserves the responses of their followers and creates three additional links from leaders to followers. Because individual productivity is positive, moving first raises player 6's effort and induces further increases among the followers. Comparing the possible groups gives $\{1, 2, 6\}$ as an optimal group for sufficiently small network effects. Panels (a) and (b) of Figure 4 illustrate these different choices at $\lambda = 0.20$, with $\alpha_A = 2$ and $\alpha_B = 1$.

When the taste for conformity is sufficiently large, Proposition 4(ii) implies that any single player in A forms an optimal leader group, since these players have the highest productivity in the network. Panel (c) of Figure 4 illustrates this choice at $\lambda = 0.80$, with $\alpha_A = 2$ and $\alpha_B = 1$. At the same parameter values, the local-aggregate model selects $\{1, 6\}$, so the optimal group continues to include a less productive player from B, as shown in Panel (d).

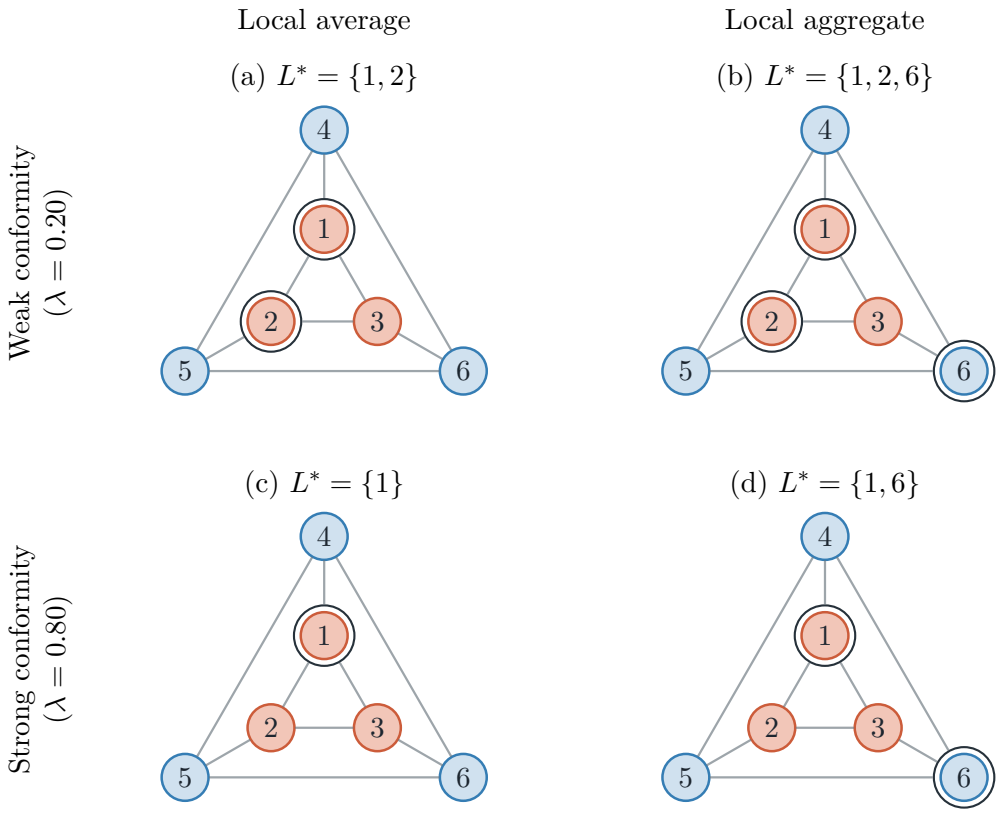


Figure 4: Key groups in a 3-regular network

Note: Red and blue nodes indicate high and low individual productivities, respectively; an outer ring identifies a leader. Columns compare key groups in the local-average game and the local-aggregate benchmark under weak conformity ($\lambda = 0.20$, panels (a) and (b)) and strong conformity ($\lambda = 0.80$, panels (c) and (d)).

3.4 Special Networks

Although solving the key group problem is NP-hard in general networks, regular interaction structures make group selection tractable and isolate the roles of productivity and group size. We examine two canonical architectures: complete networks and complete bipartite networks.

3.4.1 Complete networks

In small project teams, partnerships, or flat committees, members interact directly with one another and observe the average effort of the rest of the team. We model such an environment as the complete network K_n, where $g_{ij} = 1$ for all $i \neq j$. Even though network positions are symmetric, heterogeneous productivities $\alpha_1 > \cdots > \alpha_n > 0$ differentiate members' incentives to lead. Define $\bar{\alpha} := n^{-1} \sum_{i=1}^{n} \alpha_i$ as the average productivity and

$L_m := \{1, \ldots, m\}$ as the group of the m most productive players, and write $\bar{m} := \lfloor n/2 \rfloor$ and $\eta := (n - 1 + \lambda)/\lambda$.

Proposition 5 (Leader selection in complete networks)**.** *In the complete network K_n with $\alpha_1 > \cdots > \alpha_n > 0$:*

(i) Key leader. *Player 1 is the unique key leader.*

(ii) Key group. *An optimal leader group consists of the m^* most productive players, L_{m^*}, where $1 \leq m^* \leq \bar{m}$ is given by solving*

$$\max_{m \in \{1,\ldots,\bar{m}\}} \ell(L_m) = \frac{n(n-m)}{(n-1)\eta(\eta - n + m) - (n-m)^2} \left(\sum_{i=1}^{m} \alpha_i - m\bar{\alpha} \right). \tag{26}$$

(iii) Contraction with conformity. *For $0 < \lambda < \lambda' < 1$, optimal leader groups satisfy $L^*(\lambda') \subseteq L^*(\lambda)$.*

Proof: See Appendix A.15.

In part (i), because all members occupy identical positions in K_n, candidates share the same capacity to propagate effort to followers and receive the same norm response c_i, so single-leader selection reduces to choosing the member whose productivity exceeds the social norm (and hence Nash effort) by the largest margin. In part (ii), any fixed group size m selects the m most productive members L_m, and the objective in (26) captures how choosing m balances the cumulative productivity advantage $\sum_{i=1}^{m}(\alpha_i - \bar{\alpha})$ of early movers against the size $n - m$ of the follower group. Appointing an additional leader extends the first-mover incentive to another productive member, but also converts a responsive follower into a simultaneous first mover and dampens the induced shift in existing leaders' social norms; an above-average member ($\alpha_i > \bar{\alpha}$) may therefore contribute more by following more productive colleagues than by moving in Stage 1, and the optimal leader group never exceeds half the team ($m^* \leq \bar{m}$).

Comparing consecutive group sizes in (26) shows that marginal additions to the leader group satisfy a single-crossing property: once adding the next most productive player fails to raise aggregate effort, any further expansion lowers aggregate effort.

Claim 2. *For $1 \leq m < \bar{m}$, including player $m+1$ in the leader group L_m increases aggregate effort, $\ell(L_{m+1}) > \ell(L_m)$, if and only if*

$$\alpha_{m+1} > \bar{\alpha} + \frac{\ell(L_m)}{n}\left[1 + \frac{(n-1)\eta^2}{(n-m)(n-m-1)}\right], \tag{27}$$

where $\ell(L_m)$ is given by (26). Moreover, if $\ell(L_{m+1}) \leq \ell(L_m)$, then $\ell(L_{s+1}) < \ell(L_s)$ for all $m < s < \bar{m}$.

Proof: See Appendix A.16.

Condition (27) requires candidate $m+1$'s productivity advantage over the team mean $\bar{\alpha}$ to exceed the per-capita leadership gain $\ell(L_m)/n$ scaled by the marginal loss of followers' responses. Because the right-hand side of (27) increases with m while α_{m+1} decreases, the following greedy algorithm identifies an optimal leader group in $O(n)$ arithmetic operations.

Algorithm 1.

Step 1. Set $L_1 = \{1\}$.

Step 2. For $m = 1, \ldots, \bar{m} - 1$ in order, if condition (27) holds, set $L_{m+1} = L_m \cup \{m+1\}$; otherwise, stop with $L^* = L_m$.

Step 3. If condition (27) holds for all $1 \leq m < \bar{m}$, stop with $L^* = L_{\bar{m}}$.

Part (iii) of Proposition 5 shows that a stronger taste for conformity monotonically shrinks the optimal leader group. Although one might expect stronger peer pressure to call for more leaders to shift the social norm, a higher λ makes followers more responsive to the leaders' efforts while tightening mutual norm constraints among co-leaders. Retaining more followers therefore becomes more valuable, and the planner reassigns the least productive leaders to the follower group.

This erosion of followers' responses is sharper under social conformity than in the local-aggregate benchmark of Zhou and Chen (2015), where homogeneous complete networks obey a "50–50 rule" ($m^* = \lfloor n/2 \rfloor$). In K_n, the local-aggregate game (19) admits a finite equilibrium for all leader groups when $\lambda < \bar{\lambda} := \left(\sqrt{n^2 + 4n - 4} - n + 2\right)/4$.[16] Within this common parameter range, both models rank candidates by productivity, but social conformity leads the planner to appoint weakly fewer leaders.

[16] Across leader groups in K_n, the binding admissibility condition is the singleton requirement $1 - 2\lambda c_i > 0$. Since a single leader has $c_i = 1/(\eta - n + 1)$ with $\eta = (n - 1 + \lambda)/\lambda$, the condition $\eta - n + 1 > 2\lambda$ is equivalent to $\lambda < \bar{\lambda}$.

Corollary 3 (Comparison with local-aggregate interactions)**.** *In the complete network K_n with $\alpha_1 > \cdots > \alpha_n > 0$ and $0 < \lambda < \bar{\lambda}$, an optimal leader group in the local-average model is a subset of an optimal leader group in the local-aggregate model.*

Proof: See Appendix A.17.

3.4.2 Complete bipartite networks

Many organizations divide naturally into two interacting groups, such as managers and workers or upstream and downstream divisions, where links run exclusively across sides. In the complete bipartite network K_{n_1,n_2}, the player set N is partitioned into parts N_1 and N_2 of sizes $n_1 := |N_1|$ and $n_2 := |N_2|$, with $g_{ij} = 1$ if and only if players i and j belong to different parts; the star network corresponds to the special case $\min\{n_1, n_2\} = 1$. Each player's social norm is then the average effort of the opposite part. For $p \in \{1, 2\}$, let $\bar{\alpha}_p := n_p^{-1} \sum_{i \in N_p} \alpha_i$ and $\hat{\alpha}_p := \max_{i \in N_p} \alpha_i$ denote the average and maximum productivities in part N_p.

Proposition 6 (Leader selection in complete bipartite networks)**.** *In the complete bipartite network K_{n_1,n_2} with parts N_1 and N_2, suppose that productivities are heterogeneous.*

(i) Key leader. *A key leader is the most productive player in either N_1 or N_2. If $n_1 = n_2$, a key leader is the most productive player in a part N_p that maximizes $(1 + \lambda)\hat{\alpha}_p + (1 - \lambda)\bar{\alpha}_p$. If $\hat{\alpha}_1 = \hat{\alpha}_2$ and $\bar{\alpha}_1 = \bar{\alpha}_2$, a key leader is the most productive player in a part of smaller size.*

(ii) Key group. *Suppose that players in each part N_p have productivity α_p. An optimal leader group consists only of players in the more productive part N_p. Define $\lambda_p := \sqrt{n_p/(n_p + 1)}$. If $0 < \lambda < \lambda_p$, the unique optimal leader group is the entire part N_p. If $\lambda_p < \lambda < 1$, the optimal leader groups are the singleton subsets of N_p. Otherwise, any nonempty subset of N_p is optimal.*

Proof: See Appendix A.18.

Within each part, players occupy symmetric positions and face the same social norm, so the leading index increases with individual productivity and the key leader must be the most productive member of N_1 or N_2. When the two parts have the same size ($n_1 = n_2$), part (i) shows that selection between the two sides depends on both the maximum productivity $\hat{\alpha}_p$ and the average productivity $\bar{\alpha}_p$. Because each player's social

norm is the average effort of the opposite part, a lower average productivity in N_2 than in N_1 depresses equilibrium effort in N_2 and lowers the social norm faced by members of N_1. That wider gap between productivity and the opposite part's social norm (and hence Nash effort) can make the most productive member of N_1 a more effective leader than a more productive individual in N_2 ($\hat{\alpha}_2 > \hat{\alpha}_1$), although a stronger taste for conformity shifts weight in $(1+\lambda)\hat{\alpha}_p + (1-\lambda)\bar{\alpha}_p$ from the part average $\bar{\alpha}_p$ toward the individual maximum $\hat{\alpha}_p$. When the two parts instead share the same average and maximum productivities, part (i) implies that the planner should "start small" (Zhou and Chen, 2015): each follower in the larger part has fewer neighbors and places greater weight on a leader from the smaller part, which induces a larger shift in the leader's social norm. In particular, in a star network ($\min\{n_1, n_2\} = 1$), the central hub moves first.

Part (ii) turns to group selection when each part N_p has a common productivity α_p. Only members of the more productive part N_p then have productivity above the social norm formed by the opposite part ($\alpha_p > \bar{x}_i^N$, and hence $\alpha_p > x_i^N$), so Proposition 3(i) rules out leaders from the less productive part. In the local-aggregate model of Zhou and Chen (2015), the absence of links within N_p implies that the planner should use the natural bipartite partition and appoint the entire part N_p. Under social conformity, however, leaders on the same side still compete indirectly for followers' responses: moving additional members of N_p to Stage 1 leaves fewer followers in N_p to reinforce the Stage 2 responses of the opposite part, which dampens the induced shift in the social norms of leaders in N_p. When conformity is moderate ($\lambda < \lambda_p$), the direct effect of having more high-productivity players move first prevails, and the entire part N_p leads. When conformity is relatively strong ($\lambda > \lambda_p$), the loss of followers' indirect responses outweighs the gain from additional leaders, and the optimal leader group reduces to a single player in N_p.

4 Optimal hierarchy of leadership

In Section 3, restricting leadership to two stages requires every member either to move in Stage 1 or to wait until Stage 2, which rules out sequential responses among early movers. Allowing additional stages enables an intermediate mover to respond to earlier peers while leading later ones, yet also gives more members an opportunity to alter their own social norms through subsequent responses.

Following Zhou and Chen (2015), we model a general hierarchy as a *sequence* $\mathcal{L} = (L_1, \ldots, L_K)$, defined as an ordered partition of N into $K \in \{1, \ldots, n\}$ nonempty stage groups with $L_t \cap L_s = \varnothing$ for $t \neq s$ and $\bigcup_{t=1}^{K} L_t = N$. At stage t, players in L_t observe all efforts chosen in stages $1, \ldots, t-1$ and simultaneously choose their efforts $\mathbf{x}_{L_t}$; each player acts once, and payoffs in (1) are evaluated at the final effort profile. The case $K = 1$ reproduces the simultaneous-move game in Section 2.3, whereas $K = 2$ recovers the two-stage leader–follower game in Section 3.

A sequence $\tilde{\mathcal{L}}$ is a *refinement* of $\mathcal{L} = (L_1, \ldots, L_K)$ if there exists $t \in \{1, \ldots, K\}$ such that $\tilde{\mathcal{L}} = (L_1, \ldots, L_{t-1}, A, B, L_{t+1}, \ldots, L_K)$, where A and B are nonempty sets satisfying $A \cup B = L_t$ and $A \cap B = \varnothing$; that is, a refinement splits one group L_t with $|L_t| \geq 2$ into two consecutive nonempty groups while leaving the order of all other groups unchanged. Repeated refinement leads to a *chain*, defined as a sequence with $K = n$ in which players move one at a time.

4.1 Equilibrium Characterization

For a sequence $\mathcal{L} = (L_1, \ldots, L_K)$, we solve for equilibrium efforts by backward induction. Equilibrium incentives are governed by each player k's *norm response* $c_k(\mathcal{L})$, defined as the marginal rate at which x_k shifts player k's social norm via subsequent movers' equilibrium responses. In the final stage K, players in L_K face no subsequent movers and choose best responses as in (4), which corresponds to (10) with $c_k(\mathcal{L}) = 0$ for all $k \in L_K$. For each stage $t < K$, let $\mathcal{F}_t := \bigcup_{s=t+1}^{K} L_s$ denote the set of subsequent movers; given efforts chosen before stage t, let $\mathbf{x}_{\mathcal{F}_t}(\mathbf{x}_{L_t})$ denote the continuation equilibrium efforts on $\mathcal{F}_t$ and $\bar{x}_k(\mathbf{x}_{L_t})$ the resulting social norm of player $k \in L_t$. By (10) on $\mathcal{F}_t$, subsequent movers respond to stage-t efforts according to

$$\frac{\partial \mathbf{x}_{\mathcal{F}_t}(\mathbf{x}_{L_t})}{\partial \mathbf{x}_{L_t}} = \lambda \mathbf{U}_{\mathcal{F}_t}(\mathbf{I}_{\mathcal{F}_t} - \mathbf{C}^{\mathrm{D}}_{\mathcal{F}_t})\widehat{\mathbf{G}}_{\mathcal{F}_t L_t},$$

where $\mathbf{U}_{\mathcal{F}_t} := \Big[(1-\lambda)\mathbf{I}_{\mathcal{F}_t} + \lambda(\mathbf{I}_{\mathcal{F}_t} - \mathbf{C}^{\mathrm{D}}_{\mathcal{F}_t})(\mathbf{I}_{\mathcal{F}_t} - \widehat{\mathbf{G}}_{\mathcal{F}_t\mathcal{F}_t})\Big]^{-1}$ and $\mathbf{C}^{\mathrm{D}}_{\mathcal{F}_t} := \operatorname{diag}\Big(c_j(\mathcal{L})\Big)_{j\in\mathcal{F}_t}$ collects the norm responses already determined on $\mathcal{F}_t$. Because $g_{kk} = 0$ and players in L_t take earlier and same-stage efforts as given, each player $k \in L_t$ affects $\bar{x}_k(\mathbf{x}_{L_t})$ exclusively through subsequent movers:

$$c_k(\mathcal{L}) := \frac{\partial \bar{x}_k(\mathbf{x}_{L_t})}{\partial x_k} = \lambda\Big(\widehat{\mathbf{G}}_{L_t\mathcal{F}_t}\mathbf{U}_{\mathcal{F}_t}(\mathbf{I}_{\mathcal{F}_t} - \mathbf{C}^{\mathrm{D}}_{\mathcal{F}_t})\widehat{\mathbf{G}}_{\mathcal{F}_t L_t}\Big)_{kk}. \tag{28}$$

Each player $k \in L_t$ then chooses effort according to (10) with $c_k(\mathcal{L})$ in place of c_k. Backward induction from stage K to stage 1 uniquely determines all norm responses, with $0 \leq c_k(\mathcal{L}) \leq \lambda < 1$ for every $k \in N$. Set $\mathbf{C}^{\mathrm{D}}(\mathcal{L}) := \operatorname{diag}\Big(c_1(\mathcal{L}), \ldots, c_n(\mathcal{L})\Big)$, and let $\mathbf{x}^S(\mathcal{L}) = \Big(x_1^S(\mathcal{L}), \ldots, x_n^S(\mathcal{L})\Big)^\top$ denote the equilibrium effort profile. In vector form, (10) across $k \in N$ yields the equilibrium characterization.

Theorem 2 (Equilibrium characterization under general sequencing)**.** *For any sequence $\mathcal{L}$, the sequential-move game admits a unique subgame-perfect Nash equilibrium. Equilibrium efforts are given by*

$$\mathbf{x}^S(\mathcal{L}) = \mathbf{M}^S(\mathcal{L})\boldsymbol{\alpha},$$

where $\mathbf{M}^S(\mathcal{L}) = [m_{ij}^S(\mathcal{L})]$ is an $(n \times n)$ sequential-move influence matrix defined as

$$\mathbf{M}^S(\mathcal{L}) := (1-\lambda)\left[(1-\lambda)\mathbf{I} + \lambda\Big(\mathbf{I} - \mathbf{C}^{\mathrm{D}}(\mathcal{L})\Big)(\mathbf{I} - \widehat{\mathbf{G}})\right]^{-1}. \tag{29}$$

Proof: See Appendix A.19.

The entry $m_{ij}^S(\mathcal{L}) = \partial x_i^S(\mathcal{L})/\partial \alpha_j$ measures the marginal effect of player j's productivity on player i's equilibrium effort. For $K = 1$, $\mathbf{C}^{\mathrm{D}}(\mathcal{L}) = \mathbf{0}$ and $\mathbf{M}^S(\mathcal{L})$ coincides with $\mathbf{M}^N$ in (6); for a two-stage sequence $\mathcal{L} = (L, F)$ with leaders ordered first, $\mathbf{M}^S(\mathcal{L})$ reduces to the two-stage influence matrix $\mathbf{M}^S(L)$ in (14). Moreover, $\mathbf{M}^S(\mathcal{L})$ retains the structural properties in Theorem 1: for $0 < \lambda < 1$, all entries are strictly positive and every row sums to one, so each equilibrium effort $x_i^S(\mathcal{L}) = \sum_{j=1}^n m_{ij}^S(\mathcal{L})\alpha_j$ is a convex combination of individual productivities bounded between $\min_{j \in N} \alpha_j$ and $\max_{j \in N} \alpha_j$. If players are homogeneous ($\alpha_i = \alpha$ for all $i \in N$), then $x_i^S(\mathcal{L}) = x_i^N = \alpha$ for every player and every sequence $\mathcal{L}$, so hierarchy alters behavior only when productivities are heterogeneous.

When productivities are heterogeneous, (29) shows how sequencing reallocates equilibrium weights across members. Moving ahead of neighbors generates a positive norm response $c_k(\mathcal{L}) > 0$, which attenuates player k's effective pressure to conform through $\mathbf{I} - \mathbf{C}^{\mathrm{D}}(\mathcal{L})$ and shifts weight toward α_k; at the same time, assigning subsequent movers to separate stages endows those later movers with positive norm responses of their own in (28) and alters how earlier efforts propagate through the network.

Remark 2. For any sequence $\mathcal{L}$, the sequential-move influence matrix is row-stochastic, that is, $\mathbf{M}^S(\mathcal{L})\mathbf{1} = \mathbf{1}$.

Proof: See Appendix A.20.

Remark 2 suggests a convenient guideline. A refinement of $\mathcal{L}$ changes whose productivity determines equilibrium effort, while preserving the total weight in each player's effort choice. An increase in the weight on one player's productivity must therefore be offset by a decrease in the weights on others. Consequently, distinct influence matrices cannot be ranked entrywise. This differs from the local-aggregate model of Zhou and Chen (2015), where refinement weakly increases every entry of the influence matrix and hence weakly raises equilibrium efforts. Under social conformity, the effect on aggregate effort instead depends on how refinement redistributes influence across players with different productivities.

4.2 Optimal Sequencing

We now allow the organizational planner to choose a sequence $\mathcal{L} = (L_1, \ldots, L_K)$ to maximize aggregate equilibrium effort $X^S(\mathcal{L}) := \sum_{i=1}^n x_i^S(\mathcal{L})$, or equivalently the leadership gain $\ell(\mathcal{L}) := X^S(\mathcal{L}) - X^N$; we refer to a sequence $\mathcal{L}^*$ that solves this problem as an *optimal sequence* (or an *optimal hierarchy*). For each player $i \in N$, let $\tau_i \in \{1, \ldots, K\}$ denote player i's stage, so $i \in L_{\tau_i}$. Rather than partitioning members into a single leader group and a single follower group, a general hierarchy determines along every link whether one neighbor precedes the other ($\tau_i < \tau_j$) or both move in the same stage ($\tau_i = \tau_j$). Proposition 7 characterizes optimal hierarchies when conformity is weak and when conformity is strong.

Proposition 7 (Optimal group under general sequencing)**.** *Suppose that productivities are heterogeneous.*

(i) *For sufficiently small conformity, an optimal sequence maximizes*

$$\Lambda(\mathcal{L}) := \sum_{\tau_i < \tau_j} r_{ij}(\alpha_i - \bar{\alpha}_i). \tag{30}$$

This objective is maximized by placing the players in N_+ individually in weakly decreasing order of $\alpha_i - \bar{\alpha}_i$, followed by $N \setminus N_+$ in a single stage.

(ii) *For sufficiently large conformity, an optimal sequence begins with a singleton consisting of a player with the highest productivity, with all players in each connected component of the remaining network moving in the same stage. In particular, an optimal sequence can be chosen with two stages.*

Proof: See Appendix A.21.

In part (i), when the taste for conformity is sufficiently small ($\lambda \to 0$), player i's social norm responds at first order through direct neighbors who move in later stages, $c_i(\mathcal{L}) = \lambda \sum_{j:\tau_j > \tau_i} r_{ij} + \mathcal{O}(\lambda^2)$, and combining (29) with (6) yields $\ell(\mathcal{L}) = \lambda^2 \Lambda(\mathcal{L}) + \mathcal{O}(\lambda^3)$. Each link $\{i, j\}$ contributes $r_{ij}(\alpha_i - \bar{\alpha}_i)$ if $\tau_i < \tau_j$, $r_{ij}(\alpha_j - \bar{\alpha}_j)$ if $\tau_j < \tau_i$, and 0 if $\tau_i = \tau_j$, so maximizing $\Lambda(\mathcal{L})$ compares the two sequential orders against simultaneous moves link by link:

$$\max_{\mathcal{L}} \Lambda(\mathcal{L}) = \sum_{i<j} r_{ij} \max\{0, \alpha_i - \bar{\alpha}_i, \alpha_j - \bar{\alpha}_j\}. \tag{31}$$

Whereas the two-stage problem in (22) is NP-hard because a single cut cannot orient links among candidates independently (Claim 1), the ordering in Proposition 7(i) attains every link's upper bound in (31) simultaneously and yields an explicit characterization in the limiting case of weak conformity.[17]

Intermediate stages can raise aggregate effort by allowing a moderately productive player to follow a neighbor whose productivity exceeds the neighborhood average by a larger margin while leading another neighbor with a smaller margin. Specifically, whenever a player $j \in N_+$ has a neighbor $i \in N_+$ with $\alpha_i - \bar{\alpha}_i > \alpha_j - \bar{\alpha}_j$ and another neighbor $k \in N$ with $\alpha_j - \bar{\alpha}_j > \max\{\alpha_k - \bar{\alpha}_k, 0\}$, attaining (31) requires $\tau_i < \tau_j < \tau_k$ and hence at least three stages, which avoids the two-stage pairwise loss in (23) without forfeiting player j's leadership gain over player k. Adjacent players in N_+ whose productivities exceed the neighborhood average by the same margin still move in separate stages, in either order, whereas unlinked players in N_+ without conflicting precedence relations can share a stage without losing any link contribution.

Simultaneous moves, by contrast, are optimal among players whose productivities fall below the neighborhood average. Across any link $\{i, j\}$ with $\alpha_i < \bar{\alpha}_i$ and $\alpha_j < \bar{\alpha}_j$, placing either player ahead of the other produces a negative term in (30), because the earlier mover faces a social norm above own productivity and uses the neighbor's subsequent response to adjust effort further downward. Grouping all players in $N \setminus N_+$ into the final stage prevents such downward adjustments; consequently, whenever two adjacent players satisfy $\alpha_i < \bar{\alpha}_i$ and $\alpha_j < \bar{\alpha}_j$, every chain is strictly suboptimal for all sufficiently small $\lambda > 0$, in contrast to Zhou and Chen (2015) and Zhou (2016).

[17] The construction in part (i) uses $|N_+| + 1$ stages, which need not be the minimum number of stages that maximizes $\Lambda(\mathcal{L})$. In particular, whenever N_+ is an *independent set*, placing the entire set N_+ in Stage 1 and the remaining players $N \setminus N_+$ in Stage 2 already maximizes $\Lambda(\mathcal{L})$.

In part (ii), when the taste for conformity is sufficiently large ($\lambda \to 1$), the same logic as in Proposition 4(ii) requires a single player i^* with the highest productivity to move alone in Stage 1 so that limiting efforts across the network can reach α_{i^*}. The key design question is therefore how the remaining players in $N \setminus \{i^*\}$ should be arranged across stages. Separating any two adjacent players in $N \setminus \{i^*\}$ into different stages gives the earlier mover j a positive norm response $c_j(\mathcal{L}) > 0$; in (10), this response shifts weight from the social norm toward $\alpha_j \leq \alpha_{i^*}$ and dampens followers' responses to x_{i^*}, which strictly lowers aggregate effort for sufficiently large conformity.

Every link within $N \setminus \{i^*\}$ must therefore remain within a single stage, so all players in each connected component of the remaining network move simultaneously. By contrast, distinct connected components of $N \setminus \{i^*\}$ share no direct links and, once x_{i^*} is chosen, transmit no further responses through player i^*, so distinct components can either be merged into a single stage or ordered across separate stages without changing equilibrium efforts.[18] Specifically, pooling all remaining players into Stage 2 achieves the optimal outcome with the minimum number of stages, so two stages suffice to form an optimal hierarchy, which recovers a direct leader–follower architecture. When i^* is the unique player with the highest productivity and $N \setminus \{i^*\}$ is connected, a two-stage arrangement $(\{i^*\}, N \setminus \{i^*\})$ is the unique optimal sequence for sufficiently large conformity.

Example 3. Consider the diamond network in Figure 5, where players 1 and 4 are each connected to players 2 and 3, who are also directly connected. Player 1 has the highest productivity, $\alpha_1 = 7$, followed by player 2 with $\alpha_2 = 4$, while players 3 and 4 have the lowest productivity, $\alpha_3 = \alpha_4 = 2$.

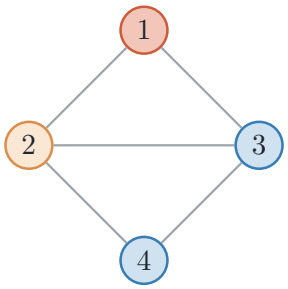


Figure 5: A diamond network

Note: Red, orange, and blue nodes indicate high, medium, and low individual productivities, respectively.

[18] For instance, in a star network with the hub player moving first, any ordering or grouping of the spoke players yields the same equilibrium efforts.

By Remark 2, each row of $\mathbf{M}^S(\mathcal{L})$ sums to one, so it is readily shown that aggregate effort under any sequence satisfies[19]

$$X^S(\mathcal{L}) = 8 + 5\sum_{i=1}^{4} m_{i1}^S(\mathcal{L}) + 2\sum_{i=1}^{4} m_{i2}^S(\mathcal{L}). \tag{32}$$

Only players 1 and 2 have productivity above their neighbors' average, with a larger margin for player 1, so Proposition 7(i) gives the unique optimal sequence $\mathcal{L}^* = (\{1\}, \{2\}, \{3, 4\})$ for sufficiently small conformity. Panel A of Table 2 shows why player 2 occupies an intermediate stage at $\lambda = 0.20$: relative to $\mathcal{L}^*$, merging $\{1, 2\}$ removes player 2's response to player 1 and lowers the weight on player 1's productivity in (32), whereas merging $\{2, 3, 4\}$ gives up player 2's lead over players 3 and 4, so the lower weight on player 2's productivity outweighs the higher weight on player 1's. Keeping $\{3, 4\}$ together avoids the downward adjustment from separating below-average neighbors, whereas the local-aggregate benchmark selects the four-stage chain $(\{1\}, \{2\}, \{3\}, \{4\})$.

For sufficiently large conformity, Proposition 7(ii) instead selects the two-stage sequence $(\{1\}, \{2, 3, 4\})$, since the subnetwork induced by $\{2, 3, 4\}$ is connected. Panel B of Table 2 shows that this two-stage hierarchy is already optimal at $\lambda = 0.60$: relative to $(\{1\}, \{2\}, \{3, 4\})$, the higher aggregate weight on player 1's productivity in (32) now outweighs the lower weight on player 2's productivity. At $\lambda = 0.60$, the local-aggregate benchmark instead retains four stages, with the optimal sequence $(\{2\}, \{1\}, \{3\}, \{4\})$.

Table 2: Influence weights and equilibrium efforts under different sequences

Sequence $\mathcal{L}$	$\sum_i m_{i1}^S$	$\sum_i m_{i2}^S$	x_1^S	x_2^S	X^S
Panel A. $\lambda = 0.20$					
$(\{1\}, \{2\}, \{3, 4\})$	0.9500	1.0647	6.2800	3.9332	**14.8795**
$(\{1, 2\}, \{3, 4\})$	0.9433	1.0679	6.2566	3.9316	14.8525
$(\{1\}, \{2, 3, 4\})$	0.9542	1.0527	6.2809	3.9298	14.8764
Panel B. $\lambda = 0.60$					
$(\{1\}, \{2\}, \{3, 4\})$	0.9525	1.2144	5.1933	3.8768	15.1913
$(\{1, 2\}, \{3, 4\})$	0.8651	1.2545	4.9844	3.8224	14.8343
$(\{1\}, \{2, 3, 4\})$	1.0099	1.0792	5.2178	3.8680	**15.2079**

Notes: The table reports local-average equilibrium outcomes for $\boldsymbol{\alpha} = (7, 4, 2, 2)^\top$. The column sum $\sum_i m_{ij}^S$ measures the aggregate effect of player j's productivity on equilibrium effort. Bold entries identify the highest aggregate effort in each panel.

[19] See Appendix A.22 for the derivation of (32).

5 Extensions

In this section, we develop several extensions of the baseline model. We consider incomplete information, heterogeneous tastes for conformity and weighted directed interactions, anti-conformism, and shifted social norms. The corresponding results and proofs are presented in Appendix B.

5.1 Incomplete Information

Leaders may have superior information about the returns to effort (Hermalin, 1998; Zhou and Chen, 2016). A nonnegative common state $\theta \in [0, \bar{\theta}]$, with $0 < \bar{\theta} < \infty$, scales individual productivity. Player i's payoff is

$$u_i(\mathbf{x}; \theta) = \alpha_i \theta x_i - \frac{x_i^2}{2} - \frac{\gamma}{2}(x_i - \bar{x}_i)^2. \tag{33}$$

The planner selects the leader group before the state is realized to maximize expected aggregate effort. Leaders then observe the state, while followers infer it from public early efforts. For every group size, we study linear separating equilibria in which followers infer the lowest state implied by the leaders' actions, capped by the state support. We select the equilibrium with the highest effort for every player at every state.

The selected equilibrium exists if each leader's effort is at least as high as the resulting social norm when leaders choose the efforts they would choose without conformity and followers know the state (Appendix B.1). Information then weakly increases every player's effort relative to complete information with the same leader group. The additional incentive operates through the social norm: a downward deviation signals a less favorable state, reduces followers' efforts, and lowers the leader's norm. For a leader whose effort exceeds that norm, this response increases the conformity cost and discourages the deviation.

Under weak conformity, group selection among players whose productivity exceeds the average productivity of their neighbors remains a weighted cut problem. The weight on each leader–follower link now includes the follower-to-leader productivity ratio. A more productive follower responds more strongly to information about the state and thereby strengthens the leader's incentive to exert effort. Appointing that follower sacrifices this response. The planner therefore values both the productive example provided by a leader and the productive responses retained among followers.

This trade-off changes the identity and number of leaders. In a complete network, leaving the most productive player as a follower can maximize total effort. Information can also increase or decrease the optimal group size. In a complete bipartite network, an intermediate number of leaders can be optimal: Appendix B.1 gives a star with three equally productive leaves and a less productive center in which two leaves lead under incomplete information, whereas all three lead under complete information.

As conformity becomes strong, the effort difference from complete information vanishes. Followers' efforts increasingly reflect the early actions of leaders, and the response to inferred productivity becomes weaker. The strong-conformity result of Section 3 therefore continues to hold: with heterogeneous productivities, every optimal group consists of a single highest-productivity player when conformity is sufficiently strong.

5.2 Heterogeneous Conformism and Interactions

In practice, peer networks are often directed because reference relationships such as friendship nominations need not be mutual (Bramoullé et al., 2009), and individuals also differ in susceptibility to peer pressure and the desire to conform (Hsieh and van Kippersluis, 2018). With heterogeneous tastes for conformity and weighted, directed interactions, player i's payoff is

$$u_i(\mathbf{x}; \mathbf{W}, \boldsymbol{\gamma}) = \alpha_i x_i - \frac{x_i^2}{2} - \frac{\gamma_i}{2}\left(x_i - \sum_{j=1}^{n} w_{ij} x_j\right)^2, \tag{34}$$

where $\mathbf{W} = [w_{ij}]$ is a nonnegative, irreducible, row-stochastic matrix ($\sum_{j=1}^{n} w_{ij} = 1$ for all $i \in N$) with zero diagonal, in which w_{ij} represents the weight assigned to player j's effort in player i's social norm; and $\boldsymbol{\gamma} := (\gamma_i)_{i \in N}$ is the vector of individual conformity parameters, in which $\gamma_i > 0$ represents player i's taste for conformity. The timing of effort choices and the planner's objective remain unchanged.

Appendix B.2 shows that every leadership arrangement retains a unique, strictly positive equilibrium. A single leader raises every player's effort precisely when individual productivity exceeds own Nash effort, as in Section 3. Heterogeneous conformity and asymmetric attention change the magnitude of this gain; with identical productivities, every leadership arrangement remains neutral.

Directed interactions change the role of followers in group selection. With unequal productivities, an optimal group of minimum size consists of players with positive single-leader gains. Each leader must also have a directed return path through followers: the

induced responses must eventually affect that leader's own norm. In an undirected network, having at least one follower as a neighbor suffices; in a directed network, several followers may be needed to complete the return path.

To compare weak and strong conformity, scale all conformity strengths proportionally, holding their ratios fixed. Under weak conformity, reciprocal attention determines the leading gain from a leader–follower link. The leader's conformity strengthens the incentive to influence the norm, while the follower's conformity strengthens the response. Both therefore enter the link's weight. Without reciprocal links, longer directed paths can generate leadership gains at higher orders.

Under strong conformity, the concentration result from Section 3 survives. When productivities differ, every optimal group eventually consists of a single highest-productivity player. Heterogeneous conformity changes the common effort approached under simultaneous moves. At a given network position, weaker conformity gives a player more influence on that common effort. A single leader, however, still determines the network's limiting effort through own productivity.

5.3 Anti-conformism

In some settings, such as creative work, innovation, or status differentiation, individuals benefit from distinguishing their behavior from that of peers rather than conforming to the group norm (Clark and Oswald, 1998; Bramoullé, 2007). Following Ushchev and Zenou (2020), we capture anti-conformist attitudes by rewarding departures from the social norm in either direction, so player i's payoff is

$$u_i(\mathbf{x};\xi) = \alpha_i x_i - \frac{x_i^2}{2} + \frac{\xi}{2}(x_i - \bar{x}_i)^2, \tag{35}$$

where $\xi > 0$ is the anti-conformity parameter that measures the benefit from departing from the social norm. Although each player still evaluates own effort against the local average $\bar{x}_i$, efforts become *strategic substitutes* (Bramoullé et al., 2014; Ushchev and Zenou, 2020): higher effort by a neighbor raises $\bar{x}_i$ and reduces the marginal return to player i's effort. The network, timing, and aggregate-effort objective remain unchanged.

As Ushchev and Zenou (2020) show in the simultaneous-move game, strong anti-conformism can drive low-productivity players to zero effort or lead to equilibrium multiplicity or nonexistence. For sufficiently small ξ, however, Appendix B.3 establishes a unique subgame-perfect equilibrium with positive effort for every leader group. In this range, followers' responses offset less than a single leader's effort increase, so aggregate

effort moves in the same direction as the leader's effort. As in Section 3, a candidate raises aggregate effort by moving first if and only if productivity exceeds the Nash social norm. The mechanism changes: an increase in the leader's effort lowers the leader's norm, making differentiation more rewarding. Own productivity continues to raise the value of leadership.

When interactions are weak, group selection is governed by the same weighted cut objective as under conformity. A leader's local productivity advantage is weighted by reciprocal influence on the remaining followers. If this objective has a unique maximizer, the same group is optimal under sufficiently weak conformity and anti-conformism.

In complete networks with distinct productivities, the most productive players remain the optimal leaders at any given group size. However, the optimal group weakly expands as anti-conformism strengthens within the stated range, and it may contain more than half the population. Appointing more leaders leaves fewer followers to reduce effort, so a larger fraction of the leaders' effort increase remains in aggregate effort.

The source of leadership can also change in complete bipartite networks. Suppose productivities vary within each part, but two unequal parts have the same mean and maximum productivity. Conformity favors the smaller part, whereas anti-conformism can favor the larger part. A leader in the smaller part reaches more followers, whose aggregate response now offsets the leader's effort increase. This reduction can outweigh the smaller part's stronger effect on the leader's own norm, even while leadership still raises aggregate effort.

5.4 Shifted Social Norms

Players may evaluate their efforts against a benchmark above or below the average effort of peers, reflecting either ambition to outperform neighbors (Ghiglino and Goyal, 2010; Ushchev and Zenou, 2020) or pressure to meet a discounted peer standard (Kandel and Lazear, 1992). To isolate the effect of a shifted norm, all players have the same productivity $\alpha > 0$, and player i's payoff is

$$u_i(\mathbf{x}; \beta) = \alpha x_i - \frac{x_i^2}{2} - \frac{\gamma}{2}(x_i - \beta \bar{x}_i)^2, \tag{36}$$

where $\beta > 0$ is a common shift parameter that scales the reference level to $\beta \bar{x}_i$. We retain $\gamma > 0$ and assume $\lambda\beta < 1$, which ensures a unique simultaneous-move equilibrium (Ushchev and Zenou, 2020).

A shifted norm creates leadership effects even with identical productivities. Appendix B.4 shows that a pure-strategy equilibrium always exists when some players lead and others follow. A downward shift gives a unique equilibrium and raises every player's effort relative to simultaneous moves. An upward shift lowers every player's effort in every equilibrium. At $\beta = 1$, leadership remains neutral, as in Section 3. The direction of the shift therefore determines the sign of leadership gains throughout the network.

The mechanism operates through the leader's response to peer pressure. A downward shift places simultaneous effort below productivity. Higher early effort induces followers to work harder, and the resulting increase in the leader's reference offsets part of the conformity cost of raising effort. An upward shift places simultaneous effort above productivity. Lower early effort then induces followers to reduce effort, lowering the leader's reference and easing the pressure to sustain high effort. Thus the same response of followers can support either an increase or a reduction in early effort.

These comparisons determine whether the planner should introduce leadership. With a downward shift, an optimal arrangement includes both leaders and followers; with an upward shift, simultaneous moves maximize aggregate effort.

Strong conformity also has a different implication from Section 3. For any fixed downward shift, all efforts converge to zero as conformity strengthens, under both simultaneous moves and leadership. Leadership therefore raises effort at finite conformity strengths but no longer anchors positive effort under strong conformity.

6 Conclusion

This paper studies leading by example in networks when players incur a cost for deviating from neighbors' average effort. Because followers adjust toward a leader's effort, moving first shifts the leader's own social norm and alters the pressure to conform. A single leader raises aggregate effort if and only if individual productivity exceeds Nash effort, and the resulting leading index can select a different key leader from the local-aggregate benchmark. An optimal leader group consists only of candidates whose productivities exceed Nash effort and leaves a dominating set of followers; as conformity strengthens, optimal selection shifts from maximizing a weighted directed cut to appointing a single player with the highest productivity.

In multi-stage hierarchies, strategic complementarity in efforts does not justify ever-finer sequencing under social conformity: refining the order of moves redistributes

influence across productivities, lowers aggregate effort when below-average neighbors are separated, and leaves two stages sufficient under strong conformity. Extensions to incomplete information, heterogeneous and directed interactions, anti-conformism, and shifted social norms confirm that leadership design turns on how followers' responses shift leaders' reference points. For instance, private information strengthens Stage 1 incentives while rewarding the retention of productive followers who respond strongly to the inferred state, whereas shifted norms cause sequential moves to raise or lower effort even under homogeneous productivities.

Our findings carry direct implications for organizational design. When early movers are themselves bound by social norms, appointing all high-productivity members to lead or imposing deep hierarchies can lower aggregate effort by depleting the follower responses that shift leaders' norms. Effective leadership design must therefore coordinate who acts first with who remains available to respond: a productive member can contribute more to collective performance by waiting and responding to colleagues than by moving first.

Several directions for future research remain open. Allowing players to observe only the prior efforts of direct neighbors, rather than all earlier actions, would link sequential effort propagation and signaling directly to network distance. Another promising avenue is to examine how an organizational designer coordinates the timing of moves with targeted incentives or network design, and to test empirically how peer pressure shapes first-mover behavior in teams.

References

Acemoglu, Daron and Matthew O. Jackson (2015). "History, Expectations, and Leadership in the Evolution of Social Norms". In: *Review of Economic Studies* 82.2, pp. 423–56.

Akerlof, George A. and Rachel E. Kranton (2000). "Economics and Identity". In: *Quarterly Journal of Economics* 115.3, pp. 715–53.

Andreoni, James and B. Douglas Bernheim (2009). "Social Image and the 50–50 Norm: A Theoretical and Experimental Analysis of Audience Effects". In: *Econometrica* 77.5, pp. 1607–36.

Ballester, Coralio, Antoni Calvó-Armengol, and Yves Zenou (2006). "Who's Who in Networks. Wanted: The Key Player". In: *Econometrica* 74.5, pp. 1403–17.

Bandiera, Oriana, Iwan Barankay, and Imran Rasul (2010). "Social Incentives in the Workplace". In: *Review of Economic Studies* 77.2, pp. 417–58.

Bernheim, B. Douglas (1994). "A Theory of Conformity". In: *Journal of Political Economy* 102.5, pp. 841–77.

Block, Juan I., Rohan Dutta, and David K. Levine (2025). "Leaders and Social Norms: On the Emergence of Consensus or Conflict". In: *Journal of Economic Behavior & Organization* 233, 106970.

Bolton, Patrick, Markus K. Brunnermeier, and Laura Veldkamp (2010). "Economists' Perspectives on Leadership". In: *Handbook of Leadership Theory and Practice*. Ed. by Nitin Nohria and Rakesh Khurana. Boston, MA: Harvard Business Press, pp. 239–64.

— (2013). "Leadership, Coordination, and Corporate Culture". In: *Review of Economic Studies* 80.2, pp. 512–37.

Boucher, Vincent, Michelle Rendall, Philip Ushchev, and Yves Zenou (2024). "Toward a General Theory of Peer Effects". In: *Econometrica* 92.2, pp. 543–65.

Bramoullé, Yann (2007). "Anti-coordination and Social Interactions". In: *Games and Economic Behavior* 58.1, pp. 30–49.

Bramoullé, Yann, Habiba Djebbari, and Bernard Fortin (2009). "Identification of Peer Effects through Social Networks". In: *Journal of Econometrics* 150.1, pp. 41–55.

Bramoullé, Yann, Rachel Kranton, and Martin D'Amours (2014). "Strategic Interaction and Networks". In: *American Economic Review* 104.3, pp. 898–930.

Bursztyn, Leonardo, Alessandra L. González, and David Yanagizawa-Drott (2020). "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia". In: *American Economic Review* 110.10, pp. 2997–3029.

Bursztyn, Leonardo and Robert Jensen (2017). "Social Image and Economic Behavior in the Field: Identifying, Understanding, and Shaping Social Pressure". In: *Annual Review of Economics* 9, pp. 131–53.

Cabrales, Antonio and Esther Hauk (2024). "Norms and the Evolution of Leaders' Followership". In: *Games and Economic Behavior* 148, pp. 138–61.

Clark, Andrew E. and Andrew J. Oswald (1998). "Comparison-concave Utility and Following Behaviour in Social and Economic Settings". In: *Journal of Public Economics* 70.1, pp. 133–55.

Costa, Dora L. (2014). "Leaders: Privilege, Sacrifice, Opportunity, and Personnel Economics in the American Civil War". In: *Journal of Law, Economics, and Organization* 30.3, pp. 437–62.

Dewan, Torun and David P. Myatt (2008). "The Qualities of Leadership: Direction, Communication, and Obfuscation". In: *American Political Science Review* 102.3, pp. 351–68.

Festinger, Leon (1954). "A Theory of Social Comparison Processes". In: *Human Relations* 7.2, pp. 117–40.

Fischer, Paul and Steven Huddart (2008). "Optimal Contracting with Endogenous Social Norms". In: *American Economic Review* 98.4, pp. 1459–75.

Frey, Bruno S. and Stephan Meier (2004). "Social Comparisons and Pro-social Behavior: Testing "Conditional Cooperation" in a Field Experiment". In: *American Economic Review* 94.5, pp. 1717–22.

Galeotti, Andrea, Benjamin Golub, and Sanjeev Goyal (2020). "Targeting Interventions in Networks". In: *Econometrica* 88.6, pp. 2445–71.

Galeotti, Andrea, Sanjeev Goyal, Matthew O. Jackson, Fernando Vega-Redondo, and Leeat Yariv (2010). "Network Games". In: *Review of Economic Studies* 77.1, pp. 218–44.

Ghiglino, Christian and Sanjeev Goyal (2010). "Keeping Up with the Neighbors: Social Interaction in a Market Economy". In: *Journal of the European Economic Association* 8.1, pp. 90–119.

Ghuge, Rohan, Joseph Kwon, Viswanath Nagarajan, and Adetee Sharma (2022). "Constrained Assortment Optimization Under the Paired Combinatorial Logit Model". In: *Operations Research* 70.2, pp. 786–804.

Goldbaum, David (2019). "Conformity and Influence". In: *The B.E. Journal of Theoretical Economics* 19.1, 20170066.

Güth, Werner, M. Vittoria Levati, Matthias Sutter, and Eline C. M. van der Heijden (2007). "Leading by Example with and without Exclusion Power in Voluntary Contribution Experiments". In: *Journal of Public Economics* 91.5–6, pp. 1023–42.

Hermalin, Benjamin E. (1998). "Toward an Economic Theory of Leadership: Leading by Example". In: *American Economic Review* 88.5, pp. 1188–206.

— (2012). "Leadership and Corporate Culture". In: *Handbook of Organizational Economics*. Ed. by Robert Gibbons and John Roberts. Princeton, NJ: Princeton University Press, pp. 432–78.

Hsieh, Chih-Sheng and Hans van Kippersluis (2018). "Smoking Initiation: Peers and Personality". In: *Quantitative Economics* 9.2, pp. 825–63.

Huck, Steffen and Pedro Rey-Biel (2006). "Endogenous Leadership in Teams". In: *Journal of Institutional and Theoretical Economics* 162.2, pp. 253–61.

Jack, B. Kelsey and María P. Recalde (2015). "Leadership and the Voluntary Provision of Public Goods: Field Evidence from Bolivia". In: *Journal of Public Economics* 122, pp. 80–93.

Jelínek, Vít, Eva Jelínková, and Jan Kratochvíl (2026). "On the Hardness of Switching to a Small Number of Edges". In: *Journal of Graph Theory* 113.2, pp. 237–51.

Kandel, Eugene and Edward P. Lazear (1992). "Peer Pressure and Partnerships". In: *Journal of Political Economy* 100.4, pp. 801–17.

Komai, Mana, Mark Stegeman, and Benjamin E. Hermalin (2007). "Leadership and Information". In: *American Economic Review* 97.3, pp. 944–7.

Landa, Dimitri and Scott A. Tyson (2017). "Coercive Leadership". In: *American Journal of Political Science* 61.3, pp. 559–74.

Mas, Alexandre and Enrico Moretti (2009). "Peers at Work". In: *American Economic Review* 99.1, pp. 112–45.

Matsuno, Shunsuke (2026). "How Does Corporate Culture Create Value? Conformity and the Coordination of Dispersed Information". Working paper, January 9 version. Previously circulated as "Conformity and Leadership in Organizations".

Meng, Dawen (2025). "Targeting Network Intervention with Social Norm". In: *Journal of Economic Theory* 230, 106092.

Meng, Dawen, Guoqiang Tian, and Jie Zheng (2026). "Mechanism Design and Relative Network Intervention with Social Norm". Working paper, April 25 version.

Olcina, Gonzalo, Fabrizio Panebianco, and Yves Zenou (2024). "Conformism, Social Pressure, and the Dynamics of Integration". In: *Journal of Economic Behavior & Organization* 220, pp. 279–304.

Paluck, Elizabeth Levy and Hana Shepherd (2012). "The Salience of Social Referents: A Field Experiment on Collective Norms and Harassment Behavior in a School Social Network". In: *Journal of Personality and Social Psychology* 103.6, pp. 899–915.

Paluck, Elizabeth Levy, Hana Shepherd, and Peter M. Aronow (2016). "Changing Climates of Conflict: A Social Network Experiment in 56 Schools". In: *Proceedings of the National Academy of Sciences* 113.3, pp. 566–71.

Rajan, Raghuram G. and Julie Wulf (2006). "The Flattening Firm: Evidence from Panel Data on the Changing Nature of Corporate Hierarchies". In: *Review of Economics and Statistics* 88.4, pp. 759–73.

Rajan, Raghuram G. and Luigi Zingales (2001). "The Firm as a Dedicated Hierarchy: A Theory of the Origins and Growth of Firms". In: *Quarterly Journal of Economics* 116.3, pp. 805–51.

Shang, Jen and Rachel Croson (2009). "A Field Experiment in Charitable Contribution: The Impact of Social Information on the Voluntary Provision of Public Goods". In: *Economic Journal* 119.540, pp. 1422–39.

Sun, Yang, Wei Zhao, and Junjie Zhou (2026). "Sequential Network Design". In: *Journal of Economic Theory* 232, p. 106145.

Tankard, Margaret E. and Elizabeth Levy Paluck (2016). "Norm Perception as a Vehicle for Social Change". In: *Social Issues and Policy Review* 10.1, pp. 181–211.

Ushchev, Philip and Yves Zenou (2020). "Social Norms in Networks". In: *Journal of Economic Theory* 185, 104969.

Varian, Hal R. (1994). "Sequential Contributions to Public Goods". In: *Journal of Public Economics* 53.2, pp. 165–86.

Winter, Eyal (2006). "Optimal Incentives for Sequential Production Processes". In: *RAND Journal of Economics* 37.2, pp. 376–90.

— (2010). "Transparency and Incentives among Peers". In: *RAND Journal of Economics* 41.3, pp. 504–23.

Zhou, Junjie (2016). "Economics of Leadership and Hierarchy". In: *Games and Economic Behavior* 95, pp. 88–106.

Zhou, Junjie and Ying-Ju Chen (2015). "Key Leaders in Social Networks". In: *Journal of Economic Theory* 157, pp. 212–35.

Zhou, Junjie and Ying-Ju Chen (2016). "Targeted Information Release in Social Networks". In: *Operations Research* 64.3, pp. 721–35.

Zhou, Junjie, Chen-Nan Liao, and Ying-Ju Chen (2023). "Optimal selling scheme in social networks: hierarchical signaling, sequential selling, and chain structure". In: *Production and Operations Management* 32.7, pp. 2138–53.

A Proofs of the Main Results

A.1 Proof of Lemma 1

Proof. Since $\widehat{\mathbf{G}}$ is row-stochastic and $0 \leq \lambda < 1$, the inverse $(\mathbf{I} - \lambda\widehat{\mathbf{G}})^{-1} = \sum_{t=0}^{\infty} \lambda^t \widehat{\mathbf{G}}^t$ exists and is nonnegative. The best-response system has the unique solution (5). This solution is strictly positive because $\boldsymbol{\alpha} > \mathbf{0}$ and the series includes the identity matrix. Each row of $(1-\lambda)(\mathbf{I} - \lambda\widehat{\mathbf{G}})^{-1}$ sums to one. For $0 < \lambda < 1$, connectedness provides a positive-weight path between any two players, so the influence matrix has strictly positive entries. □

A.2 Proof of Theorem 1

We first establish the followers' response and the bounds on norm feedback used in the leaders' equilibrium system.

Lemma 2. *For a leader group L with $0 < |L| < n$, $0 \leq \lambda < 1$, and nonnegative $\mathbf{x}_L$, the follower subgame admits a unique strictly positive equilibrium, given by (8). The leaders' final social norms are given by (11). Both $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}$ and $\mathbf{C}_L$ are nonnegative, $0 \leq c_i \leq \lambda$, and $\mathbf{C}_L \mathbf{1}_L \leq \mathbf{1}_L$.*

Proof. For given leading efforts $\mathbf{x}_L$, the follower best responses form the linear system

$$(\mathbf{I}_F - \lambda\widehat{\mathbf{G}}_{FF})\mathbf{x}_F = (1-\lambda)\boldsymbol{\alpha}_F + \lambda\widehat{\mathbf{G}}_{FL}\mathbf{x}_L.$$

Since $\widehat{\mathbf{G}}_{FF}$ is substochastic and $\lambda < 1$, the inverse exists and equals the nonnegative series $\mathbf{U}_F = \sum_{t\geq 0} \lambda^t \widehat{\mathbf{G}}_{FF}^t$. The identity term and positive productivities make the unique solution strictly positive for any nonnegative $\mathbf{x}_L$. Strict concavity of each follower's payoff then gives the unique follower equilibrium in (8). Its derivative with respect to $\mathbf{x}_L$ gives the followers' response matrix. Substitution into $\bar{\mathbf{x}}_L = \widehat{\mathbf{G}}_{LL}\mathbf{x}_L + \widehat{\mathbf{G}}_{LF}\mathbf{x}_F$ gives the leaders' social norms in (11) and the norm-response matrix in (12).

To bound norm feedback, increase one leading effort by one unit and hold the other leading efforts fixed. Each derivative of a follower's effort is nonnegative. Each response equation expresses that derivative as λ times a weighted average of derivatives of followers' efforts and derivatives of leaders' efforts in $\{0, 1\}$. If the largest derivative of a follower's effort exceeded one, the corresponding response equation would contradict $\lambda < 1$. All derivatives of followers' efforts are therefore at most one. Substitution into the response

equations sharpens the bound to λ. Because the network has no self-loops, c_i is a weighted sum of these derivatives of followers' efforts, with nonnegative weights that sum to at most one. Hence $0 \leq c_i \leq \lambda$.

Finally, an increase of one unit in all leading efforts yields derivatives of followers' efforts at most λ. The resulting change in each leader's social norm is therefore at most one, so $\mathbf{C}_L \mathbf{1}_L \leq \mathbf{1}_L$. □

With Lemma 2 established, we solve the leaders' equilibrium system and characterize the influence weights.

Proof. Step 1: Equilibrium efforts. For a leader group L, substitute the followers' response (8) into the leaders' payoffs. Leader i's deviation from the social norm changes at rate $1 - c_i > 0$ with respect to own effort. The reduced payoff is strictly concave, with second derivative $-1 - \gamma(1 - c_i)^2 < 0$. At zero effort, the marginal payoff is positive: the direct term is $\alpha_i > 0$, and the norm generated by other leaders' efforts and followers' productivities is nonnegative. Each best response is therefore interior, so an equilibrium must satisfy (10).

By (11), the leaders' first-order conditions become

$$\begin{aligned}&\left[(1-\lambda)\mathbf{I}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L)\right]\mathbf{x}_L^S \\ &\qquad = (1-\lambda)\left[\boldsymbol{\alpha}_L + \lambda(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F\right].\end{aligned}$$

The coefficient matrix has nonpositive off-diagonal entries. In row i, the diagonal exceeds the sum of the absolute off-diagonal entries by

$$(1-\lambda) + \lambda(1-c_i)\left(1 - \sum_{j \in L}(\mathbf{C}_L)_{ij}\right) > 0.$$

To obtain the inverse, divide each row by its positive diagonal entry. The original coefficient matrix therefore factors as a positive diagonal matrix times $\mathbf{I} - \mathbf{B}$, where $\mathbf{B} \geq \mathbf{0}$ has row sums below one. The convergent series for $(\mathbf{I} - \mathbf{B})^{-1}$ gives a nonnegative inverse. Since the right-hand side is strictly positive, the system has a unique positive solution. The followers' response evaluated at this solution gives (14).

Step 2: Influence weights. Let $\mathbf{E}_L$ be diagonal, with entry $1 - c_i$ for a leader and entry one for a follower. We can write the full equilibrium system as

$$\mathbf{Q}_L \mathbf{x}^S(L) = \boldsymbol{\alpha}, \qquad \mathbf{Q}_L := \mathbf{I} + \gamma \mathbf{E}_L(\mathbf{I} - \widehat{\mathbf{G}}). \tag{A.1}$$

The row-dominance argument in Step 1 also gives a nonnegative inverse for $\mathbf{Q}_L$. For strict positivity, suppose that $0 < \lambda < 1$. After division of row i by $1 + \gamma(\mathbf{E}_L)_{ii}$, the coefficient matrix becomes $\mathbf{I} - \mathbf{T}\widehat{\mathbf{G}}$, where $\mathbf{T} := \operatorname{diag}(t_{11}, \ldots, t_{nn})$ has diagonal entries

$$t_{ii} := \frac{\gamma(\mathbf{E}_L)_{ii}}{1 + \gamma(\mathbf{E}_L)_{ii}} \in (0, 1).$$

Connectedness makes the series $\sum_{t \geq 0} (\mathbf{T}\widehat{\mathbf{G}})^t$ strictly positive, and hence $\mathbf{M}^S = \mathbf{Q}_L^{-1}$ is strictly positive. To obtain the row sums, use $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{1} = \mathbf{0}$ in (A.1). We obtain $\mathbf{Q}_L\mathbf{1} = \mathbf{1}$, so $\mathbf{M}^S\mathbf{1} = \mathbf{1}$. At $\lambda = 0$, $\mathbf{M}^S = \mathbf{I}$. Strict concavity and interior best responses establish that the resulting outcome is the unique subgame-perfect equilibrium. □

A.3 Proof of Proposition 1

Proof. To compare the two effort profiles, use $\mathbf{E}_L$ and $\mathbf{Q}_L$ from (A.1), with $\mathbf{M}^S = \mathbf{Q}_L^{-1}$. The simultaneous-move first-order conditions give $\gamma(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^N = \boldsymbol{\alpha} - \mathbf{x}^N$. The difference between the equilibrium systems is therefore

$$\begin{aligned} \mathbf{Q}_L(\mathbf{x}^S - \mathbf{x}^N) &= \boldsymbol{\alpha} - \mathbf{x}^N - \gamma\mathbf{E}_L(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^N \\ &= (\mathbf{I} - \mathbf{E}_L)(\boldsymbol{\alpha} - \mathbf{x}^N). \end{aligned}$$

After multiplication by $\mathbf{M}^S$, coordinate j is

$$x_j^S - x_j^N = \sum_{i \in L} m_{ji}^S c_i (\alpha_i - x_i^N),$$

because the diagonal entries of $\mathbf{I} - \mathbf{E}_L$ are c_i for leaders and zero for followers. The identity $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$ expresses the gap in terms of productivities. Comparison with zero gives (15). □

A.4 Proof of Remark 1

Proof. Given a single leader $L = \{i\}$, the identity in the proof of Proposition 1 becomes $x_j^S - x_j^N = m_{ji}^S c_i (\alpha_i - x_i^N)$ for each player $j \in N$. Theorem 1 gives $m_{ji}^S > 0$. Connectedness ensures that leader i has a neighbor, and all of the leader's neighbors are followers. Since the network is undirected, (13) implies

$$c_i \geq \lambda \sum_{j \in F} \hat{g}_{ij}\hat{g}_{ji} > 0,$$

if $\lambda > 0$. Thus the effort changes all have the sign of $\alpha_i - x_i^N$. Equation (7) gives the equivalent comparison with $x_i^N - \bar{x}_i^N$.

To obtain (16), we use the Nash equilibrium representation in (5). Note that $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$, we get

$$\alpha_i - x_i^N = (1 - m_{ii}^N) \left(\alpha_i - \sum_{j \neq i} \frac{m_{ij}^N}{1 - m_{ii}^N} \alpha_j \right).$$

Since $\mathbf{M}^N$ is strictly positive and row-stochastic, we have $0 < m_{ii}^N < 1$. The inequality $\alpha_i - x_i^N > 0$ is therefore equivalent to (16). □

A.5 Proof of Proposition 2

Proof. Step 1: Followers' responses and norm feedback. Let player i be the sole leader, so $F = N \setminus \{i\}$. By Lemma 2, the followers' response satisfies

$$\frac{\partial \mathbf{x}_{-i}}{\partial x_i} = \lambda (\mathbf{I}_{-i} - \lambda \widehat{\mathbf{G}}_{-i})^{-1} \widehat{\mathbf{G}}_{-i,i}.$$

Restricting column i of $(\mathbf{I} - \lambda \widehat{\mathbf{G}})\mathbf{M}^N = (1 - \lambda)\mathbf{I}$ to the follower rows $N \setminus \{i\}$ gives

$$(\mathbf{I}_{-i} - \lambda \widehat{\mathbf{G}}_{-i}) \mathbf{M}_{-i,i}^N = \lambda \widehat{\mathbf{G}}_{-i,i} m_{ii}^N.$$

The followers' response vector is therefore $\partial \mathbf{x}_{-i} / \partial x_i = \mathbf{M}_{-i,i}^N / m_{ii}^N$. Row i of $(\mathbf{I} - \lambda \widehat{\mathbf{G}})\mathbf{M}^N = (1 - \lambda)\mathbf{I}$ gives $m_{ii}^N - \lambda \widehat{\mathbf{G}}_{i,-i} \mathbf{M}_{-i,i}^N = 1 - \lambda$, which yields the feedback formula in (17). Connectedness and undirectedness give a positive return through a neighboring follower, so $c_i > 0$. Lemma 2 bounds the feedback coefficient below one.

Step 2: The leading index. We evaluate the leader's incentive to adjust effort at the Nash benchmark. At x_i^N, the derivative of the reduced payoff is

$$\alpha_i - x_i^N - \gamma(1 - c_i)(x_i^N - \bar{x}_i^N) = c_i(\alpha_i - x_i^N),$$

where the equality follows from (7). The reduced payoff has constant second derivative $-1 - \gamma(1 - c_i)^2$. The leader's effort adjustment and the followers' response from Step 1 therefore give

$$\mathbf{x}^S(\{i\}) - \mathbf{x}^N = \frac{c_i(\alpha_i - x_i^N)}{1 + \gamma(1 - c_i)^2} \frac{\mathbf{M}_{\cdot i}^N}{m_{ii}^N}. \tag{A.2}$$

The sum of the coordinates in (A.2), together with $\gamma = \lambda/(1 - \lambda)$, gives (18). With $c_i = (m_{ii}^N + \lambda - 1)/(\lambda m_{ii}^N)$, the index can also be written as

$$\ell_i = \frac{(m_{ii}^N + \lambda - 1) \sum_{j=1}^n m_{ji}^N}{\lambda (m_{ii}^N)^2 + (1 - \lambda)(1 - m_{ii}^N)^2} (\alpha_i - x_i^N).$$

Since the benchmark X^N is common to all candidates, a key leader maximizes this index.

Step 3: The final social norm. We also express the index in terms of the leader's final social norm:

$$\ell_i = \gamma c_i \left(\sum_{j=1}^{n} m_{ji}^N\right)(x_i^S - \bar{x}_i^S) = \left(\frac{1}{1-\lambda} - \frac{1}{m_{ii}^N}\right)\left(\sum_{j=1}^{n} m_{ji}^N\right)(x_i^S - \bar{x}_i^S), \tag{A.3}$$

where all sequential quantities refer to $L = \{i\}$. The followers' response implies

$$x_i^S - \bar{x}_i^S = x_i^N - \bar{x}_i^N + (1 - c_i)(x_i^S - x_i^N).$$

By (A.2) and $\alpha_i - x_i^N = \gamma(x_i^N - \bar{x}_i^N)$, we obtain

$$x_i^S - \bar{x}_i^S = \frac{\lambda m_{ii}^N}{\lambda (m_{ii}^N)^2 + (1-\lambda)(1-m_{ii}^N)^2}(x_i^N - \bar{x}_i^N), \qquad x_i^S - x_i^N = \frac{m_{ii}^N + \lambda - 1}{1-\lambda}(x_i^S - \bar{x}_i^S).$$

The aggregate effort change in (A.2) is

$$\ell_i = \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N}(x_i^S - x_i^N),$$

which proves (A.3). □

A.6 Proof of Corollary 1

Proof. Part (i): Weak conformity. As $\lambda \downarrow 0$, the absence of self-loops gives

$$m_{ii}^N + \lambda - 1 = \lambda^2 r_i + \mathcal{O}(\lambda^3), \qquad \sum_{j=1}^{n} m_{ji}^N = 1 + \mathcal{O}(\lambda).$$

The productivity gap satisfies $\alpha_i - x_i^N = \lambda(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$. Substituting these expansions into (18) yields

$$\ell_i(\lambda) = \lambda^2 r_i(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^3), \tag{A.4}$$

so a key leader maximizes $r_i(\alpha_i - \bar{\alpha}_i)$ for sufficiently small $\lambda > 0$.

Part (ii): Strong conformity. The random-walk matrix of a connected undirected graph is similar to a symmetric matrix and has a simple eigenvalue one. The stationary vector is $\boldsymbol{\pi}$, with $\pi_i = d_i / \sum_{j=1}^n d_j$. The remaining eigenvalues differ from one, including a possible eigenvalue minus one. By spectral decomposition,

$$\mathbf{M}^N = \mathbf{1}\boldsymbol{\pi}^\top + \mathcal{O}(1-\lambda). \tag{A.5}$$

Thus $m_{ii}^N = \pi_i + \mathcal{O}(1-\lambda)$ and $\sum_{j=1}^n m_{ji}^N = n\pi_i + \mathcal{O}(1-\lambda)$. By (5), the productivity gap is $\alpha_i - x_i^N = \alpha_i - \bar{\alpha}_g + \mathcal{O}(1-\lambda)$. The coefficient on this gap in (18) is $n + \mathcal{O}(1-\lambda)$, which proves the limiting gain in part (ii).

The same expansions also give the individual consensus limit used in the group comparison. Equation (17) implies $c_i = 1 + \mathcal{O}(1-\lambda)$, while $m_{ji}^N/m_{ii}^N = 1 + \mathcal{O}(1-\lambda)$ and $\gamma(1-c_i)^2 = \mathcal{O}(1-\lambda)$. Equations (A.2) and (A.5) then yield

$$\mathbf{x}^S(\{i\}) = \alpha_i \mathbf{1} + \mathcal{O}(1-\lambda). \tag{A.6}$$

There are finitely many candidates, so a strict leading-order ranking near either endpoint remains strict on a common neighborhood. This gives the stated selection implications, with the most productive players tied in the strong-conformity limit. □

A.7 Proof of Corollary 2

Proof. Part (i): Productivities. In (18), ℓ_i is a positive network coefficient, independent of productivities, multiplied by $\alpha_i - x_i^N$. Since $x_i^N = \sum_{j=1}^n m_{ij}^N \alpha_j$ with $0 < m_{ii}^N < 1$ and $m_{ij}^N > 0$ for $j \neq i$, $\partial \ell_i / \partial \alpha_i$ is proportional to $1 - m_{ii}^N > 0$ and $\partial \ell_i / \partial \alpha_j$ $(j \neq i)$ is proportional to $-m_{ij}^N < 0$.

Part (ii): Conformity. By Corollary 1, ℓ_i extends continuously to $[0,1]$. If $\bar{\alpha}_i < \alpha_i \leq \bar{\alpha}_g$, then $\ell_i(0) = 0$, ℓ_i is strictly positive for sufficiently small positive λ, and $\ell_i(1) \leq 0$. Compactness gives a global maximum, and these signs place the maximum in $(0,1)$. The index cannot be nonincreasing from zero or nondecreasing toward a nonpositive endpoint, so the index is nonmonotone. Under the reversed inequalities, the same argument with reversed signs gives an interior global minimum and proves the second conclusion.

Part (iii): Link dilution. Consider a new link $\{j,k\}$ between a neighbor $j \in N(i)$ and another follower $k \neq i,j$. The link leaves $N(i)$, d_i, and $\alpha_i - \bar{\alpha}_i > 0$ unchanged. The two endpoint degrees rise by one. The change in d_k enters r_i only if $k \in N(i)$. Hence local reciprocity $r_i = d_i^{-1} \sum_{u \in N(i)} d_u^{-1}$ decreases by $d_i^{-1}[d_j^{-1}(d_j+1)^{-1} + \mathbf{1}_{\{k \in N(i)\}} d_k^{-1}(d_k+1)^{-1}] > 0$. By (A.4), $\ell_i(\lambda)$ strictly decreases for all sufficiently small $\lambda > 0$. □

A.8 Derivation of (20)

Under the local-aggregate payoff u_i^{agg} in (19), Stage 2 best responses coincide with (4). When player i moves alone in Stage 1 ($L = \{i\}$), each follower $j \neq i$ adjusts effort at rate $\partial x_j(x_i)/\partial x_i = m_{ji}^N / m_{ii}^N$ as in Step 1 of the proof of Proposition 2, so $\bar{x}_i(x_i) =$

$\bar{x}_i^N + c_i(x_i - x_i^N)$. Substituting $\bar{x}_i(x_i)$ into $u_i^{\text{agg}}\big(x_i, \mathbf{x}_{-i}(x_i); g\big) = \alpha_i x_i - \frac{1+\gamma}{2}x_i^2 + \gamma x_i \bar{x}_i(x_i)$ and differentiating with respect to x_i yields

$$\frac{du_i^{\text{agg}}}{dx_i} = \alpha_i - (1+\gamma)x_i + \gamma \bar{x}_i(x_i) + \gamma c_i x_i, \qquad \frac{d^2 u_i^{\text{agg}}}{dx_i^2} = -(1+\gamma)(1-2\lambda c_i),$$

where $\gamma/(1+\gamma) = \lambda$. When $1 - 2\lambda c_i > 0$, the reduced payoff is strictly concave in x_i. Dividing the first-order condition by $1+\gamma$ and using $(1-\lambda)\alpha_i + \lambda \bar{x}_i^N = x_i^N$ from (4) gives

$$(1-\lambda)\alpha_i + \lambda\Big[\bar{x}_i^N + c_i(x_i - x_i^N)\Big] + \lambda c_i x_i - x_i = (1 - \lambda c_i)x_i^N - (1 - 2\lambda c_i)x_i = 0.$$

Since $x_i^N > 0$ and $0 < c_i \leq \lambda < 1$, the unique Stage 1 equilibrium effort $x_i^{S,\text{agg}} = \frac{1-\lambda c_i}{1-2\lambda c_i} x_i^N$ is strictly positive and satisfies $x_i^{S,\text{agg}} - x_i^N = \frac{\lambda c_i}{1-2\lambda c_i} x_i^N$. Summing $x_j^{S,\text{agg}} - x_j^N = (m_{ji}^N / m_{ii}^N)(x_i^{S,\text{agg}} - x_i^N)$ across $j \in N$ yields

$$\ell_i^{\text{agg}} = \left(1 + \sum_{j \neq i} \frac{m_{ji}^N}{m_{ii}^N}\right)(x_i^{S,\text{agg}} - x_i^N) = \frac{\lambda c_i}{1 - 2\lambda c_i} \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N} x_i^N,$$

which gives (20). Moreover, substituting $(1-\lambda)(\alpha_i - x_i^N) = \lambda(x_i^N - \bar{x}_i^N)$ from (7) and $1 - \lambda c_i(2 - c_i) = 1 - 2\lambda c_i + \lambda c_i^2$ into (18) yields (21).

A.9 Derivation of the ratio in Example 1

By symmetry of the bridge network in Figure 2, simultaneous-move equilibrium efforts satisfy $x_1^N = x_6^N$, $x_2^N = x_5^N$, and $x_3^N = x_4^N$. Because $d_1 = d_2 = 2$ and $d_3 = 3$ with $N(1) = \{2,3\}$, $N(2) = \{1,3\}$, and $N(3) = \{1,2,4\}$, the best-response equations in (4) for players 1, 2, and 3 are

$$\begin{aligned}
x_1^N &= (1-\lambda)\alpha_B + \frac{\lambda}{2}(x_2^N + x_3^N), \\
x_2^N &= (1-\lambda)\alpha_A + \frac{\lambda}{2}(x_1^N + x_3^N), \\
x_3^N &= (1-\lambda)\alpha_A + \frac{\lambda}{3}(x_1^N + x_2^N + x_3^N).
\end{aligned}$$

For $i \in \{1,2,3\}$, set $u_i := \alpha_A - x_i^N$. Subtracting the equations for x_2^N and x_3^N from α_A gives $u_2 = \frac{\lambda}{2}(u_1 + u_3)$ and $(3-\lambda)u_3 = \lambda(u_1 + u_2)$. Eliminating $\lambda u_1 = 2u_2 - \lambda u_3$ from the second equation yields $(3-\lambda)u_3 = (2+\lambda)u_2 - \lambda u_3$, so $3u_3 = (\lambda+2)u_2$. Combining this relation with $u_1 = (1-\lambda)(\alpha_A - \alpha_B) + \frac{\lambda}{2}(u_2 + u_3)$ gives

$$u_2 = \frac{6\lambda}{(\lambda+2)(\lambda+6)}(\alpha_A - \alpha_B) > 0, \qquad u_3 = \frac{2\lambda}{\lambda+6}(\alpha_A - \alpha_B) > 0,$$

whence $x_3^N - x_2^N = u_2 - u_3 = \frac{2\lambda(1-\lambda)}{(\lambda+2)(\lambda+6)}(\alpha_A - \alpha_B) > 0$ for all $\lambda \in (0,1)$ and $\alpha_A > \alpha_B$. Since $\alpha_2 = \alpha_3 = \alpha_A$, (7) implies $x_i^N - \bar{x}_i^N = u_i/\gamma$ for $i \in \{2,3\}$, and therefore

$$\frac{x_2^N - \bar{x}_2^N}{x_3^N - \bar{x}_3^N} = \frac{u_2}{u_3} = \frac{3}{\lambda+2} > 1.$$

A.10 Proof of Proposition 3

The following lemma gives the effort comparison used to establish leader eligibility in Proposition 3. Throughout this subsection, $0 < \lambda < 1$. For the empty group, set $\mathbf{E}_\varnothing = \mathbf{I}$ and $\mathbf{x}^S(\varnothing) = \mathbf{x}^N$.

Lemma 3. *For a leader group L with $0 < |L| < n$, let*

$$B := \left\{ i \in L : \alpha_i \le \sum_{j=1}^{n} m_{ij}^S(L)\alpha_j \right\}, \qquad A := L \setminus B.$$

If A and B are both nonempty, then $\mathbf{x}^S(A) > \mathbf{x}^S(L)$ coordinatewise. If $A = \varnothing$, then $\mathbf{x}^S(L) \le \mathbf{x}^N$ coordinatewise.

Proof. We evaluate the equilibrium system for A at the effort profile under L. By the proof of Theorem 1, the matrix in (A.1) has a strictly positive inverse. For a retained leader $i \in A$, norm feedback under L has the return-walk representation

$$c_i(L) = \sum_{t \ge 0} \lambda^{t+1} \widehat{\mathbf{G}}_{iF} \widehat{\mathbf{G}}_{FF}^t \widehat{\mathbf{G}}_{Fi}. \tag{A.7}$$

When the players in B become followers, the return walks available under L remain available under A, and the additional walks have nonnegative weights. Thus $c_i(A) \ge c_i(L)$ for $i \in A$.

By (A.1), the difference between the two systems at $\mathbf{x}^S(L)$ is

$$(\mathbf{Q}_A \mathbf{x}^S(L) - \boldsymbol{\alpha})_i = \gamma\Big[(\mathbf{E}_A)_{ii} - (\mathbf{E}_L)_{ii}\Big](x_i^S(L) - \bar{x}_i^S(L)).$$

For $i \in B$, the coefficient change is $c_i(L) \ge 0$, and the deviation from the social norm under L is nonpositive by (10). For $i \in A$, the coefficient change is nonpositive and the deviation under L is strictly positive. The coefficients of the original followers remain unchanged. Hence $\mathbf{Q}_A \mathbf{x}^S(L) \le \boldsymbol{\alpha}$. Since the inverse is positive, we obtain $\mathbf{x}^S(A) \ge \mathbf{x}^S(L)$.

It remains to establish strict improvement when both sets are nonempty. Take a shortest path between A and B; all internal vertices lie in F. From its endpoint $i \in A$,

follow the path to B and return along the same path. Since the network is undirected and $0 < \lambda < 1$, this return walk has positive weight and is available under A but not under L. Hence $c_i(A) > c_i(L)$. The retained leader's deviation from the social norm under L is strictly positive, so at least one coordinate of $\boldsymbol{\alpha} - \mathbf{Q}_A \mathbf{x}^S(L)$ is strictly positive. The strictly positive inverse then gives a strict increase in each player's effort. If $A = \varnothing$, the weak comparison gives $\mathbf{x}^N \geq \mathbf{x}^S(L)$. □

We now return to the proof of Proposition 3.

Proof. We first show that an optimal leader group is nonempty and proper. For a player i with the highest productivity, positivity of the simultaneous-move influence weights and heterogeneous productivities imply

$$\alpha_i - x_i^N = \sum_{j=1}^{n} m_{ij}^N (\alpha_i - \alpha_j) > 0.$$

By Proposition 2, $\ell_i > 0$. The empty and full groups yield zero gain, so neither can be optimal.

Part (i): Leader eligibility. Lemma 3 excludes an optimal group in which all leaders exert effort at or above productivity, as well as a group combining such leaders with leaders whose efforts are below productivity. Thus $\alpha_i - x_i^S(L^*) > 0$ for each leader $i \in L^*$. To compare equilibrium efforts with the Nash benchmark, we subtract the simultaneous-move system from (A.1). The leaders' first-order conditions then give

$$\mathbf{x}^S(L^*) - \mathbf{x}^N = \sum_{i \in L^*} \frac{c_i(L^*)}{1 - c_i(L^*)} \mathbf{M}_{\cdot i}^N [\alpha_i - x_i^S(L^*)]. \tag{A.8}$$

The terms in (A.8) are nonnegative. Connectedness ensures that at least one leader has at least one follower as a neighbor, and the two-step return through that neighbor gives strictly positive norm feedback. Since $\mathbf{M}^N$ has positive entries, $\mathbf{x}^S(L^*) > \mathbf{x}^N$. Together with the upper bound $x_i^S(L^*) < \alpha_i$ and (7), this yields $\bar{x}_i^N < x_i^N < x_i^S(L^*) < \alpha_i$ for each leader $i \in L^*$, so Proposition 2 implies $\ell_i > 0$. This proves part (i).

Part (ii): Neighbors in the follower group. Suppose, to the contrary, that a leader $i \in L^*$ has no followers among her neighbors. Then $c_i(L^*) = 0$. By connectedness, player i has a neighbor $j \in L^*$, so $R := L^* \setminus \{i\}$ is nonempty. Under R, player i becomes a follower. The corresponding diagonal entry of $\mathbf{E}_R$ remains one, and the diagonal entries of the original followers do not change. The return walks available to a retained leader under L^* remain available under R, so $c_k(R) \geq c_k(L^*)$. The additional walk $j \to i \to j$ makes the inequality strict for j.

By the upper effort bound in part (i) and (10), $(x_k^S(L^*) - \bar{x}_k^S(L^*)) > 0$ for all retained leaders. With the changed coefficients, the original effort profile satisfies

$$\mathbf{Q}_R \mathbf{x}^S(L^*) \leq \boldsymbol{\alpha}, \qquad (\mathbf{Q}_R \mathbf{x}^S(L^*))_j < \alpha_j.$$

Since $\mathbf{Q}_R^{-1} > \mathbf{0}$, the nonnegative residual with a strictly positive coordinate implies $\mathbf{x}^S(R) > \mathbf{x}^S(L^*)$. This contradicts the optimality of L^*. Thus each leader in an optimal group has at least one follower as a neighbor.

Under a positive upper bound on group size, the singleton led by a player with the highest productivity and each leader deletion remain feasible. The preceding comparisons therefore establish the same restrictions and effort bounds under that constraint. □

A.11 Proof of Proposition 4

The following lemma identifies the limiting effort profile for a fixed leader group. We use this limit to compare groups in part (ii) of Proposition 4.

Lemma 4. *For a leader group L with $|L| \geq 2$, each $c_i(1) < 1$, and equilibrium efforts converge to the common limit in (25). A singleton instead satisfies (A.6). The empty and full groups converge to $\bar{\alpha}_g \mathbf{1}$.*

Proof. For a nonempty proper leader group, each follower has a positive-probability random-walk path to a leader. Finiteness gives a common finite horizon and a positive lower bound on the probability of leaving the follower set. Thus the substochastic matrix $\widehat{\mathbf{G}}_{FF}$ is transient and satisfies $\rho(\widehat{\mathbf{G}}_{FF}) < 1$. Hence

$$\mathbf{C}_L(1) := \lim_{\lambda \to 1} \mathbf{C}_L(\lambda) = \widehat{\mathbf{G}}_{LL} + \widehat{\mathbf{G}}_{LF}(\mathbf{I}_F - \widehat{\mathbf{G}}_{FF})^{-1} \widehat{\mathbf{G}}_{FL}$$

is well defined. Entry (i, j) is the probability that a walk starting at leader i next reaches the leader set at j. If there are at least two leaders, each leader has a path to another leader with no internal leader. Since that path has positive probability, $1 - c_i(1) > 0$. For a singleton, the walk returns to the sole leader with probability one, so $c_i(1) = 1$.

We first establish convergence to a common effort level for $2 \leq |L| < n$. The leader and follower equilibrium conditions can be written as

$$\begin{aligned} \alpha_i - x_i^S &= \gamma(1 - c_i)(x_i^S - \bar{x}_i^S), \quad i \in L, \\ \alpha_j - x_j^S &= \gamma(x_j^S - \bar{x}_j^S), \qquad\qquad j \in F. \end{aligned}$$

By Theorem 1, equilibrium efforts lie between the minimum and maximum productivities. Since $c_i(1) < 1$, the first-order conditions imply that $x_k^S - \bar{x}_k^S \to 0$ for every player as $\lambda \to 1$. Any convergent subsequence therefore has a limit $\mathbf{y}$ satisfying $\mathbf{y} = \widehat{\mathbf{G}}\mathbf{y}$. Connectedness implies that all entries of $\mathbf{y}$ are equal.

To determine the common level, multiply each leader's condition by $\pi_i/(1-c_i)$ and each follower's condition by π_j, and sum across players. For every nonempty proper group and $0 < \lambda < 1$, this gives

$$\begin{aligned}\sum_{i\in L} \frac{\pi_i}{1-c_i}(\alpha_i - x_i^S) + \sum_{j\in F} \pi_j(\alpha_j - x_j^S) \\ = \gamma \sum_{k\in N} \pi_k(x_k^S - \bar{x}_k^S) = 0.\end{aligned}$$

The last equality follows from $\boldsymbol{\pi}^\top \widehat{\mathbf{G}} = \boldsymbol{\pi}^\top$, which implies

$$\sum_{k\in N} \pi_k \bar{x}_k^S = \sum_{k\in N} \pi_k \sum_{j\in N} \hat{g}_{kj} x_j^S = \sum_{j\in N} \pi_j x_j^S.$$

Rearranging yields the weighted identity

$$\sum_{i\in L} \frac{\pi_i}{1-c_i} x_i^S + \sum_{j\in F} \pi_j x_j^S = \sum_{i\in L} \frac{\pi_i}{1-c_i}\alpha_i + \sum_{j\in F} \pi_j \alpha_j. \tag{A.9}$$

For $2 \le |L| < n$, let z denote the common effort along a convergent subsequence. Taking limits in (A.9) gives

$$z\left[\sum_{i\in L} \frac{\pi_i}{1-c_i(1)} + \sum_{j\in F} \pi_j\right] = \sum_{i\in L} \frac{\pi_i}{1-c_i(1)}\alpha_i + \sum_{j\in F} \pi_j \alpha_j.$$

Division by the strictly positive coefficient on z gives the first expression in (25). The second follows from $\pi_i/[1-c_i(1)] = \pi_i + \pi_i c_i(1)/[1-c_i(1)]$, $\sum_{k\in N} \pi_k = 1$, and $\sum_{k\in N} \pi_k \alpha_k = \bar{\alpha}_g$. Every convergent subsequence has this same limit, so the entire equilibrium effort profile converges.

For a singleton $L = \{i\}$, (A.6) establishes convergence to $\alpha_i \mathbf{1}$. The empty and full groups reproduce the Nash equilibrium, whose limit is given by (A.5). For the full group, $c_i(1) = 0$ for every player, so (25) also gives $\bar{\alpha}_g$.

Under heterogeneous productivities, the limit for any proper group with at least two leaders assigns strictly positive weight to every player's productivity and is therefore strictly below the maximum productivity. The same strict inequality holds for the empty and full groups. A singleton has limiting effort α_i, so equality with the maximum

productivity holds if and only if the group consists of a single player with the highest productivity. □

With Lemma 4 established, we prove the weak- and strong-conformity statements in turn.

Proof. Part (i): Weak conformity. The expansion gives a common objective for group selection. We then use that objective to screen candidates and characterize when appointing all candidates is optimal.

Step 1: The aggregate gain. For a nonempty proper group L, define

$$\mathbf{H}_L := \mathbf{I}_L + \gamma(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L).$$

The leader system and the Nash equilibrium can be evaluated along the same response of followers. Their difference is

$$\mathbf{x}_L^S - \mathbf{x}_L^N = \mathbf{H}_L^{-1}\mathbf{C}_L^{\mathrm{D}}(\boldsymbol{\alpha}_L - \mathbf{x}_L^N).$$

The difference in followers' efforts equals $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}$ times the difference in leaders' efforts. To obtain the leading term in the aggregate gain, we use the following expansions near zero:

$$\begin{aligned}
\mathbf{H}_L^{-1} &= \mathbf{I}_L + \mathcal{O}(\lambda), \\
\mathbf{C}_L^{\mathrm{D}} &= \lambda(\widehat{\mathbf{G}}_{LF}\widehat{\mathbf{G}}_{FL})^{\mathrm{D}} + \mathcal{O}(\lambda^2), \\
\boldsymbol{\alpha}_L - \mathbf{x}_L^N &= \lambda[(\mathbf{I} - \widehat{\mathbf{G}})\boldsymbol{\alpha}]_L + \mathcal{O}(\lambda^2), \\
\mathbf{1}_L^\top + \lambda \mathbf{1}_F^\top \mathbf{U}_F \widehat{\mathbf{G}}_{FL} &= \mathbf{1}_L^\top + \mathcal{O}(\lambda).
\end{aligned}$$

The product of these expressions gives $\ell(L) = \lambda^2 \Lambda(L) + \mathcal{O}(\lambda^3)$, where $\Lambda(L)$ is defined in (22). The empty and full groups have zero gain and zero coefficient. Since the number of groups is finite, the remainder is uniform across groups, and a strict difference in the leading coefficients remains strict for sufficiently small positive λ.

Step 2: Candidate screening. Let $A := L \cap N_+$ and $B := L \setminus N_+$. Removing B eliminates its nonpositive contributions to followers and makes its members available to respond to leaders in A. The change in the objective is

$$\Lambda(A) - \Lambda(L) = \sum_{i \in A, j \in B} r_{ij}(\alpha_i - \bar{\alpha}_i) - \sum_{i \in B, j \notin L} r_{ij}(\alpha_i - \bar{\alpha}_i) \geq 0.$$

Hence restricting a group to N_+ weakly increases the objective. If all local productivity gaps are nonzero and B is nonempty, each gap in B is negative. Heterogeneous productivities and $\boldsymbol{\pi}^\top(\mathbf{I} - \widehat{\mathbf{G}})\boldsymbol{\alpha} = 0$ guarantee a positive-gap player outside B. Connectedness then gives an edge from B to a player either in A or outside L. That edge makes a strictly positive contribution to the displayed difference. Thus a leading-order optimum is contained in N_+ when no local productivity gap is zero.

Step 3: Interactions among candidates. For $L \subseteq N_+$, the sum of the singleton values $r_i(\alpha_i - \bar{\alpha}_i)$ counts both directions of each link within L. Subtracting those contributions gives the pairwise-loss formula (23). For $S, T \subseteq N_+$, the singleton contributions cancel in the submodularity difference, and we obtain

$$\Lambda(S) + \Lambda(T) - \Lambda(S \cup T) - \Lambda(S \cap T) = \sum_{\substack{i \in S \setminus T \\ j \in T \setminus S}} r_{ij}[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)] \geq 0.$$

Thus Λ is submodular on 2^{N_+}. The inequality is strict precisely when an edge joins $S \setminus T$ to $T \setminus S$. In particular, for $L \subseteq N_+$ and distinct $i, j \in N_+ \setminus L$, appointing j reduces the marginal contribution of appointing i by $r_{ij}[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)]$, which is strictly positive when the candidates are adjacent. Comparing each candidate's appointment with leaving that candidate as a follower gives

$$\Lambda(N_+) - \Lambda(N_+ \setminus \{i\}) = (\alpha_i - \bar{\alpha}_i) \sum_{j \notin N_+} r_{ij} - \sum_{j \in N_+} r_{ij}(\alpha_j - \bar{\alpha}_j).$$

If N_+ is optimal, each deletion margin must be nonnegative. Conversely, suppose all deletion margins are nonnegative. By submodularity, the marginal contribution of a candidate to a smaller group is at least as large as the corresponding margin at N_+. Adding the missing candidates therefore cannot lower the objective. Step 2 extends this comparison from candidate groups to all groups.

If all deletion margins are strictly positive, N_+ is uniquely optimal among candidate groups. A group omitting a candidate has a strictly smaller value after screening and hence also before screening. By the uniform expansion in Step 1, an exact optimum near zero contains N_+. If no local productivity gap is zero, the strict screening comparison also excludes additional noncandidates, so the exact optimum is unique. If a deletion margin is negative, removing that candidate strictly improves on N_+, whereas screening bounds the value of any group containing N_+ by $\Lambda(N_+)$. The strict difference persists near zero, so no exact optimum contains all candidates. Zero deletion margins allow leading-order ties.

Part (ii): Strong conformity. A singleton led by a player i with the highest productivity yields limiting aggregate effort $n \max_j \alpha_j$. We compare this limit with those of singletons with lower productivity and groups of other sizes. A singleton with lower productivity has a strictly lower limit. By Lemma 4, a proper group with at least two members has a limit equal to n times an average that assigns positive weight to each player's productivity. Heterogeneity makes that limit strictly lower as well. The empty and full groups have limit $n\bar{\alpha}_g < n \max_j \alpha_j$. There are finitely many groups of these types, so the singleton led by i strictly outperforms all of them on a common neighborhood below one. An optimal group in that neighborhood must therefore consist of a single player with the highest productivity.

For the individual effort comparison, a singleton led by a player with the highest productivity makes all efforts converge to $\max_j \alpha_j$. For a group of at least two such leaders, the common limit in Lemma 4 assigns positive weight to all productivities, including a lower-productivity follower, and is therefore strictly smaller. The number of singletons, groups and players is finite, so all coordinatewise comparisons hold on a common neighborhood below one. □

A.12 Proof of Claim 1

Proof. The submodularity claim follows from the pairwise-loss calculation in the proof of Proposition 4. We reduce MAX-CUT on connected regular graphs to (22). This source problem is NP-hard: Jelínek et al. (2026) establish NP-completeness of Large-Deg-Max-Cut, whose input graphs on n vertices are $(n-4)$-regular with connected cubic complements. For $n \geq 8$, degree $n - 4 \geq n/2$ also makes the input graph connected; excluding finitely many smaller instances does not affect hardness.

Let H be such a connected d-regular graph, with adjacency matrix $\mathbf{A}$, and fix a vertex r. Write $\boldsymbol{\mathcal{L}} = d\mathbf{I} - \mathbf{A}$ for its Laplacian, with grounded Laplacian $\boldsymbol{\mathcal{L}}_{-r}$. Keep H as the interaction network and set

$$\alpha_r = 1, \qquad \boldsymbol{\alpha}_{-r} = \mathbf{1}_{-r} + d\boldsymbol{\mathcal{L}}_{-r}^{-1}\mathbf{1}_{-r}.$$

The grounded Laplacian is positive definite: extending a vector by zero at r gives the quadratic form $\sum_{\{i,j\}\in E(H)}(z_i - z_j)^2$, which vanishes only at zero by connectedness. The vector $\mathbf{u}_{-r} = d\boldsymbol{\mathcal{L}}_{-r}^{-1}\mathbf{1}_{-r}$ is strictly positive. Otherwise a nonpositive minimum, with $u_r = 0$,

would satisfy $\sum_{j\in N(i)}(u_i - u_j) \leq 0$, contradicting its equation $\sum_{j\in N(i)}(u_i - u_j) = d$. Thus all productivities are positive. The construction gives

$$\alpha_i - \bar{\alpha}_i = \begin{cases} 1, & i \neq r, \\ -(n-1), & i = r. \end{cases}$$

The second equality follows because local productivity gaps sum to zero on a regular undirected graph. The rational linear system can be solved in polynomial time, and determinant bounds give polynomial bit length for the resulting productivities.

It remains to relate the leadership objective to the cut value. An optimal group leaves r as a follower: if $r \in L$, and a and b count its neighbors in the follower and leader groups, respectively, removing r raises the objective by $[(n-1)a + b]/d^2 > 0$, since $a + b = d$. For $L \subseteq N \setminus \{r\}$, each cut edge has a leading endpoint with local productivity gap one and link reciprocity $1/d^2$. Therefore

$$\Lambda(L) = \frac{|\partial_H L|}{d^2}, \qquad \max_{L\subseteq N} \Lambda(L) = \frac{\mathrm{MAXCUT}(H)}{d^2}.$$

The second equality uses the symmetry of an undirected cut: either side of a maximum cut can be designated as leaders, so requiring r to follow loses no cut value. An instance with cut threshold k is therefore equivalent to the leadership instance with threshold k/d^2. The reduction preserves the network and has polynomial encoding length, proving the claim. □

A.13 Proof of the claim on leaving candidates as followers

Proof. If $N_+ \subseteq L^*$, the candidate i with $N(i) \subseteq N_+$ is a leader with no followers among her neighbors, contradicting Proposition 3(ii). □

A.14 Derivation of the objective in Example 2

In the 3-regular network of Figure 4, every player has degree $d_i = 3$, so every link $\{i, j\}$ has link reciprocity $r_{ij} = 1/(d_i d_j) = 1/9$ and every player $i \in N$ has local reciprocity $r_i = \sum_{j\in N(i)} r_{ij} = 1/3$. Each player $i \in A = \{1, 2, 3\}$ has two neighbors in A with productivity α_A and one neighbor in B with productivity α_B, which gives $\bar{\alpha}_i = (2\alpha_A + \alpha_B)/3$ and $\alpha_i - \bar{\alpha}_i = (\alpha_A - \alpha_B)/3 > 0$. Because A forms a triangle, any

nonempty leader group $L \subseteq A$ of size $l = |L| \in \{1, 2, 3\}$ contains $l(l-1)/2$ links with both endpoints in L. Substituting these quantities into (23) yields

$$\Lambda(L) = \frac{l}{3} \cdot \frac{\alpha_A - \alpha_B}{3} - \frac{l(l-1)}{2} \cdot \frac{2(\alpha_A - \alpha_B)}{27} = \frac{\alpha_A - \alpha_B}{27} \Big[3l - l(l-1)\Big] = \frac{\alpha_A - \alpha_B}{27}\, l(4-l).$$

A.15 Proof of Proposition 5

Proof. Parts (i) and (ii): Leader selection and group size. We first express the gain in terms of group size and the sum of leading productivities. For a group L of size m, let $f := n - m$, $a_L := \sum_{i \in L} \alpha_i$ and $a_N := \sum_{i=1}^{n} \alpha_i$. Each follower responds to a given leader at rate $\gamma/(n - 1 + m\gamma)$, so each leader's norm feedback is

$$t_m := \frac{f\gamma}{(n-1)(n-1+m\gamma)}.$$

Let $s := 1 - t_m$, $v := 1 + \gamma n/(n-1)$ and $v_L := 1 + \gamma s n/(n-1)$. The equilibrium conditions are $v x_j = \alpha_j + \gamma X/(n-1)$ for followers and $v_L x_i = \alpha_i + \gamma s X/(n-1)$ for leaders. The sum over both groups gives

$$X^S(L) = \frac{a_L v + (a_N - a_L) v_L}{1 + \gamma(m + sf)/(n-1)}.$$

Since $X^N = a_N$ on a complete network, subtraction and simplification give

$$\ell(L) = \frac{\gamma t_m}{n - 1 + \gamma(n - f t_m)} (n a_L - m a_N). \tag{A.10}$$

At a prescribed size, only a_L depends on group membership. Its coefficient is strictly positive, so the strict productivity ordering uniquely selects L_m and, at $m = 1$, makes player 1 the unique key leader. This establishes part (i). Substitution of t_m into (A.10) gives

$$\ell(L_m) = \frac{n\gamma^2 (n-m) \sum_{j=1}^{m} (\alpha_j - \bar{\alpha})}{(n-1)^3 + (n-1)^2(n+m)\gamma + [n(n-1)m - (n-m)^2]\gamma^2}. \tag{A.11}$$

To compare group sizes, let $T_m := \sum_{j=1}^{m} (\alpha_j - \bar{\alpha})$ and $A_m := (n-m) T_m$. The strict productivity ordering implies $T_m > 0$ for $1 \leq m \leq n-1$. Write the denominator of (A.11) as

$$\begin{aligned} D_m(\gamma) &:= d_0 + b_m \gamma + a_m \gamma^2, \\ d_0 &:= (n-1)^3, \qquad b_m := (n-1)^2 (n+m), \\ a_m &:= n(n-1)m - (n-m)^2. \end{aligned}$$

Then $\ell(L_m) = n\gamma^2 A_m/D_m$. Both D_m and its coefficients are positive. Since $\eta = (n-1+n\gamma)/\gamma$,

$$\frac{D_m}{\gamma^2} = (n-1)\eta(\eta - n + m) - (n-m)^2,$$

so (A.11) equals the objective in (26). If $\alpha_m \leq \bar{\alpha}$, then $m \geq 2$, $A_m < A_{m-1}$, and $D_m > D_{m-1}$. Such a group has $\ell(L_m) < \ell(L_{m-1})$ and cannot be optimal.

We next show that the optimal size is at most $\bar{m}$. Player 1 alone yields a positive gain, whereas the empty and full groups reproduce simultaneous play, so an optimal group has size between 1 and $n-1$. Suppose a top-m group has $m > \bar{m}$ and $\alpha_m > \bar{\alpha}$. Since $T_{m-1} \geq (m-1)(\alpha_m - \bar{\alpha})$,

$$A_m - A_{m-1} = (n-m)(\alpha_m - \bar{\alpha}) - T_{m-1} \leq (n-2m+1)(\alpha_m - \bar{\alpha}) \leq 0.$$

Moreover, for $2 \leq m \leq n-1$ and $\gamma > 0$,

$$D_m - D_{m-1} = (n-1)^2\gamma + [n(n+1) - 2m + 1]\gamma^2 > 0.$$

As $A_{m-1} > 0$, these inequalities imply $\ell(L_m) < \ell(L_{m-1})$. Hence no size above $\bar{m}$ can be optimal. For $n = 2$, the only nontrivial size is one. Fixed-size productivity sorting and the bound $1 \leq m^* \leq \bar{m}$ therefore establish the exact reduction in part (ii).

Part (iii): Contraction with conformity. For $m < s$, it suffices to show that $\ell(L_s)/\ell(L_m)$ strictly decreases with γ. Since A_s/A_m does not depend on γ, we compare the denominators. The numerator of the derivative of D_s/D_m is

$$D_s' D_m - D_s D_m' = d_0(b_s - b_m) + 2d_0(a_s - a_m)\gamma + (a_s b_m - a_m b_s)\gamma^2.$$

All three coefficients are strictly positive, because

$$\begin{aligned} a_s - a_m &= (s-m)[n(n+1) - s - m] > 0, \\ a_s b_m - a_m b_s &= (n-1)^2(s-m)\{n^2(n+2) - n(s+m) - sm\} > 0. \end{aligned}$$

For $m, s \leq n-1$, the last bracket is at least $n^3 - n^2 + 4n - 1$. Hence D_s/D_m strictly increases with γ, and $\ell(L_s)/\ell(L_m) = (A_s/A_m)D_m/D_s$ strictly decreases. If size m is optimal at one parameter, no larger size can be optimal at a larger parameter. Since $\gamma = \lambda/(1-\lambda)$ strictly increases with λ and optimal groups share the same strict productivity ranking, $L^*(\lambda') \subseteq L^*(\lambda)$ for $\lambda' > \lambda$. This proves part (iii). □

A.16 Proof of Claim 2

Proof. We compare adjacent group sizes and then show that the gain increments change sign at most once. For $1 \le m \le \bar{m}$, define

$$T_m := \sum_{i=1}^{m} (\alpha_i - \bar{\alpha}), \qquad Q_m := \frac{(n-1)\eta^2}{n-m} - (n-1)\eta - (n-m).$$

The objective in (26) is $\ell(L_m) = nT_m/Q_m$. Strict productivity ordering gives $T_m > 0$. Since $\eta > n$, $Q_1 = \eta^2 - (n-1)\eta - (n-1) > 1$, and for $1 \le m < \bar{m}$,

$$\Delta Q_m := Q_{m+1} - Q_m = 1 + \frac{(n-1)\eta^2}{(n-m)(n-m-1)} > 0.$$

Thus all Q_m are positive, and the increments ΔQ_m strictly increase with m. Subtracting adjacent gains yields

$$\ell(L_{m+1}) - \ell(L_m) = \frac{nH_m}{Q_m Q_{m+1}}, \qquad H_m := (\alpha_{m+1} - \bar{\alpha})Q_m - T_m \Delta Q_m.$$

Since $T_m/Q_m = \ell(L_m)/n$, the inequality $H_m > 0$ is exactly (27).

It remains to compare the signs of successive gain increments. For $1 \le m < \bar{m} - 1$, the identities $T_{m+1} = T_m + \alpha_{m+1} - \bar{\alpha}$ and $Q_{m+1} = Q_m + \Delta Q_m$ imply

$$H_{m+1} - H_m = (\alpha_{m+2} - \alpha_{m+1})Q_{m+1} - T_{m+1}(\Delta Q_{m+1} - \Delta Q_m) < 0.$$

Hence the signs of the gain increments change at most once, from positive to nonpositive, and all increments after the first nonpositive one are strictly negative. Algorithm 1 therefore stops at an optimal size. If no appointment is rejected, the gain increases through $\bar{m}$, so $L_{\bar{m}}$ is optimal. The same conclusion holds directly when $\bar{m} = 1$. Computing $\bar{\alpha}$ takes $O(n)$ operations, and updating T_m and evaluating each appointment takes constant time. At most $\bar{m} - 1$ appointments are tested, giving $O(n)$ arithmetic operations for the stated productivity ordering. □

A.17 Proof of Corollary 3

Proof. We first identify the parameter range in which all leader groups are admissible in both models. Let $d := n - 1$, $H := d + n\gamma$ and $f := n - m$. Under the local-aggregate payoff (19), followers' responses coincide with those in the local-average model, so norm

feedback remains $t_m = f\gamma/[d(d+m\gamma)]$. The reduced local-aggregate leader system has nonpositive off-diagonal entries and row sum

$$\frac{dH - \gamma^2 f}{d(d+m\gamma)}.$$

The inequality $\gamma^2 < n\gamma + n - 1$ makes the row sums positive for any m, giving a unique positive equilibrium by strict diagonal dominance and positivity of the right-hand side. The bound is tight across groups: at or above the boundary, the singleton leader's reduced payoff has nonnegative curvature and a positive linear coefficient, hence no finite optimum. Thus $0 < \gamma < (n+\sqrt{n^2+4(n-1)})/2$ permits unrestricted group selection in both models; substituting $\gamma = \lambda/(1-\lambda)$ into $\gamma^2 < n\gamma + n - 1$ gives $2\lambda^2 + (n-2)\lambda - (n-1) < 0$, which holds for $\lambda \in (0,1)$ if and only if $0 < \lambda < \bar{\lambda} := \left(\sqrt{n^2+4n-4} - n + 2\right)/4$.

Let $\ell^{\text{agg}}(L_m)$ be the gain from the top m players in the corresponding local-aggregate model. Summing the leaders' first-order conditions $(1 + n\gamma/d - \gamma t_m)x_i = \alpha_i + \gamma X/d$ and the followers' first-order conditions $(1 + n\gamma/d)x_j = \alpha_j + \gamma X/d$ gives

$$\ell^{\text{agg}}(L_m) = \frac{\gamma^2(n-m)\left[d\sum_{i=1}^{m}\alpha_i + m\gamma n\bar{\alpha}\right]}{E_m}, \qquad E_m := (d+m\gamma)[dH - \gamma^2(n-m)] > 0. \tag{A.12}$$

For an arbitrary group of size m, the same formula replaces the top-productivity sum by the sum over that group. Its coefficient is positive, so an optimal group at a fixed size selects the highest productivities in both models. Both gains are positive for top groups of sizes $1, \ldots, n-1$, whereas the empty and full groups have zero gain.

To compare optimal group sizes, we use D_m and T_m from the proof of Proposition 5. Define $z_m := m(n-m)/D_m > 0$ and $\sigma_m := T_m/m$. Heterogeneity implies $\sigma_m > 0$, and the productivity ordering makes σ_m weakly decreasing in m. Since

$$D_m = dH(d+m\gamma) - \gamma^2(n-m)^2, \qquad \frac{E_m}{D_m} = 1 - \gamma^2 z_m\left(1 + \gamma - \frac{1}{m}\right) > 0,$$

equations (A.11) and (A.12) imply

$$\ell(L_m) = n\gamma^2 z_m \sigma_m, \qquad \ell^{\text{agg}}(L_m) = \frac{(d/n)\ell(L_m) + \gamma^2 H\bar{\alpha} z_m}{1 - \gamma^2 z_m(1 + \gamma - 1/m)}. \tag{A.13}$$

Suppose $s > r$ and $\ell(L_s) \geq \ell(L_r)$. Because $\sigma_s \leq \sigma_r$, the first identity gives $z_s \geq z_r$. In the second identity the numerator therefore weakly increases, while the positive denominator strictly decreases, since $1 + \gamma - 1/s > 1 + \gamma - 1/r$. Hence

$$\ell(L_s) \geq \ell(L_r) \quad \Longrightarrow \quad \ell^{\text{agg}}(L_s) > \ell^{\text{agg}}(L_r) \qquad (s > r).$$

If an optimal local-average size exceeded an optimal local-aggregate size, this implication would contradict local-aggregate optimality. Thus an optimal local-average size is no larger than any optimal local-aggregate size. Since both models choose the most productive players at a prescribed size, an optimal local-average group is contained in any optimal local-aggregate group. □

A.18 Proof of Proposition 6

Proof. Part (i): Key leader. To compare candidates, we first express the leading index in terms of individual productivity and the average productivity in each part. The mean Nash efforts in the two parts satisfy $\bar{x}_1^N = (1-\lambda)\bar{\alpha}_1 + \lambda\bar{x}_2^N$ and $\bar{x}_2^N = (1-\lambda)\bar{\alpha}_2 + \lambda\bar{x}_1^N$. The solution is

$$\bar{x}_1^N = \frac{\bar{\alpha}_1 + \lambda\bar{\alpha}_2}{1+\lambda}, \qquad \bar{x}_2^N = \frac{\bar{\alpha}_2 + \lambda\bar{\alpha}_1}{1+\lambda}.$$

For a candidate $i \in N_1$, the equations for followers' responses imply that an increase in x_i raises each effort in N_2 at rate $\lambda/[n_1 - \lambda^2(n_1-1)]$ and each follower's effort in N_1 at rate $\lambda^2/[n_1 - \lambda^2(n_1-1)]$. These responses determine the norm feedback and the network propagation multiplier:

$$c_i = \frac{\lambda}{n_1 - \lambda^2(n_1-1)}, \qquad \frac{\sum_{j\in N} m_{ji}^N}{m_{ii}^N} = \frac{n_1 + \lambda n_2}{n_1 - \lambda^2(n_1-1)}.$$

Let $H_r := r^2(1-\lambda)(1+\lambda)^2 + \lambda^3 > 0$ for $r \in \{n_1, n_2\}$. By (18) and $x_i^N = (1-\lambda)\alpha_i + \lambda\bar{x}_2^N$, we obtain

$$\ell_i = \frac{\lambda^2(n_1 + \lambda n_2)}{H_{n_1}} \left(\alpha_i - \frac{\bar{\alpha}_2 + \lambda\bar{\alpha}_1}{1+\lambda} \right), \qquad i \in N_1.$$

Interchanging n_1, n_2 and the average productivities of the two parts gives the expression for $j \in N_2$. Within either part, the network coefficient is strictly positive and the average productivities of the two parts are common to all candidates. The leading index therefore ranks candidates by individual productivity. A key leader must have the highest productivity in the corresponding part, and candidates tied for the highest productivity have the same index. Let ℓ_{N_1} and ℓ_{N_2} denote the maximal indices in the two parts.

For equal part sizes, $n_1 = n_2$ implies $H_{n_1} = H_{n_2}$. The difference between the maximal indices is then

$$\ell_{N_1} - \ell_{N_2} = \frac{\lambda^2 n_1}{H_{n_1}} \left[(1+\lambda)(\hat{\alpha}_1 - \hat{\alpha}_2) + (1-\lambda)(\bar{\alpha}_1 - \bar{\alpha}_2) \right].$$

Since the coefficient is strictly positive, the optimal part maximizes $(1+\lambda)\hat{\alpha}_p + (1-\lambda)\bar{\alpha}_p$ over $p \in \{1,2\}$. Both parts are eligible at equality.

If the two parts share the same average and maximum productivities, let $\bar{\alpha}_1 = \bar{\alpha}_2 = \mu$ and $\hat{\alpha}_1 = \hat{\alpha}_2 = a$. Heterogeneous productivities imply $a > \mu$. The maximal indices are $\ell_{N_1} = \lambda^2(n_1 + \lambda n_2)(a-\mu)/H_{n_1}$ and $\ell_{N_2} = \lambda^2(n_2 + \lambda n_1)(a-\mu)/H_{n_2}$, with difference

$$\ell_{N_1} - \ell_{N_2} = \frac{\lambda^2(a-\mu)(n_2-n_1)(1-\lambda)}{H_{n_1}H_{n_2}} \left\{(1+\lambda)^2\left[n_1n_2 + \lambda(n_1^2 + n_1n_2 + n_2^2)\right] - \lambda^3\right\}.$$

The term in braces is strictly positive: $n_1, n_2 \geq 1$ and $0 < \lambda < 1$ imply $(1+\lambda)^2 n_1 n_2 > \lambda^3$. Hence $\ell_{N_1} - \ell_{N_2}$ has the sign of $n_2 - n_1$, and a part of smaller size is optimal. This proves part (i).

Part (ii): Key group. Suppose $\alpha_i = \alpha_p$ for $i \in N_p$ ($p \in \{1,2\}$). Heterogeneity gives $\alpha_1 \neq \alpha_2$. Let N_p be the more productive part and N_q the other part, so $\alpha_p > \alpha_q$. The Nash efforts satisfy $x_i^N = (\alpha_p + \lambda\alpha_q)/(1+\lambda) < \alpha_p$ for $i \in N_p$ and $x_j^N = (\alpha_q + \lambda\alpha_p)/(1+\lambda) > \alpha_q$ for $j \in N_q$. By Proposition 3(i), no player in N_q belongs to an optimal leader group. A singleton in N_p yields a strictly positive gain, so an optimal group is a nonempty subset $L \subseteq N_p$.

It remains to determine how many players from N_p should lead. For a nonempty group $L \subseteq N_p$ of size $k := |L| \in \{1, \ldots, n_p\}$, let $D := n_p - \lambda^2(n_p - k) > 0$. The system of equations for followers' responses (8) gives $(\mathbf{C}_L)_{ij} = \lambda/D$ for all $i, j \in L$. The sum of the leaders' first-order conditions and the aggregate response of followers give

$$\sum_{i\in L}(x_i^S(L) - x_i^N) = \frac{(\lambda/D)\sum_{i\in L}(\alpha_i - x_i^N)}{1 + \gamma(1-\lambda/D)(1-k\lambda/D)},$$
$$\ell(L) = \frac{n_p + \lambda n_q}{D}\sum_{i\in L}(x_i^S(L) - x_i^N).$$

For each $i \in L$, $\alpha_i - x_i^N = \lambda(\alpha_p - \alpha_q)/(1+\lambda)$. This identity and $\gamma = \lambda/(1-\lambda)$ yield

$$\ell(L) = \frac{\lambda^2(n_p + \lambda n_q)(\alpha_p - \alpha_q)}{1+\lambda}\frac{k}{a_p k + b_p}, \tag{A.14}$$

where $a_p := \lambda^2[n_p(1+\lambda) + \lambda] > 0$ and $b_p := n_p(1+\lambda)[n_p - (n_p+1)\lambda^2]$, with $a_p k + b_p > 0$ for all $1 \leq k \leq n_p$.

For $1 \leq k < n_p$, the difference $(k+1)/[a_p(k+1) + b_p] - k/(a_p k + b_p) = b_p/[(a_p k + b_p)(a_p(k+1) + b_p)]$ has the sign of b_p. Since $b_p \gtrless 0$ according as $\lambda \lessgtr \sqrt{n_p/(n_p+1)} = \lambda_p$, $\ell(L)$ strictly increases with k when $0 < \lambda < \lambda_p$ and strictly decreases with k when $\lambda_p < \lambda < 1$. At $\lambda = \lambda_p$, the gain is independent of k. When $n_p = 1$, the entire part N_p

and its singleton coincide. These comparisons establish the optimal leader groups in all three regimes. □

A.19 Proof of Theorem 2

Proof. We use backward induction. Let $d_k := 1 - c_k$ and $\mathbf{D}_S := \operatorname{diag}(d_k)_{k \in S}$ for any set S.

Step 1: Continuation responses. Suppose the coefficients $d_j \in (0, 1]$ are determined for the subsequent movers F, and let $H = N \setminus F$. Their first-order conditions are

$$\mathbf{B}_F \mathbf{x}_F = \boldsymbol{\alpha}_F + \gamma \mathbf{D}_F \widehat{\mathbf{G}}_{FH} \mathbf{x}_H, \qquad \mathbf{B}_F := \mathbf{I}_F + \gamma \mathbf{D}_F (\mathbf{I}_F - \widehat{\mathbf{G}}_{FF}).$$

The off-diagonal entries of $\mathbf{B}_F$ are nonpositive, and each diagonal exceeds the sum of their absolute values by at least one. After division by its diagonal, $\mathbf{B}_F$ is an identity matrix minus a nonnegative matrix with row sums at most $\lambda < 1$. The geometric series therefore gives $\mathbf{B}_F^{-1} \geq \mathbf{0}$ with positive diagonal. Continuation efforts are positive at every nonnegative history, and their response to $\mathbf{x}_H$ is

$$\mathbf{R}_{FH} := \mathbf{B}_F^{-1} \gamma \mathbf{D}_F \widehat{\mathbf{G}}_{FH} \geq \mathbf{0}, \qquad \mathbf{R}_{FH} \mathbf{1}_H = \mathbf{1}_F - \mathbf{B}_F^{-1} \mathbf{1}_F < \mathbf{1}_F.$$

The row equations then give $(\mathbf{R}_{FH} \mathbf{1}_H)_j \leq \gamma d_j / (1 + \gamma d_j) \leq \lambda$.

For the current group L, the induced norms are $\mathbf{a} + \mathbf{C}_L \mathbf{x}_L$, where $\mathbf{a} \geq \mathbf{0}$ collects productivities and earlier efforts, and

$$\mathbf{C}_L = \widehat{\mathbf{G}}_{LL} + \widehat{\mathbf{G}}_{LF} \mathbf{R}_{FL} \geq \mathbf{0}, \qquad \mathbf{C}_L \mathbf{1}_L \leq \mathbf{1}_L.$$

Since $\widehat{g}_{kk} = 0$, the own norm response is $c_k = (\mathbf{C}_L)_{kk} \in [0, \lambda]$, so $d_k \in [1 - \lambda, 1]$. At the final stage, $F = \varnothing$ and $c_k = 0$, which starts the induction. Multiplication of $\mathbf{B}_F$ by $1 - \lambda$ gives the continuation matrix used in (28).

Step 2: Stage equilibrium and the influence matrix. Given the induced norms, player k's marginal payoff and curvature are

$$\frac{du_k}{dx_k} = \alpha_k - x_k - \gamma d_k (x_k - \bar{x}_k), \qquad \frac{d^2 u_k}{dx_k^2} = -1 - \gamma d_k^2 < 0.$$

The marginal payoff at zero is $\alpha_k + \gamma d_k \bar{x}_k > 0$, so best responses are interior. The current group's first-order conditions are

$$[\mathbf{I}_L + \gamma \mathbf{D}_L (\mathbf{I}_L - \mathbf{C}_L)] \mathbf{x}_L = \boldsymbol{\alpha}_L + \gamma \mathbf{D}_L \mathbf{a}.$$

The coefficient matrix has nonpositive off-diagonal entries and row-dominance margin $1+\gamma d_k\big(1-\sum_{j\in L}(\mathbf{C}_L)_{kj}\big)\geq 1$. The same row-scaling argument as in Step 1 gives a nonnegative inverse with positive diagonal. Thus the unique solution is positive and, by strict concavity, is the unique stage equilibrium. Since the coefficients do not depend on earlier efforts, backward induction establishes uniqueness in every subgame.

Finally, let $\mathbf{D}=\mathbf{I}-\mathbf{C}^{\mathrm{D}}(\mathcal{L})$. The first-order conditions across stages give

$$[\mathbf{I}+\gamma\mathbf{D}(\mathbf{I}-\widehat{\mathbf{G}})]\mathbf{x}^S=\boldsymbol{\alpha}.$$

The coefficient matrix is invertible by the same argument, and its inverse is (29). For $0<\lambda<1$, connectedness and $d_k>0$ supply a positive term for every pair of players in the inverse series, so $\mathbf{M}^S(\mathcal{L})>\mathbf{0}$. At $\lambda=0$, $\mathbf{M}^S(\mathcal{L})=\mathbf{I}$. □

A.20 Proof of Remark 2

Proof. By Theorem 2, $\mathbf{M}^S(\mathcal{L})$ is nonnegative (with $\mathbf{M}^S(\mathcal{L})=\mathbf{I}$ at $\lambda=0$). Since $\widehat{\mathbf{G}}\mathbf{1}=\mathbf{1}$, we have $(\mathbf{I}-\widehat{\mathbf{G}})\mathbf{1}=\mathbf{0}$. The matrix inverted in (29) therefore maps $\mathbf{1}$ to $(1-\lambda)\mathbf{1}$, so

$$\mathbf{M}^S(\mathcal{L})\mathbf{1}=(1-\lambda)\left[(1-\lambda)\mathbf{I}+\lambda\Big(\mathbf{I}-\mathbf{C}^{\mathrm{D}}(\mathcal{L})\Big)(\mathbf{I}-\widehat{\mathbf{G}})\right]^{-1}\mathbf{1}=\frac{1-\lambda}{1-\lambda}\mathbf{1}=\mathbf{1}.$$

□

A.21 Proof of Proposition 7

Proof. Part (i): Weak conformity. The continuation matrices in (28) are analytic near $\lambda=0$, where the matrices being inverted equal identity matrices. Hence

$$c_i(\mathcal{L})=\lambda\sum_{j:\,\tau_j>\tau_i}r_{ij}+\mathcal{O}(\lambda^2).$$

By the inverse identity,

$$\mathbf{x}^S(\mathcal{L})-\mathbf{x}^N=\gamma\mathbf{M}^S(\mathcal{L})\mathbf{C}^{\mathrm{D}}(\mathcal{L})(\mathbf{I}-\widehat{\mathbf{G}})\mathbf{x}^N.$$

Since both influence matrices equal $\mathbf{I}+\mathcal{O}(\lambda)$ and $\gamma=\lambda+\mathcal{O}(\lambda^2)$, we obtain

$$x_i^S(\mathcal{L})-x_i^N=\lambda^2(\alpha_i-\bar{\alpha}_i)\sum_{j:\,\tau_j>\tau_i}r_{ij}+\mathcal{O}(\lambda^3).$$

Summing over players gives $\ell(\mathcal{L})=\lambda^2\Lambda(\mathcal{L})+\mathcal{O}(\lambda^3)$. There are finitely many sequences, so the remainder is uniform and a strictly positive coefficient gap separates maximizers of

Λ from nonmaximizers. Consequently, every optimal sequence maximizes Λ for sufficiently small $\lambda > 0$.

Each link $\{i, j\}$ contributes either $r_{ij}(\alpha_i - \bar{\alpha}_i)$, $r_{ij}(\alpha_j - \bar{\alpha}_j)$, or zero. The contribution is therefore at most $r_{ij} \max\{0, \alpha_i - \bar{\alpha}_i, \alpha_j - \bar{\alpha}_j\}$. The stated ordering attains this bound on every link: a positive gap precedes any smaller gap, equal positive gaps can be ordered either way, and nonpositive gaps share the final stage. This final group is nonempty because a player with minimum productivity belongs to $N \setminus N_+$. Thus the ordering attains (31), proving part (i).

Part (ii): Strong conformity. Let $\varepsilon := 1/\gamma = (1-\lambda)/\lambda$, $d_j := 1 - c_j$, and $\mathbf{D}_S := \operatorname{diag}(d_j)_{j \in S}$.

Step 1: Limiting efforts. For a proper subset $F \subset N$, let $\mathbf{R}_F := (\mathbf{I}_F - \widehat{\mathbf{G}}_{FF})^{-1}$. Connectedness ensures that a random walk exits F with probability one, so $\mathbf{R}_F = \sum_{q \geq 0} \widehat{\mathbf{G}}_{FF}^{\,q} \geq \mathbf{0}$.

We first show that every d_j, except possibly that of a singleton first mover, has a positive limit

$$\bar{d}_j = 1 - \widehat{\mathbf{G}}_{j\mathcal{F}_t} \mathbf{R}_{\mathcal{F}_t} \widehat{\mathbf{G}}_{\mathcal{F}_t j} > 0, \qquad j \in L_t. \tag{A.15}$$

At the final stage, $d_j = 1$. For the backward induction, suppose the subsequent coefficients have positive limits. Their response to $H_t = N \setminus \mathcal{F}_t$ is

$$[\mathbf{I}_{\mathcal{F}_t} - \widehat{\mathbf{G}}_{\mathcal{F}_t \mathcal{F}_t} + \varepsilon \mathbf{D}_{\mathcal{F}_t}^{-1}]^{-1} \widehat{\mathbf{G}}_{\mathcal{F}_t H_t}.$$

This matrix is analytic at zero and converges to $\mathbf{R}_{\mathcal{F}_t} \widehat{\mathbf{G}}_{\mathcal{F}_t H_t}$, giving (A.15). The subtracted term is the probability of leaving j through $\mathcal{F}_t$ and next reaching H_t at j. Unless j is the only first mover, H_t contains another vertex; a simple path to that vertex reaches a different boundary vertex without returning to j. The probability is therefore below one. This completes the induction and establishes analyticity of the subsequent coefficients.

If $|L_1| \geq 2$, all d_j have positive limits. The bounds in Theorem 2 and $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^S = \varepsilon \mathbf{D}_N^{-1}(\boldsymbol{\alpha} - \mathbf{x}^S)$ imply that every subsequential limit is constant. Let $\boldsymbol{\pi}$ be the positive stationary distribution of $\widehat{\mathbf{G}}$. Multiplication by $\boldsymbol{\pi}^\top$ gives $\sum_j \pi_j x_j^S / d_j = \sum_j \pi_j \alpha_j / d_j$, so every player's effort converges to

$$\frac{\sum_j \pi_j \alpha_j / \bar{d}_j}{\sum_j \pi_j / \bar{d}_j} < \max_j \alpha_j,$$

where strictness follows from heterogeneity and positive weights.

Step 2: A singleton first mover. Suppose $L_1 = \{i\}$ and let $F = N \setminus \{i\}$. Given the first effort y, the continuation equilibrium satisfies

$$\begin{aligned}\mathbf{x}_F(y) &= [\mathbf{I}_F - \widehat{\mathbf{G}}_{FF} + \varepsilon \mathbf{D}_F^{-1}]^{-1}(\widehat{\mathbf{G}}_{Fi} y + \varepsilon \mathbf{D}_F^{-1} \boldsymbol{\alpha}_F) \\ &= y\mathbf{1}_F - \varepsilon \mathbf{R}_F \bar{\mathbf{D}}_F^{-1}(y\mathbf{1}_F - \boldsymbol{\alpha}_F) + \mathcal{O}(\varepsilon^2),\end{aligned}$$

where $\bar{\mathbf{D}}_F = \operatorname{diag}(\bar{d}_j)_{j \in F}$ and the expansion uses $(\mathbf{I}_F - \widehat{\mathbf{G}}_{FF})\mathbf{1}_F = \widehat{\mathbf{G}}_{Fi}$. Define

$$\sigma := \widehat{\mathbf{G}}_{iF} \mathbf{R}_F \bar{\mathbf{D}}_F^{-1} \mathbf{1}_F, \qquad \mathbf{g} := \mathbf{R}_F \bar{\mathbf{D}}_F^{-1}(\alpha_i \mathbf{1}_F - \boldsymbol{\alpha}_F).$$

The continuation map is affine in y, with analytic coefficients, so differentiation gives $d_i = \varepsilon\sigma + \mathcal{O}(\varepsilon^2)$. Since equilibrium efforts are bounded, the first mover's condition $\alpha_i - y = \gamma d_i (y - \bar{x}_i)$ first implies $y = \alpha_i + \mathcal{O}(\varepsilon)$ and then yields

$$y = \alpha_i - \varepsilon\sigma \widehat{\mathbf{G}}_{iF}\mathbf{g} + \mathcal{O}(\varepsilon^2), \qquad \mathbf{x}_F = y\mathbf{1}_F - \varepsilon \mathbf{g} + \mathcal{O}(\varepsilon^2).$$

Thus all efforts converge to α_i, and

$$X^S(\mathcal{L}) = n\alpha_i - \varepsilon \beta_i(\mathcal{L}) + \mathcal{O}(\varepsilon^2), \qquad \beta_i(\mathcal{L}) := n\sigma \widehat{\mathbf{G}}_{iF}\mathbf{g} + \mathbf{1}_F^\top \mathbf{g}. \tag{A.16}$$

Together with Step 1, this excludes multiple first movers and any first mover below the maximum productivity for all sufficiently large λ: their limiting aggregate efforts are strictly smaller, and there are finitely many sequences.

Step 3: The subsequent stages. Fix a first mover i with maximum productivity and compare $\mathcal{L}$ with $\mathcal{L}_0 = (\{i\}, F)$. Under $\mathcal{L}_0$, $\bar{\mathbf{D}}_F = \mathbf{I}_F$; use subscripts zero for its quantities in (A.16). Since $0 < \bar{d}_j \leq 1$ for all $j \in F$ (so $\mathbf{0} \leq \bar{\mathbf{D}}_F \leq \mathbf{I}_F$ and $\bar{\mathbf{D}}_F^{-1} \geq \mathbf{I}_F$) and $\alpha_i \mathbf{1}_F - \boldsymbol{\alpha}_F \geq \mathbf{0}$, we have $\mathbf{g} \geq \mathbf{g}_0 \geq \mathbf{0}$ and $\sigma \geq \sigma_0 > 0$. Every entry of $\widehat{\mathbf{G}}_{iF}\mathbf{R}_F$ is positive, because a simple path from i to any player in F contributes to its series. Hence heterogeneity gives $\widehat{\mathbf{G}}_{iF}\mathbf{g}_0 > 0$. If $\bar{\mathbf{D}}_F \neq \mathbf{I}_F$, then

$$\sigma - \sigma_0 = \widehat{\mathbf{G}}_{iF}\mathbf{R}_F(\bar{\mathbf{D}}_F^{-1} - \mathbf{I}_F)\mathbf{1}_F > 0,$$

and therefore

$$\beta_i(\mathcal{L}) - \beta_i(\mathcal{L}_0) \geq n(\sigma - \sigma_0)\widehat{\mathbf{G}}_{iF}\mathbf{g}_0 > 0.$$

Equation (A.16) implies that $\mathcal{L}_0$ yields higher aggregate effort for sufficiently large λ.

By (A.15), $\bar{d}_j < 1$ exactly when j has a later neighbor: such a neighbor k contributes $\widehat{g}_{jk}\widehat{g}_{kj} > 0$; without one, the response is zero. Hence $\bar{\mathbf{D}}_F = \mathbf{I}_F$ exactly when each connected component of the remaining subnetwork lies within one stage. Then $d_j = 1$

for all $j \in F$ at every finite γ, so the continuation equilibrium and first effort coincide with those under $\mathcal{L}_0$.

Finiteness gives a common threshold for these strict comparisons. Choosing the best two-stage sequence among players with maximum productivity proves part (ii). If that player is unique and the remaining subnetwork is connected, the optimal sequence is unique. □

A.22 Derivation of (32)

For $\boldsymbol{\alpha} = (7, 4, 2, 2)^\top$, write the productivity vector as $\boldsymbol{\alpha} = 2\mathbf{1} + 5\mathbf{e}_1 + 2\mathbf{e}_2$, where $\mathbf{e}_i$ has entry one at coordinate i and zero elsewhere. By Theorem 2 and Remark 2, the equilibrium effort profile under any sequence $\mathcal{L}$ satisfies

$$\mathbf{x}^S(\mathcal{L}) = \mathbf{M}^S(\mathcal{L})(2\mathbf{1} + 5\mathbf{e}_1 + 2\mathbf{e}_2) = 2\mathbf{1} + 5\mathbf{M}^S(\mathcal{L})\mathbf{e}_1 + 2\mathbf{M}^S(\mathcal{L})\mathbf{e}_2.$$

Summing across players $i \in \{1, 2, 3, 4\}$ yields

$$X^S(\mathcal{L}) = \sum_{i=1}^{4} x_i^S(\mathcal{L}) = 8 + 5\sum_{i=1}^{4} m_{i1}^S(\mathcal{L}) + 2\sum_{i=1}^{4} m_{i2}^S(\mathcal{L}),$$

which gives (32).

B Model Extension

B.1 Incomplete Information

Leaders may have superior information about the returns to effort (Hermalin, 1998; Zhou and Chen, 2016). We retain the baseline network and introduce a common state θ drawn from a prior distribution with full support on $[0, \bar{\theta}]$, where $0 < \bar{\theta} < \infty$. Individual productivities are public, and player i's payoff is

$$u_i(\mathbf{x}; \theta) = \alpha_i \theta x_i - \frac{1}{2} x_i^2 - \frac{\gamma}{2}(x_i - \bar{x}_i)^2, \tag{B.1}$$

where $\bar{x}_i = \sum_{j=1}^n \widehat{g}_{ij} x_j$ represents player i's social norm.

Throughout this subsection, $0 < \lambda = \gamma/(1+\gamma) < 1$, and we consider a leader group L with $0 < |L| < n$ and follower group $F := N \setminus L$ (we omit the argument L when the leader group is fixed). When θ is publicly observed, payoff (B.1) scales all individual productivities by θ, so simultaneous-move Nash efforts and two-stage equilibrium efforts are $\mathbf{x}^N(\theta) := \theta \mathbf{x}^N$ and $\mathbf{x}^S(L; \theta) := \theta \mathbf{x}^S(L)$, with aggregate efforts θX^N and $\theta X^S(L)$, where $X^N = \sum_{i=1}^n x_i^N$ and $X^S(L) = \sum_{i=1}^n x_i^S(L)$. Under private information, in Stage 1, leaders in L observe θ and simultaneously choose efforts $\mathbf{x}_L$; in Stage 2, followers in F observe all leaders' efforts $\mathbf{x}_L$ and simultaneously choose efforts $\mathbf{x}_F$. Before θ is realized, the planner selects a leader group L to maximize expected aggregate effort. Along any equilibrium path $\mathbf{x}^I(L; \theta) = \big(x_i^I(L; \theta)\big)_{i \in N}$ that is linear in θ, aggregate effort $X^I(L; \theta) := \sum_{i=1}^n x_i^I(L; \theta)$ satisfies $X^I(L; \theta) - \theta X^N = \sum_{i=1}^n \big[x_i^I(L; \theta) - \theta x_i^N\big] = \widehat{\ell}(L)\theta$, so expected aggregate effort equals $\mathbb{E}[\theta]\big(X^N + \widehat{\ell}(L)\big)$. Because $\mathbb{E}[\theta] > 0$, maximizing expected aggregate effort is equivalent to maximizing the leadership gain $\widehat{\ell}(L)$; for a single leader $L = \{i\}$, we refer to $\widehat{\ell}_i := \widehat{\ell}(\{i\})$ as player i's *signaling leading index*.

Followers' responses and separating equilibrium. Given leaders' Stage 1 efforts $\mathbf{x}_L = (x_i)_{i \in L} \geq \mathbf{0}$, followers share the same public observation and form the common posterior expectation $\mu(\mathbf{x}_L) := \mathbb{E}[\theta \mid \mathbf{x}_L]$ conditional on $\mathbf{x}_L$. Because followers do not observe θ, Stage 2 pure strategies $\mathbf{x}_F = (x_j)_{j \in F}$ depend only on $\mathbf{x}_L$. At any profile $(\mathbf{x}_L, \mathbf{x}_{F \setminus \{k\}})$, the social norm $\bar{x}_k = \sum_{i \in L} \widehat{g}_{ki} x_i + \sum_{j \in F} \widehat{g}_{kj} x_j$ is therefore nonrandom

conditional on $\mathbf{x}_L$, and each follower $k \in F$ chooses $x_k \geq 0$ to maximize the conditional expected payoff

$$\begin{aligned}\mathbb{E}\Big[u_k(\mathbf{x};\theta) \mid \mathbf{x}_L\Big] &= \alpha_k \mathbb{E}[\theta \mid \mathbf{x}_L] x_k - \frac{1}{2}x_k^2 - \frac{\gamma}{2}(x_k - \bar{x}_k)^2 \\ &= \alpha_k \mu(\mathbf{x}_L) x_k - \frac{1}{2}x_k^2 - \frac{\gamma}{2}(x_k - \bar{x}_k)^2.\end{aligned}$$

Strict concavity in x_k and $\mu(\mathbf{x}_L) \geq 0$ yield follower k's unique best response $x_k = (1-\lambda)\alpha_k \mu(\mathbf{x}_L) + \lambda \sum_{i \in L} \widehat{g}_{ki} x_i + \lambda \sum_{j \in F} \widehat{g}_{kj} x_j$. In vector form, with $\mathbf{U}_F := (\mathbf{I}_F - \lambda \widehat{\mathbf{G}}_{FF})^{-1}$, the Stage 2 best-response system across F admits the unique equilibrium solution

$$\mathbf{x}_F(\mathbf{x}_L) = \mathbf{U}_F\Big[(1-\lambda)\mu(\mathbf{x}_L)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL}\mathbf{x}_L\Big]. \tag{B.2}$$

In Stage 1, each leader $k \in L$ observes θ and evaluates (B.1) at the deterministic Stage 2 responses $\mathbf{x}_F(\mathbf{x}_L) = \big(x_j(\mathbf{x}_L)\big)_{j \in F}$ in (B.2), where $\mathbf{x}_L$ affects followers both through the posterior expectation $\mu(\mathbf{x}_L)$ and through direct conformity $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}\mathbf{x}_L$. Leader k therefore faces the induced social norm

$$\bar{x}_k(\mathbf{x}_L) = \sum_{i \in L} \widehat{g}_{ki} x_i + \sum_{j \in F} \widehat{g}_{kj} x_j(\mathbf{x}_L) = s_k \mu(\mathbf{x}_L) + \sum_{i \in L} c_{ki} x_i,$$

where $s_k := (1-\lambda)\big(\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F\big)_k$ and $c_{ki} := \widehat{g}_{ki} + \lambda\big(\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\big)_{ki}$. In vector form, the leaders' social norms are

$$\bar{\mathbf{x}}_L(\mathbf{x}_L) = \mu(\mathbf{x}_L)\mathbf{s}_L + \mathbf{C}_L\mathbf{x}_L, \qquad \mathbf{s}_L := (s_i)_{i \in L} = (1-\lambda)\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F, \tag{B.3}$$

where $\mathbf{s}_L$ records how a unit increase in the inferred state $\mu(\mathbf{x}_L)$ raises each leader's social norm through followers' efforts, and $\mathbf{C}_L = [c_{ij}]_{i,j \in L} = \widehat{\mathbf{G}}_{LL} + \lambda \widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}$ is the norm-response matrix from Section 3.1, with diagonal entries $c_i := (\mathbf{C}_L)_{ii}$. As in Section 3.1, the Neumann series $\mathbf{U}_F = \sum_{t=0}^{\infty} \lambda^t \widehat{\mathbf{G}}_{FF}^{\,t} \geq \mathbf{I}_F$ and $\boldsymbol{\alpha}_F > \mathbf{0}$ imply $(1-\lambda)\mathbf{U}_F\boldsymbol{\alpha}_F > \mathbf{0}$, $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL} \geq \mathbf{0}$, $\mathbf{s}_L \geq \mathbf{0}$, and $\mathbf{C}_L \geq \mathbf{0}$, with $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL}\mathbf{1}_L \leq \lambda \mathbf{1}_F$, $\mathbf{C}_L\mathbf{1}_L \leq \mathbf{1}_L$, $0 \leq c_i \leq \lambda < 1$, and $(\mathbf{I}_L - \mathbf{C}_L)^{-1} \geq \mathbf{I}_L$.

As in Zhou and Chen (2016), we focus on linear separating strategies in which each leader $i \in L$ chooses

$$x_i(\theta) = \kappa_i \theta, \qquad \theta \in [0, \bar{\theta}],$$

where $\kappa_i \in \mathbb{R}$ is a constant coefficient. Because efforts are nonnegative on $[0, \bar{\theta}]$ with $\bar{\theta} > 0$, any linear strategy satisfies $\kappa_i \geq 0$; moreover, $\kappa_i = 0$ pools all states at zero effort, whereas individual separation requires a one-to-one mapping from θ to $x_i(\theta)$ for

each leader $i \in L$. Every such linear separating strategy therefore satisfies $\kappa_i > 0$, so we restrict attention to coefficient vectors $\boldsymbol{\kappa} = (\kappa_i)_{i \in L} > \mathbf{0}$. On the equilibrium path, each leader's effort reveals $x_i(\theta)/\kappa_i = \theta$. To specify follower beliefs after arbitrary Stage 1 histories $\mathbf{x}_L \geq \mathbf{0}$, we adopt the belief rule of Zhou and Chen (2016), capped at the upper endpoint of the state support.

Assumption B.1 (Posterior beliefs)**.** For a coefficient vector $\boldsymbol{\kappa} = (\kappa_i)_{i \in L} > \mathbf{0}$, after observing leaders' efforts $\mathbf{x}_L = (x_i)_{i \in L} \geq \mathbf{0}$, followers assign probability one to the state

$$\mu_{\boldsymbol{\kappa}}(\mathbf{x}_L) := \min \left\{ \bar{\theta}, \min_{i \in L} \frac{x_i}{\kappa_i} \right\}. \tag{B.4}$$

Under complete information in Section 3, moving first raises a single leader i's effort above the Nash benchmark when $\alpha_i > x_i^N$, or equivalently $\alpha_i > \bar{x}_i^N$. Under private information, unlike settings with positive payoff externalities (Zhou and Chen, 2016) where higher efforts by followers always benefit leaders, social conformity implies $\partial u_i / \partial \bar{x}_i = \gamma(x_i - \bar{x}_i)$: a higher inferred state $\mu(\mathbf{x}_L)$ raises $\bar{x}_i$ through followers' responses, which reduces leader i's conformity cost when $x_i \geq \bar{x}_i$ but increases that cost when $x_i < \bar{x}_i$. A leader with $x_i < \bar{x}_i$ can therefore lower conformity costs by cutting effort to depress $\mu(\mathbf{x}_L)$ and $\bar{x}_i$, which undermines linear separation. To ensure that leaders in a group L remain in the regime $x_i \geq \bar{x}_i$, suppose all leaders choose standalone efforts $\mathbf{x}_L = \theta \boldsymbol{\alpha}_L$ and followers infer $\mu(\mathbf{x}_L) = \theta$. By (B.2) and (B.3), followers choose $\mathbf{x}_F = \mathbf{U}_F \big[(1-\lambda)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL} \boldsymbol{\alpha}_L \big] \theta$, and the leaders face resulting social norms

$$\widehat{\mathbf{G}}_{LL}(\theta \boldsymbol{\alpha}_L) + \widehat{\mathbf{G}}_{LF} \mathbf{x}_F = \theta \mathbf{s}_L + \mathbf{C}_L(\theta \boldsymbol{\alpha}_L) = (\mathbf{s}_L + \mathbf{C}_L \boldsymbol{\alpha}_L)\theta.$$

Standalone efforts $\theta \boldsymbol{\alpha}_L$ lie weakly above these resulting social norms for all $\theta \in (0, \bar{\theta}]$ if and only if

$$\boldsymbol{\alpha}_L \geq \mathbf{s}_L + \mathbf{C}_L \boldsymbol{\alpha}_L. \tag{B.5}$$

In simultaneous play, $\bar{\mathbf{x}}_L^N = \mathbf{s}_L + \mathbf{C}_L \mathbf{x}_L^N$ and $\mathbf{x}_L^N = (1-\lambda)\boldsymbol{\alpha}_L + \lambda \bar{\mathbf{x}}_L^N$ give $(\mathbf{I}_L - \lambda \mathbf{C}_L)(\boldsymbol{\alpha}_L - \mathbf{x}_L^N) = \lambda(\boldsymbol{\alpha}_L - \mathbf{s}_L - \mathbf{C}_L \boldsymbol{\alpha}_L)$. For a single leader $L = \{i\}$, condition (B.5) reduces to $\alpha_i \geq s_i + c_i \alpha_i$, which is equivalent to $\alpha_i \geq x_i^N$ and hence $\alpha_i \geq \bar{x}_i^N$. With multiple leaders, $(\mathbf{I}_L - \lambda \mathbf{C}_L)^{-1} \geq \mathbf{0}$ implies that (B.5) entails $\alpha_i \geq x_i^N \geq \bar{x}_i^N$ for all $i \in L$, while ensuring that each leader's standalone effort $\theta \alpha_i$ remains at least the resulting social norm even when co-leaders raise Stage 1 efforts from $\theta \mathbf{x}_L^N$ to $\theta \boldsymbol{\alpha}_L$.

Theorem B.1 (Separating equilibrium)**.** *For a leader group L, under Assumption B.1 and condition (B.5), there exists a separating perfect Bayesian equilibrium with followers' responses (B.2). On the equilibrium path, for all $\theta \in [0, \bar{\theta}]$, leaders choose $\mathbf{x}_L^I(\theta) = \boldsymbol{\kappa}^* \theta$ and followers choose*

$$\mathbf{x}_F^I(\theta) = \mathbf{U}_F \Big[(1-\lambda)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL} \boldsymbol{\kappa}^* \Big] \theta, \tag{B.6}$$

where the coefficients $\boldsymbol{\kappa}^* = (\kappa_i^*)_{i \in L}$ are uniquely determined by

$$\kappa_i^* + \gamma \left(1 - c_i - \frac{s_i}{\kappa_i^*} \right) \left(\kappa_i^* - s_i - \sum_{j \in L} c_{ij} \kappa_j^* \right) = \alpha_i, \qquad i \in L, \tag{B.7}$$

subject to $\boldsymbol{\kappa}^* \geq (\mathbf{I}_L - \mathbf{C}_L)^{-1} \mathbf{s}_L$.

Proof: See Appendix B.5.1.

Condition (B.5) connects the equilibrium in Theorem B.1 directly to the baseline effort ordering in Section 3.1 by keeping leaders' efforts between the equilibrium social norms $\bar{\mathbf{x}}_L^I(\theta) := (\mathbf{s}_L + \mathbf{C}_L \boldsymbol{\kappa}^*)\theta$ and standalone efforts $\theta \boldsymbol{\alpha}_L$, so $\bar{\mathbf{x}}_L^I(\theta) \leq \mathbf{x}_L^I(\theta) \leq \theta \boldsymbol{\alpha}_L$ for all $\theta \in [0, \bar{\theta}]$. Compared with the complete-information first-order condition in Section 3.1, private information introduces the term $-s_i/\kappa_i^*$ in (B.7): a unit reduction in x_i lowers leader i's social norm through followers' conformity responses at rate c_i and through the lower inferred state $\mu_{\boldsymbol{\kappa}^*}(\mathbf{x}_L)$ by an additional s_i/κ_i^*. Because $\bar{x}_i$ falls more steeply when leader i cuts effort, shading effort downward narrows the norm deviation $x_i - \bar{x}_i$ at the smaller rate $1 - c_i - s_i/\kappa_i^*$ rather than $1 - c_i$, which strengthens each leader's incentive to exert effort in Stage 1.

For a single leader ($|L| = 1$), both upward and downward deviations within the equilibrium range alter x_i/κ_i, so local incentive compatibility pins down a unique linear separating equilibrium (Proposition B.2). With multiple leaders ($|L| \geq 2$), by contrast, a unilateral downward deviation by leader i lowers $\min_{j \in L}(x_j/\kappa_j)$ and reduces $\mu_{\boldsymbol{\kappa}}(\mathbf{x}_L)$ at rate $1/\kappa_i$, whereas a unilateral upward deviation leaves $\min_{j \in L}(x_j/\kappa_j) = \theta$ unchanged; because $\mu_{\boldsymbol{\kappa}}(\mathbf{x}_L)$ has a kink at $\mathbf{x}_L = \boldsymbol{\kappa}\theta$, multiple coefficient vectors can satisfy the resulting one-sided incentive inequalities, and system (B.7) characterizes the equilibrium in which every leader's downward-deviation constraint binds. Let $\mathcal{K}(L)$ denote the set of coefficient vectors $\boldsymbol{\kappa} > \mathbf{0}$ that, together with followers' responses (B.2) and Assumption B.1, form a separating perfect Bayesian equilibrium, and for any $\boldsymbol{\kappa} \in \mathcal{K}(L)$ let $\mathbf{x}^I(L, \boldsymbol{\kappa}; \theta) := \Big(\boldsymbol{\kappa}, \mathbf{U}_F[(1-\lambda)\boldsymbol{\alpha}_F + \lambda \widehat{\mathbf{G}}_{FL} \boldsymbol{\kappa}] \Big) \theta$ denote the corresponding equilibrium

effort profile in state θ. As in Zhou and Chen (2016), we select the equilibrium in $\mathcal{K}(L)$ with the componentwise greatest effort levels.

Proposition B.1 (Selected equilibrium)**.** *For a leader group L, under Assumption B.1 and condition (B.5), $\boldsymbol{\kappa}^* \in \mathcal{K}(L)$ in Theorem B.1 satisfies $\boldsymbol{\kappa}^* \geq \boldsymbol{\kappa}$ and $\mathbf{x}^I(L, \boldsymbol{\kappa}^*; \theta) \geq \mathbf{x}^I(L, \boldsymbol{\kappa}; \theta)$ for all $\boldsymbol{\kappa} \in \mathcal{K}(L)$ and $\theta \in [0, \bar{\theta}]$.*

Proof: See Appendix B.5.2.

Hereafter, $\mathbf{x}^I(L; \theta) := \mathbf{x}^I(L, \boldsymbol{\kappa}^*; \theta)$ denotes the selected equilibrium effort profile in state θ, with aggregate effort $X^I(L; \theta)$ and leadership gain $\widehat{\ell}(L)$.

A single informed leader. For a single leader $L = \{i\}$, Assumption B.1 becomes $\mu_{\kappa_i}(x_i) = \min\{\bar{\theta}, x_i/\kappa_i\}$, and condition (B.5) reduces to $\alpha_i \geq s_i/(1 - c_i)$; with the leader group $\{i\}$ fixed, leader i's Stage 1 efforts under private and complete information are $x_i^I(\theta) = \kappa_i^* \theta$ and $x_i^S(\theta) = \theta x_i^S$.

Proposition B.2 (Signaling leading index)**.** *If player $i \in N$ is the single leader, Assumption B.1 admits a linear separating equilibrium if and only if $\alpha_i \geq s_i/(1 - c_i)$, or equivalently $\alpha_i \geq x_i^N$; when this condition holds, the linear separating equilibrium is unique. For all $\theta \in [0, \bar{\theta}]$, the leader's effort exceeds the complete-information benchmark by*

$$x_i^I(\theta) - x_i^S(\theta) = (\kappa_i^* - x_i^S)\,\theta = \frac{\sqrt{\alpha_i^2 + 4\gamma s_i\big[(1 - c_i)\alpha_i - s_i\big]} - \alpha_i}{2\big[1 + \gamma(1 - c_i)^2\big]}\,\theta \geq 0, \tag{B.8}$$

and the signaling leading index satisfies

$$\widehat{\ell}_i\,\theta = \left\{\ell_i + \frac{\sum_{j=1}^n m_{ji}^N}{m_{ii}^N}(\kappa_i^* - x_i^S)\right\}\theta. \tag{B.9}$$

Moreover, the signaling leading index is strictly positive ($\widehat{\ell}_i > 0$) when $\alpha_i > x_i^N$ and equals zero ($\widehat{\ell}_i = 0$) when $\alpha_i = x_i^N$.

Proof: See Appendix B.5.3.

Private information preserves the network propagation multiplier $\sum_{j=1}^n m_{ji}^N/m_{ii}^N$ from Section 3.2 and adds the leader's signaling effort increment $(\kappa_i^* - x_i^S)\theta$, which is strictly positive in every positive state whenever $\alpha_i > x_i^N$. Because s_i increases in followers'

productivities $\boldsymbol{\alpha}_F$, more productive followers respond more strongly to an increase in the inferred state, which shifts leader i's social norm more sharply and strengthens the incentive to signal through higher Stage 1 effort.

Comparison with complete information. We now compare equilibrium efforts and optimal group selection under private information with the complete-information benchmark in Section 3.

Proposition B.3 (Effort comparison)**.** *For a leader group L, every equilibrium $\boldsymbol{\kappa} \in \mathcal{K}(L)$ induces efforts weakly above the complete-information two-stage equilibrium with the same leader group for all $\theta \in [0, \bar{\theta}]$, that is, $\mathbf{x}^I(L, \boldsymbol{\kappa}; \theta) \geq \mathbf{x}^S(L; \theta)$.*

Proof: See Appendix B.5.4.

Proposition B.3 applies to every separating equilibrium in $\mathcal{K}(L)$, not only to the selected equilibrium $\boldsymbol{\kappa}^*$. At the upper endpoint $\bar{\theta}$, an upward deviation cannot raise followers' posterior expectation above $\bar{\theta}$, so sequential rationality requires each leader's coefficient κ_i to lie at or above the complete-information best-response locus; strategic complementarity among leaders and followers' nonnegative responses $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL} \geq \mathbf{0}$ then propagate higher Stage 1 efforts across the network. Under condition (B.5), the selected equilibrium $\boldsymbol{\kappa}^*$ realizes the largest effort gain within $\mathcal{K}(L)$.

For finite conformity, let $\mathcal{F}(\lambda)$ denote the set of leader groups L satisfying (B.5), and evaluate expected aggregate effort and the leadership gain $\widehat{\ell}(L)$ at the selected equilibrium $\mathbf{x}^I(L; \theta)$.

Proposition B.4 (Leader eligibility and follower domination)**.** *Suppose that productivities are heterogeneous. For any leader–follower partition (L^*, F^*) with $F^* := N \setminus L^*$ such that L^* maximizes expected aggregate effort over $\mathcal{F}(\lambda)$:*

(i) *The leader group L^* consists only of players with a positive local-average leading index ($\ell_i > 0$).*

(ii) *The follower group F^* forms a* dominating set*.*

Proof: See Appendix B.5.5.

Both membership conditions from Section 3.3 carry over to $\mathcal{F}(\lambda)$, and follower domination now reflects an informational channel alongside strategic conformity. If a leader $i \in L$ has no neighboring follower, then $s_i = c_i = 0$, and leader i's Stage 1 effort

rule coincides with the Stage 2 best response of an informed follower; reassigning i to Stage 2 restores follower i's response to the inferred state, strengthens neighboring leaders' Stage 1 incentives, and strictly raises aggregate effort.

To obtain an explicit characterization in the limiting cases of weak and strong conformity, let $\bar{\alpha}_i := \sum_{j=1}^{n} \widehat{g}_{ij}\alpha_j$ denote the average productivity of player i's neighbors and $N_+ := \{i \in N : \alpha_i > \bar{\alpha}_i\}$ the set of candidates whose productivity exceeds this average productivity.

Proposition B.5 (Key groups under weak and strong conformity)**.** *Suppose that productivities are heterogeneous.*

(i) *For sufficiently small conformity, an optimal leader group maximizes*

$$\widehat{\Lambda}(L) := \sum_{i\in L}\sum_{j\in F} \widehat{g}_{ij}\frac{\alpha_j}{\alpha_i}(\alpha_i - \bar{\alpha}_i). \tag{B.10}$$

(ii) *For sufficiently large conformity, an optimal leader group is a singleton consisting of a player with the highest productivity.*

Proof: See Appendix B.5.6.

In part (i), when the taste for conformity is sufficiently small ($\lambda \to 0$), every nonempty candidate group $L \subseteq N_+$ belongs to $\mathcal{F}(\lambda)$ and yields leadership gain $\widehat{\ell}(L) = \lambda\widehat{\Lambda}(L)+\mathcal{O}(\lambda^2)$. In particular, a single candidate $i \in N_+$ achieves $\widehat{\ell}_i = \lambda(\bar{\alpha}_i/\alpha_i)(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$, which raises the leadership gain from order λ^2 under complete information to order λ. Under weak conformity, leader i's equilibrium coefficient approaches α_i, so a unit reduction in x_i lowers the inferred state at rate $1/\alpha_i$; follower $j \in F$ responds to the inferred state at rate $(1-\lambda)\alpha_j \approx \alpha_j$, and the weight $\widehat{g}_{ij}$ maps that response into leader i's social norm. The follower-to-leader productivity ratio α_j/α_i therefore replaces the complete-information link reciprocity $r_{ij} = \widehat{g}_{ij}\widehat{g}_{ji}$ in (B.10).

For any group $L \subseteq N$, the subset $L \cap N_+$ achieves at least as high a value of $\widehat{\Lambda}$ as L, so problem (B.10) admits a maximum directed cut representation (Ghuge et al., 2022) with nonnegative directed edge weights $\widehat{g}_{ij}(\alpha_j/\alpha_i)\max\{\alpha_i - \bar{\alpha}_i, 0\}$. On the candidate domain 2^{N_+}, objective (B.10) separates standalone signaling gains from pairwise losses between adjacent leaders:

$$\widehat{\Lambda}(L) = \sum_{i\in L}\frac{\bar{\alpha}_i}{\alpha_i}(\alpha_i - \bar{\alpha}_i) - \sum_{\substack{i,j\in L\\ i<j}}\left[\widehat{g}_{ij}\frac{\alpha_j}{\alpha_i}(\alpha_i - \bar{\alpha}_i) + \widehat{g}_{ji}\frac{\alpha_i}{\alpha_j}(\alpha_j - \bar{\alpha}_j)\right].$$

Because each bracketed pair term is nonnegative for $i, j \in N_+$, $\widehat{\Lambda}$ is submodular on 2^{N_+}. Moreover, assigning a productive player to L rather than F deprives other leaders in L of that player's Stage 2 response to the inferred state, so an optimal leader group need not include a player with the highest productivity.

In part (ii), when the taste for conformity is sufficiently large ($\lambda \to 1$), followers place vanishing weight $1 - \lambda$ on the inferred state ($\mathbf{s}_L = \mathcal{O}(1-\lambda)$) and instead conform to the efforts of neighbors. The additional effort induced by private information therefore vanishes at rate $\mathcal{O}(1-\lambda)$ uniformly across $\mathcal{K}(L)$, and separating-equilibrium efforts for every group approach the complete-information limit in Section 3.3: under any multi-leader group, all efforts converge to a common strict convex combination of individual productivities, whereas a single leader i anchors effort across the network at α_i. A singleton consisting of a player with the highest productivity belongs to $\mathcal{F}(\lambda)$ for all $\lambda \in (0,1)$ and consequently maximizes expected aggregate effort for sufficiently large conformity, whether the planner optimizes over $\mathcal{F}(\lambda)$ or selects an equilibrium from $\mathcal{K}(L)$ in each feasible group.

Complete networks. To see how private information reshapes both single-leader and group selection in a symmetric network, consider the complete network K_n ($g_{ij} = 1$ for all $i \neq j$) with heterogeneous productivities $\alpha_1 > \cdots > \alpha_n > 0$, and let $\bar{\alpha} := n^{-1} \sum_{i=1}^{n} \alpha_i$ denote the average productivity.

Corollary B.1 (Leader selection in complete networks)**.** *In the complete network K_n with $\alpha_1 > \cdots > \alpha_n > 0$:*

(i) *For all $\lambda \in (0,1)$, a single leader i satisfies $\widehat{\ell}_i > 0$ if and only if $\alpha_i > \bar{\alpha}$, and any optimal leader group L^* over $\mathcal{F}(\lambda)$ consists only of players with $\alpha_i > \bar{\alpha}$.*

(ii) *For sufficiently small conformity, an optimal leader group maximizes*

$$\widehat{\Lambda}(L) = \frac{n}{(n-1)^2} \left(\sum_{j \in F} \alpha_j \right) \sum_{i \in L} \left(1 - \frac{\bar{\alpha}}{\alpha_i} \right) \tag{B.11}$$

over $L \subseteq \{i \in N : \alpha_i > \bar{\alpha}\}$. In particular, for a single leader, player 1 is the unique key leader when $n \leq 4$, whereas when $n \geq 5$, player 2 outperforms player 1 whenever $\alpha_1 \alpha_2 > n\bar{\alpha}^2$.

Proof: See Appendix B.5.7.

Part (i) contrasts with signaling under local-aggregate payoffs (Zhou and Chen, 2016, Corollary 1 and Section 7.2), which selects the least productive player ($i^* = n$) as the key leader and ranks group members in ascending order of productivity: under social conformity, below-average members ($\alpha_i \leq \bar{\alpha}$, including player n) face equilibrium social norms at or above individual effort ($x_i^N \leq \bar{x}_i^N$) and never lead at any $\lambda \in (0, 1)$. Part (ii) simultaneously departs from complete information in Proposition 5 of Section 3.4, where player 1 is always the unique key leader and every optimal leader group takes the prefix form $\{1, \ldots, m^*\}$. In (B.11), higher productivity α_i strengthens a leader's signaling incentive $1-\bar{\alpha}/\alpha_i$ at a diminishing rate while reducing followers' aggregate responsiveness $\sum_{j\in F} \alpha_j$ one for one; when $n \geq 5$ and α_1 is sufficiently large relative to $\bar{\alpha}$, retaining player 1's Stage 2 response to the inferred state outweighs appointing player 1 in Stage 1.

Example B.1. Consider the complete network K_6, with $\alpha_1 = 18$, $\alpha_2 = 14$, and $\alpha_3 = \cdots = \alpha_6 = 1$, so $\bar{\alpha} = 6$. At $\lambda = 0.1$, Table B.1 shows that player 1 is the key leader under complete information, and the optimal leader group is $\{1, 2\}$. Under private information, player 2 becomes the key leader and leads alone.

Retaining player 1 as a follower increases the sum of followers' productivities from 18 to 22 relative to appointing player 1 alone. The stronger response to the inferred state outweighs player 1's greater productivity as a leader. Appointing both players instead leaves only the four low-productivity followers, whose productivities sum to 4. Private information therefore changes both who leads and how many players lead.

Table B.1: Leadership gains under complete and incomplete information

Leader group L	Complete information, $\ell(L)$	Incomplete information, $\widehat{\ell}(L)$
$\{1\}$	0.0308	0.3326
$\{2\}$	0.0205	**0.3350**
$\{1, 2\}$	**0.0401**	0.1607

Notes: The network is K_6, with $\boldsymbol{\alpha} = (18, 14, 1, 1, 1, 1)$ and $\lambda = 0.1$. Bold entries identify the optimal leader group in each information environment.

B.2 Heterogeneous Conformism and Interactions

Players may differ both in the taste for conformity and in the weights assigned to peers' efforts when forming the social norm. Let $\mathbf{W} = [w_{ij}]$ be a nonnegative, irreducible $(n \times n)$ matrix with zero diagonal ($w_{ii} = 0$ for all $i \in N$) that is row-stochastic, that

is, $\sum_{j=1}^{n} w_{ij} = 1$ for all $i \in N$. The entry w_{ij} measures the norm weight that player i places on player j's effort, so strategic influence runs from player j to player i whenever $w_{ij} > 0$, opposite to the orientation of directed links in $\mathbf{W}$. For individual productivity $\alpha_i > 0$ and taste for conformity $\gamma_i > 0$, player i chooses $x_i \geq 0$ to maximize

$$u_i(\mathbf{x}) = \alpha_i x_i - \frac{1}{2}x_i^2 - \frac{\gamma_i}{2}(x_i - \bar{x}_i)^2, \tag{B.12}$$

where $\bar{x}_i := \sum_{j=1}^{n} w_{ij} x_j$ represents player i's social norm.

Define $\mathbf{\Gamma} := \operatorname{diag}(\gamma_i)_{i \in N}$, $\lambda_i := \gamma_i/(1+\gamma_i) \in (0,1)$, and $\mathbf{\Lambda} := \operatorname{diag}(\lambda_i)_{i \in N}$; the baseline model corresponds to $\mathbf{W} = \widehat{\mathbf{G}}$ and $\gamma_i = \gamma$ for all $i \in N$. Throughout this subsection, we retain the two-stage timing for a leader group L with $0 < |L| < n$ and follower group $F := N \setminus L$, where the planner's objective is to maximize aggregate effort, that is, $X^S(L) := \sum_{i=1}^{n} x_i^S(L)$, or equivalently the leadership gain $\ell(L) := \sum_{i=1}^{n} \left[x_i^S(L) - x_i^N\right]$ relative to simultaneous-move aggregate effort $X^N := \sum_{i=1}^{n} x_i^N$.

Equilibrium characterization. In simultaneous play, player i's best response is $x_i = (1-\lambda_i)\alpha_i + \lambda_i \bar{x}_i$. For a leader group L, define $\widehat{\mathbf{U}}_F := (\mathbf{I}_F - \mathbf{\Lambda}_F \mathbf{W}_{FF})^{-1}$. Followers' responses in Stage 2 and the resulting Stage 1 social norms of the leaders are

$$\mathbf{x}_F(\mathbf{x}_L) = \widehat{\mathbf{U}}_F \left[(\mathbf{I}_F - \mathbf{\Lambda}_F)\boldsymbol{\alpha}_F + \mathbf{\Lambda}_F \mathbf{W}_{FL} \mathbf{x}_L\right], \tag{B.13}$$

$$\bar{\mathbf{x}}_L(\mathbf{x}_L) = \mathbf{W}_{LF} \widehat{\mathbf{U}}_F (\mathbf{I}_F - \mathbf{\Lambda}_F)\boldsymbol{\alpha}_F + \widehat{\mathbf{C}}_L \mathbf{x}_L, \tag{B.14}$$

where $\widehat{\mathbf{C}}_L := \mathbf{W}_{LL} + \mathbf{W}_{LF}\widehat{\mathbf{U}}_F \mathbf{\Lambda}_F \mathbf{W}_{FL}$ is the norm-response matrix and $\widehat{c}_i := (\widehat{\mathbf{C}}_L)_{ii}$ measures the shift in leader i's social norm induced by a unit increase in x_i. At an interior Stage 1 optimum, each leader $i \in L$ satisfies the first-order condition $(1-\lambda_i)x_i^S + \lambda_i(1 - \widehat{c}_i)(x_i^S - \bar{x}_i^S) = (1-\lambda_i)\alpha_i$, or in vector form $(\mathbf{I}_L - \mathbf{\Lambda}_L)\mathbf{x}_L^S + \mathbf{\Lambda}_L(\mathbf{I}_L - \widehat{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{x}_L^S - \bar{\mathbf{x}}_L^S) = (\mathbf{I}_L - \mathbf{\Lambda}_L)\boldsymbol{\alpha}_L$. Define

$$\widehat{\mathbf{V}}_L := \left[\mathbf{I}_L - \mathbf{\Lambda}_L + \mathbf{\Lambda}_L(\mathbf{I}_L - \widehat{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{I}_L - \widehat{\mathbf{C}}_L)\right]^{-1}.$$

Proposition B.6 (Equilibrium under heterogeneous conformism)**.** *The simultaneous-move game admits a unique Nash equilibrium* $\mathbf{x}^N = \widehat{\mathbf{M}}^N \boldsymbol{\alpha} > \mathbf{0}$*, where* $\widehat{\mathbf{M}}^N = [\widehat{m}_{ij}^N]$ *is defined as*

$$\widehat{\mathbf{M}}^N := (\mathbf{I} - \boldsymbol{\Lambda}\mathbf{W})^{-1}(\mathbf{I} - \boldsymbol{\Lambda}), \tag{B.15}$$

and for any leader group L*, the two-stage game admits a unique subgame-perfect Nash equilibrium* $\mathbf{x}^S(L) > \mathbf{0}$*. With leaders ordered first,* $\mathbf{x}^S(L) = \widehat{\mathbf{M}}^S(L)\boldsymbol{\alpha}$*, where* $\widehat{\mathbf{M}}^S(L) = [\widehat{m}_{ij}^S(L)]$ *is defined as*

$$\widehat{\mathbf{M}}^S(L) := \begin{pmatrix} \widehat{\mathbf{V}}_L & \widehat{\mathbf{V}}_L \boldsymbol{\Lambda}_L (\mathbf{I}_L - \hat{\mathbf{C}}_L^{\mathrm{D}}) \mathbf{W}_{LF} \widehat{\mathbf{U}}_F \\ \widehat{\mathbf{U}}_F \boldsymbol{\Lambda}_F \mathbf{W}_{FL} \widehat{\mathbf{V}}_L & \widehat{\mathbf{U}}_F + \widehat{\mathbf{U}}_F \boldsymbol{\Lambda}_F \mathbf{W}_{FL} \widehat{\mathbf{V}}_L \boldsymbol{\Lambda}_L (\mathbf{I}_L - \hat{\mathbf{C}}_L^{\mathrm{D}}) \mathbf{W}_{LF} \widehat{\mathbf{U}}_F \end{pmatrix} (\mathbf{I} - \boldsymbol{\Lambda}). \tag{B.16}$$

Both $\widehat{\mathbf{M}}^N$ *and* $\widehat{\mathbf{M}}^S(L)$ *are strictly positive and row-stochastic, that is,* $\widehat{\mathbf{M}}^N \mathbf{1} = \widehat{\mathbf{M}}^S(L)\mathbf{1} = \mathbf{1}$*.*

Proof: See Appendix B.5.8.

Leadership gains. In simultaneous play, each player $k \in N$ satisfies $x_k^N + \gamma_k(x_k^N - \bar{x}_k^N) = \alpha_k$. Subtracting this condition from the two-stage equilibrium equations $x_i^S(L) + \gamma_i(1 - \widehat{c}_i)\big(x_i^S(L) - \bar{x}_i^S(L)\big) = \alpha_i$ $(i \in L)$ and $x_j^S(L) + \gamma_j\big(x_j^S(L) - \bar{x}_j^S(L)\big) = \alpha_j$ $(j \in F)$ shows that $\mathbf{x}^S(L) - \mathbf{x}^N$ solves the two-stage linear system with right-hand side $\widehat{c}_i\gamma_i(x_i^N - \bar{x}_i^N) = \widehat{c}_i(\alpha_i - x_i^N)$ for $i \in L$ and zero for $j \in F$. Applying $\widehat{\mathbf{M}}^S(L)$ therefore gives

$$x_j^S(L) - x_j^N = \sum_{i \in L} \widehat{m}_{ji}^S(L)\, \widehat{c}_i\, (\alpha_i - x_i^N), \qquad j \in N. \tag{B.17}$$

For a single leader $L = \{i\}$, restricting column i of $(\mathbf{I} - \boldsymbol{\Lambda}\mathbf{W})\widehat{\mathbf{M}}^N = \mathbf{I} - \boldsymbol{\Lambda}$ to the follower rows $N \setminus \{i\}$ gives $(\mathbf{I}_{-i} - \boldsymbol{\Lambda}_{-i}\mathbf{W}_{-i})\widehat{\mathbf{M}}_{-i,i}^N = \boldsymbol{\Lambda}_{-i}\mathbf{W}_{-i,i}\widehat{m}_{ii}^N$, so each follower $j \neq i$ adjusts effort at rate $\partial x_j(x_i)/\partial x_i = \widehat{m}_{ji}^N/\widehat{m}_{ii}^N$ by (B.13). At $x_i = x_i^N$, leader i's reduced Stage 1 marginal payoff equals $\widehat{c}_i(\alpha_i - x_i^N)$, and the constant second derivative $-\big[1 + \gamma_i(1 - \widehat{c}_i)^2\big]$ implies $x_i^S(\{i\}) - x_i^N = \widehat{c}_i(\alpha_i - x_i^N)/\big[1 + \gamma_i(1 - \widehat{c}_i)^2\big]$. Summing $x_j^S(\{i\}) - x_j^N = (\widehat{m}_{ji}^N/\widehat{m}_{ii}^N)\big[x_i^S(\{i\}) - x_i^N\big]$ across $j \in N$ yields the single-leader leadership gain $\ell_i := \ell(\{i\})$:

$$\ell_i = \frac{\widehat{c}_i}{1 + \gamma_i(1 - \widehat{c}_i)^2} \frac{\sum_{j=1}^n \widehat{m}_{ji}^N}{\widehat{m}_{ii}^N} (\alpha_i - x_i^N). \tag{B.18}$$

In (B.18), leader i's taste for conformity γ_i governs the curvature $1+\gamma_i(1-\widehat{c}_i)^2$ of the Stage 1 objective, whereas followers' conformity weights $\mathbf{\Lambda}_{-i}$ govern both the induced norm response $\widehat{c}_i$ and network propagation $\left(\sum_{j=1}^{n}\widehat{m}_{ji}^{N}\right)/\widehat{m}_{ii}^{N}$. Because $\mathbf{W}$ is irreducible and $\lambda_k > 0$ for all $k \in N$, a shortest closed directed walk from i to i visits only followers at intermediate steps and implies $\widehat{c}_i = \mathbf{W}_{i,-i}(\mathbf{I}_{-i} - \mathbf{\Lambda}_{-i}\mathbf{W}_{-i})^{-1}\mathbf{\Lambda}_{-i}\mathbf{W}_{-i,i} > 0$; row-stochasticity and strict positivity of $\widehat{\mathbf{M}}^N$ imply $0 < \widehat{m}_{ii}^{N} < 1$ and $\widehat{m}_{ij}^{N} > 0$ for all $j \neq i$. Consequently, a single leader i increases every player's effort (and achieves $\ell_i > 0$) if and only if $\alpha_i > x_i^N$; for fixed $\mathbf{W}$ and $\mathbf{\Gamma}$, $\alpha_i - x_i^N = (1-\widehat{m}_{ii}^{N})\alpha_i - \sum_{j\neq i}\widehat{m}_{ij}^{N}\alpha_j$ and hence ℓ_i strictly increase in α_i and strictly decrease in α_j for all $j \neq i$.

Group selection in directed networks. For a leader group L, moving first alters leader i's Stage 1 incentive only if followers' responses shift $\bar{x}_i$. A *directed cycle* in $\mathbf{W}$ is a sequence of players $(u_0, u_1, \ldots, u_m)$ with $m \geq 2$, $u_m = u_0$, distinct players $u_0, u_1, \ldots, u_{m-1}$, and $w_{u_{r-1}u_r} > 0$ for all $r = 1, \ldots, m$. The Neumann series for $\widehat{\mathbf{U}}_F$ in (B.14) yields

$$\widehat{c}_i = \sum_{t=0}^{\infty} \mathbf{W}_{iF}(\mathbf{\Lambda}_F \mathbf{W}_{FF})^t \mathbf{\Lambda}_F \mathbf{W}_{Fi}, \tag{B.19}$$

which sums the weights of closed directed walks from i to i whose intermediate players belong to F. Because a shortest such walk has distinct intermediate players in F, $\widehat{c}_i > 0$ holds if and only if leader i belongs to a directed cycle in $\mathbf{W}$ whose other players all lie in F. In an undirected network ($w_{ij} > 0 \iff w_{ji} > 0$), leader i belongs to such a directed cycle if and only if i has at least one neighbor $j \in F$, which forms the length-2 cycle (i, j, i); requiring $\widehat{c}_i > 0$ for all $i \in L$ is therefore equivalent to requiring the follower group F to form a *dominating set*, as in Proposition 3(ii) of Section 3.3. In a directed network, by contrast, an outgoing weight $w_{ij} > 0$ means that player i's social norm places weight on follower j's effort, not that follower j responds to x_i; shifting $\bar{x}_i$ requires a chain of followers in F that transmits x_i back into $\bar{x}_i$, that is, a directed cycle through i whose other players all belong to F.

Proposition B.7 (Selection with directed interactions)**.** *Suppose that productivities are heterogeneous. For any optimal leader group L^* of minimum cardinality, $\mathbf{x}^S(L^*) > \mathbf{x}^N$, and each leader $i \in L^*$ satisfies*

$$\widehat{c}_i > 0, \qquad \alpha_i > x_i^S(L^*) > x_i^N. \tag{B.20}$$

In particular, each leader $i \in L^*$ has a strictly positive leading index ($\ell_i > 0$) and belongs to a directed cycle in $\mathbf{W}$ whose other players all lie in $F^* := N \setminus L^*$.

Proof: See Appendix B.5.9.

Proposition B.7 extends Proposition 3 in Section 3.3 from follower domination to the directed-cycle condition $\widehat{c}_i > 0$. Unlike an undirected network, where reassigning a leader i with $\widehat{c}_i = 0$ to Stage 2 creates a length-2 cycle with a neighboring co-leader and strictly raises aggregate effort at any candidate optimum, in a directed network a leader $i \in L$ with $\widehat{c}_i = 0$ may belong to no other leader's directed cycle through followers. Moving such a player i between L and F leaves followers' responses and equilibrium efforts unchanged; selecting an optimal leader group of minimum cardinality excludes such payoff-irrelevant leaders.

Weak and strong conformity. To vary overall conformity while holding the ratios of players' tastes for conformity fixed, let $\gamma_i = \gamma\nu_i$, where $\gamma > 0$ is a common scale parameter and $\nu_i > 0$ is fixed for all $i \in N$. Define $\bar{\alpha}_i := \sum_{j=1}^n w_{ij}\alpha_j$ and $N_+ := \{i \in N : \alpha_i > \bar{\alpha}_i\}$.

Proposition B.8 (Key groups under weak and strong conformity)**.** *Suppose that productivities are heterogeneous.*

(i) *For sufficiently small conformity, an optimal leader group maximizes*

$$\Phi(L) := \sum_{i \in L} \sum_{j \in F} \nu_i \nu_j w_{ij} w_{ji} (\alpha_i - \bar{\alpha}_i). \tag{B.21}$$

(ii) *For sufficiently large conformity, an optimal leader group is a singleton consisting of a player with the highest productivity.*

Proof: See Appendix B.5.10.

In part (i), when the taste for conformity is sufficiently small ($\gamma \to 0$), the leadership gain satisfies

$$\ell(L) = \gamma^2 \Phi(L) + \mathcal{O}(\gamma^3). \tag{B.22}$$

As in Section 3.3, reciprocal weights $w_{ij}w_{ji}$ and the local productivity difference $\alpha_i - \bar{\alpha}_i$ govern the second-order contribution of each leader–follower pair, now scaled by $\nu_i\nu_j$: ν_i governs leader i's incentive to shift $\bar{x}_i$, whereas ν_j governs follower j's response to x_i. For any group $L \subseteq N$, the subset $L \cap N_+$ achieves at least as high a value of Φ as L, and Φ is submodular on 2^{N_+}. In particular, whenever Φ admits a unique maximizer on 2^N, that group is the unique optimal leader group for all sufficiently small $\gamma > 0$.

In part (ii), let $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_n)^\top > \mathbf{0}$ denote the unique stationary distribution of $\mathbf{W}$ ($\boldsymbol{\pi}^\top \mathbf{W} = \boldsymbol{\pi}^\top$ and $\sum_{i=1}^n \pi_i = 1$). When the taste for conformity is sufficiently large ($\gamma \to \infty$), simultaneous-move and single-leader efforts satisfy

$$\lim_{\gamma\to\infty} \mathbf{x}^N = \frac{\sum_{j=1}^n (\pi_j/\nu_j)\alpha_j}{\sum_{j=1}^n \pi_j/\nu_j}\mathbf{1}, \qquad \lim_{\gamma\to\infty} \mathbf{x}^S(\{i\}) = \alpha_i \mathbf{1}, \tag{B.23}$$

whereas under any multi-leader group ($|L| \geq 2$), equilibrium efforts converge to a common strict convex combination of all individual productivities. Conditional on stationary weight π_j, players with weaker tastes for conformity (smaller ν_j) adjust less toward peers and receive greater weight in the simultaneous-move limit in (B.23). A single leader i instead anchors effort across the network at α_i as $\gamma \to \infty$, so a singleton consisting of a player with the highest productivity maximizes aggregate effort for sufficiently large γ.

When reciprocal links are absent ($w_{ij}w_{ji} = 0$ for all $i, j \in N$), $\Phi(L)$ vanishes identically and weak-conformity gains arise at higher orders along longer directed cycles. Suppose $\nu_j = 1$ for all $j \in N$ and let $\lambda = \gamma/(1+\gamma)$. For a single leader i, let $k \geq 3$ denote the length of a shortest directed cycle through i in $\mathbf{W}$. Because no shorter closed directed walk from i to i exists and any shortest directed cycle visits i only at the endpoints, equation (B.19) gives $\widehat{c}_i = \lambda^{k-1}(\mathbf{W}^k)_{ii} + \mathcal{O}(\lambda^k)$, and equation (B.18) yields

$$\ell_i = \lambda^k (\mathbf{W}^k)_{ii}(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^{k+1}). \tag{B.24}$$

Example B.2 (A directed cycle)**.** Consider three players with $w_{12} = w_{23} = w_{31} = 1$, all other weights equal to zero, and $\nu_j = 1$ for all $j \in N$. For a single leader i, the unique directed cycle through i traverses both followers, so $\widehat{c}_i = \lambda^2$ and (B.24) yields $\ell_i = \lambda^3(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^4)$, which is nonzero of order λ^3 whenever $\alpha_i \neq \bar{\alpha}_i$ (and strictly positive when $\alpha_i > \bar{\alpha}_i$). With two leaders ($|L| = 2$), neither leader belongs to a directed cycle whose other players lie in F, so $\widehat{c}_i = 0$ for both leaders and (B.17) yields $x_j^S(L) = x_j^N$ for all $j \in N$. Multiple followers are therefore necessary in this network to transmit a leader's effort back into that leader's social norm.

B.3 Anti-conformism

Players may value differentiation from peers rather than conformity to the social norm. We retain the baseline specification of network structure g. We consider the case of positive individual productivities and efforts: for individual productivity $\alpha_i > 0$ and taste for anti-conformism $\xi > 0$, player i chooses $x_i \geq 0$ to maximize

$$u_i(\mathbf{x}) = \alpha_i x_i - \frac{1}{2}x_i^2 + \frac{\xi}{2}(x_i - \bar{x}_i)^2, \tag{B.25}$$

where $\bar{x}_i = \sum_{j=1}^n \widehat{g}_{ij} x_j$ represents player i's social norm and the positive sign on the last term rewards departures from $\bar{x}_i$. For $0 < \xi < 1$, payoff (B.25) is strictly concave in x_i. Define $\chi := \xi/(1-\xi) > 0$, so an interior simultaneous-move best response takes the affine form

$$x_i = (1+\chi)\alpha_i - \chi\bar{x}_i.$$

Efforts are *strategic substitutes*, since for any $j \neq i$,

$$\frac{\partial^2 u_i(\mathbf{x})}{\partial x_i \partial x_j} = -\xi\widehat{g}_{ij} \leq 0,$$

so higher effort by a neighbor raises $\bar{x}_i$ and reduces the marginal return to player i's effort. Throughout this subsection, we retain the two-stage timing for a leader group L with $0 < |L| < n$ and follower group $F := N \setminus L$, where the planner's objective is to maximize aggregate effort, that is, $X^S(L) := \sum_{i=1}^n x_i^S(L)$, or equivalently the leadership gain $\ell(L) := \sum_{i=1}^n \left[x_i^S(L) - x_i^N\right]$ relative to simultaneous-move aggregate effort $X^N := \sum_{i=1}^n x_i^N$, and we focus on the case where the taste for anti-conformism $\xi > 0$ (or equivalently $\chi > 0$) is sufficiently small.

Equilibrium and leadership incentives. In the simultaneous-move game, for sufficiently small $\chi > 0$, every player's best response is interior and solves $(\mathbf{I} + \chi\widehat{\mathbf{G}})\mathbf{x}^N = (1+\chi)\boldsymbol{\alpha}$, so $\mathbf{x}^N = \widetilde{\mathbf{M}}^N\boldsymbol{\alpha}$, where the simultaneous-move influence matrix $\widetilde{\mathbf{M}}^N = [\widetilde{m}_{ij}^N]$ is defined as

$$\widetilde{\mathbf{M}}^N := (1+\chi)(\mathbf{I} + \chi\widehat{\mathbf{G}})^{-1}. \tag{B.26}$$

For a leader group L, define $\widetilde{\mathbf{U}}_F := (\mathbf{I}_F + \chi\widehat{\mathbf{G}}_{FF})^{-1}$. When Stage 2 best responses are interior, followers' efforts satisfy $\mathbf{x}_F(\mathbf{x}_L) = \widetilde{\mathbf{U}}_F\left[(1+\chi)\boldsymbol{\alpha}_F - \chi\widehat{\mathbf{G}}_{FL}\mathbf{x}_L\right]$, so the leaders'

induced social norms $\bar{\mathbf{x}}_L(\mathbf{x}_L) = \widehat{\mathbf{G}}_{LL}\mathbf{x}_L + \widehat{\mathbf{G}}_{LF}\mathbf{x}_F(\mathbf{x}_L)$ equal $(1+\chi)\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\boldsymbol{\alpha}_F + \widetilde{\mathbf{C}}_L\mathbf{x}_L$, with norm-response matrix

$$\widetilde{\mathbf{C}}_L := [\widetilde{c}_{ij}]_{i,j\in L} = \widehat{\mathbf{G}}_{LL} - \chi\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL},$$

where each diagonal entry $\widetilde{c}_i := (\widetilde{\mathbf{C}}_L)_{ii} = \partial\bar{x}_i(\mathbf{x}_L)/\partial x_i$ equals $-\chi(\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL})_{ii}$ because g has no self-loops. At an interior Stage 1 optimum, each leader $i \in L$ satisfies the first-order condition $\alpha_i - x_i^S + \xi(1-\widetilde{c}_i)(x_i^S - \bar{x}_i^S) = 0$, or in vector form $(1+\chi)\mathbf{x}_L^S - \chi(\mathbf{I}_L - \widetilde{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{x}_L^S - \bar{\mathbf{x}}_L^S) = (1+\chi)\boldsymbol{\alpha}_L$. Define

$$\widetilde{\mathbf{V}}_L := \left[(1+\chi)\mathbf{I}_L - \chi(\mathbf{I}_L - \widetilde{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{I}_L - \widetilde{\mathbf{C}}_L)\right]^{-1}.$$

Proposition B.9 (Equilibrium under anti-conformism)**.** *For sufficiently small anti-conformism $\chi > 0$, the simultaneous-move game admits a unique Nash equilibrium, given by $\mathbf{x}^N = \widetilde{\mathbf{M}}^N\boldsymbol{\alpha} > \mathbf{0}$ with $\widetilde{\mathbf{M}}^N$ in (B.26), and for any leader group L, the two-stage game admits a unique subgame-perfect Nash equilibrium, with strictly positive efforts on the equilibrium path. With leaders ordered first, equilibrium efforts are given by*

$$\begin{pmatrix}\mathbf{x}_L^S\\ \mathbf{x}_F^S\end{pmatrix} = \widetilde{\mathbf{M}}^S(L)\begin{pmatrix}\boldsymbol{\alpha}_L\\ \boldsymbol{\alpha}_F\end{pmatrix},$$

where $\widetilde{\mathbf{M}}^S(L) = [\widetilde{m}_{ij}^S(L)]$ is an $(n\times n)$ two-stage influence matrix defined as

$$\widetilde{\mathbf{M}}^S(L) := (1+\chi)\begin{pmatrix}\widetilde{\mathbf{V}}_L & -\chi\widetilde{\mathbf{V}}_L(\mathbf{I}_L - \widetilde{\mathbf{C}}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\\ -\chi\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL}\widetilde{\mathbf{V}}_L & \widetilde{\mathbf{U}}_F + \chi^2\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL}\widetilde{\mathbf{V}}_L(\mathbf{I}_L - \widetilde{\mathbf{C}}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\end{pmatrix}. \quad \text{(B.27)}$$

Proof: See Appendix B.5.11.

Both influence matrices have unit row sums, that is, $\widetilde{\mathbf{M}}^N\mathbf{1} = \widetilde{\mathbf{M}}^S(L)\mathbf{1} = \mathbf{1}$, so under homogeneous productivities ($\alpha_i = \alpha$ for all $i \in N$), $\mathbf{x}^S(L) = \mathbf{x}^N = \alpha\mathbf{1}$ and $\ell(L) = 0$. When productivities are heterogeneous, strategic substitutability reverses the sign of cross-group interactions in (B.27) relative to Theorem 1 in Section 3.1. Higher efforts by leaders induce followers to reduce effort through $-\chi\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL}$, which lowers each leader i's social norm ($\widetilde{c}_i \le 0$). Conversely, an increase in followers' productivities $\boldsymbol{\alpha}_F$ raises followers' efforts in Stage 2 and elevates the leaders' social norms, which lowers leaders' efforts in Stage 1 through $-\chi\widetilde{\mathbf{V}}_L(\mathbf{I}_L - \widetilde{\mathbf{C}}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F$.

For a single leader $L = \{i\}$ with $F = N \setminus \{i\}$, let $x_j(x_i)$ denote follower j's interior equilibrium response to x_i. As in Section 3.2, each follower $j \neq i$ adjusts effort at the

constant rate $\partial x_j(x_i)/\partial x_i = \widetilde{m}_{ji}^N/\widetilde{m}_{ii}^N$, so a unit increase in leader i's effort changes aggregate effort by $1+\sum_{j\neq i}\partial x_j(x_i)/\partial x_i = \left(\sum_{j=1}^n \widetilde{m}_{ji}^N\right)/\widetilde{m}_{ii}^N$, and leader i's norm response is

$$\widetilde{c}_i = \frac{1+\chi-\widetilde{m}_{ii}^N}{\chi\widetilde{m}_{ii}^N}. \tag{B.28}$$

Proposition B.10 (Single-leader selection under anti-conformism)**.** *For sufficiently small anti-conformism $\chi > 0$, each player $i \in N$ as a single leader satisfies $\widetilde{c}_i < 0$ and $0 < \left(\sum_{j=1}^n \widetilde{m}_{ji}^N\right)/\widetilde{m}_{ii}^N < 1$, with leadership gain*

$$\ell_i = \frac{(1+\chi)\widetilde{c}_i}{1+\chi\widetilde{c}_i(2-\widetilde{c}_i)}\frac{\sum_{j=1}^n \widetilde{m}_{ji}^N}{\widetilde{m}_{ii}^N}(\alpha_i - x_i^N) = \frac{\widetilde{c}_i}{1-\xi(1-\widetilde{c}_i)^2}\frac{\sum_{j=1}^n \widetilde{m}_{ji}^N}{\widetilde{m}_{ii}^N}(\alpha_i - x_i^N). \tag{B.29}$$

Consequently, $\ell_i > 0$ if and only if $x_i^N > \alpha_i$, or equivalently $\alpha_i > \bar{x}_i^N$.

Proof: See Appendix B.5.12.

In the simultaneous-move equilibrium, the first-order condition implies $x_i^N - \alpha_i = \xi(x_i^N - \bar{x}_i^N) = \chi(\alpha_i - \bar{x}_i^N)$. Unlike the baseline model under social conformity, where peer pressure depresses Nash effort below individual productivity whenever $\alpha_i > \bar{x}_i^N$ ($\bar{x}_i^N < x_i^N < \alpha_i$), anti-conformism leads a candidate i with $\alpha_i > \bar{x}_i^N$ to exert effort above individual productivity ($x_i^N > \alpha_i > \bar{x}_i^N$) to depart further from the social norm. Moving first allows leader i to raise effort even higher: a larger Stage 1 effort induces followers to cut Stage 2 effort, which lowers $\bar{x}_i$ ($\widetilde{c}_i < 0$) and raises the marginal return to departing from the social norm. Because $0 < \left(\sum_{j=1}^n \widetilde{m}_{ji}^N\right)/\widetilde{m}_{ii}^N < 1$ for sufficiently small $\chi > 0$, followers' aggregate effort reduction offsets only part of the leader's effort increase, so aggregate effort rises.

Weak anti-conformism and group selection. We next ask how strategic substitutability affects optimal group selection in the limiting case of weak anti-conformism.

Proposition B.11 (Key groups under weak anti-conformism)**.** *Suppose that productivities are heterogeneous. For sufficiently small anti-conformism $\chi > 0$, an optimal leader group maximizes*

$$\Lambda(L) := \sum_{i \in L} \sum_{j \in F} r_{ij}(\alpha_i - \bar{\alpha}_i), \tag{B.30}$$

where $\bar{\alpha}_i := \sum_{j=1}^{n} \widehat{g}_{ij}\alpha_j$ and $r_{ij} := \widehat{g}_{ij}\widehat{g}_{ji}$.

Proof: See Appendix B.5.13.

When the taste for anti-conformism is sufficiently small ($\chi \to 0$, or equivalently $\xi \to 0$), the leadership gain satisfies $\ell(L) = \chi^2\Lambda(L) + \mathcal{O}(\chi^3) = \xi^2\Lambda(L) + \mathcal{O}(\xi^3)$. Relative to social conformity in Section 3.3, strategic substitutability reverses both components of each leader's first-mover incentive: leader i's norm response becomes negative, $\widetilde{c}_i = -\chi \sum_{j \in F} r_{ij} + \mathcal{O}(\chi^2)$, while the difference between individual productivity and Nash effort also changes sign, $\alpha_i - x_i^N = -\chi(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\chi^2)$. Because leader i's Stage 1 effort adjustment is governed by the product $\widetilde{c}_i(\alpha_i - x_i^N)$, the two sign reversals cancel and preserve the maximum directed cut objective $\Lambda(L)$ from Section 3.3, along with candidate screening to $N_+ = \{i \in N : \alpha_i > \bar{\alpha}_i\}$ and submodularity on 2^{N_+}. In particular, whenever $\Lambda(L)$ admits a unique maximizer, that group is the unique optimal leader group under both sufficiently small conformity and sufficiently small anti-conformism.

B.4 Shifted Social Norms

Players may seek to match an effort level above or below the average effort of peers. We retain the baseline specification of network structure g and nonnegative effort choices $x_i \geq 0$; in place of $\bar{x}_i$, each player evaluates effort against the shifted reference $\beta\bar{x}_i$, where the shift parameter $\beta > 0$ is common across players. For individual productivity $\alpha_i > 0$ and taste for conformity $\gamma > 0$, player i maximizes

$$u_i(\mathbf{x}) = \alpha_i x_i - \frac{1}{2}x_i^2 - \frac{\gamma}{2}(x_i - \beta\bar{x}_i)^2, \tag{B.31}$$

where $\bar{x}_i = \sum_{j=1}^{n} \widehat{g}_{ij}x_j$ represents the average effort of player i's neighbors. Throughout this subsection, $\lambda = \gamma/(1+\gamma) \in (0,1)$, we assume $\lambda\beta < 1$, and we consider a leader group L with $0 < |L| < n$ and follower group $F := N \setminus L$. The shifted reference $\beta\bar{x}_i$ lies below $\bar{x}_i$ when $0 < \beta < 1$, coincides with the baseline social norm when $\beta = 1$, and exceeds $\bar{x}_i$ when $1 < \beta < 1/\lambda$; players incur a conformity cost for deviating from $\beta\bar{x}_i$ in either direction.

Let $\mathbf{x}^N(\beta) = \left(x_i^N(\beta)\right)_{i\in N}$ denote the simultaneous-move Nash equilibrium and $X^N(\beta) := \sum_{i=1}^n x_i^N(\beta)$ the corresponding aggregate effort. For a leader group L, let $\mathbf{x}^S(L;\beta) = \left(x_i^S(L;\beta)\right)_{i\in N}$ denote a pure-strategy subgame-perfect equilibrium, with aggregate effort $X^S(L;\beta) := \sum_{i=1}^n x_i^S(L;\beta)$, and define the leadership gain relative to the simultaneous-move benchmark at the same β as $\ell(L;\beta) := X^S(L;\beta) - X^N(\beta) = \sum_{i=1}^n \left[x_i^S(L;\beta) - x_i^N(\beta)\right]$ (evaluated at the equilibrium in question if multiple equilibria exist). To isolate the effect of the shifted reference from productivity differences, our analysis focuses on homogeneous individual productivities.

Proposition B.12 (The direction of leadership under shifted norms)**.** *Suppose that individual productivities are homogeneous ($\alpha_i = \alpha$ for all $i \in N$). The simultaneous-move game admits a unique Nash equilibrium, in which each player $i \in N$ chooses*

$$x_i^N(\beta) = \frac{1-\lambda}{1-\lambda\beta}\,\alpha. \tag{B.32}$$

For any leader group L, a pure-strategy subgame-perfect equilibrium of the two-stage sequential-move game exists and satisfies the following properties:

(i) *If $0 < \beta < 1$, the equilibrium is unique and $x_i^N(\beta) < x_i^S(L;\beta) < \alpha$ for all $i \in N$.*

(ii) *If $\beta = 1$, the equilibrium is unique and $x_i^S(L;1) = x_i^N(1) = \alpha$ for all $i \in N$.*

(iii) *If $1 < \beta < 1/\lambda$, any equilibrium satisfies $0 \le x_i^S(L;\beta) < x_i^N(\beta)$ for all $i \in L$ and $0 < x_j^S(L;\beta) < x_j^N(\beta)$ for all $j \in F$.*

Proof: See Appendix B.5.14.

Summing effort differences across players in Proposition B.12 shows that a shifted reference creates an incentive to move first even under homogeneous productivities, with

$$\operatorname{sign}\Big(\ell(L;\beta)\Big) = \operatorname{sign}(1-\beta)$$

for all leader groups L: the planner prefers sequential moves to simultaneous play when $0 < \beta < 1$, is indifferent between sequential and simultaneous play when $\beta = 1$, and prefers simultaneous play when $1 < \beta < 1/\lambda$. Under a downward shift ($0 < \beta < 1$), peer pressure toward the lower benchmark $\beta\bar{x}_i$ constrains each player i in simultaneous play and depresses $x_i^N(\beta)$ below α in (B.32). Moving first allows a leader $i \in L$ to raise Stage 1 effort toward α: followers respond upward in Stage 2, which raises the shifted reference $\beta\bar{x}_i$, partially offsets the conformity cost, and increases equilibrium efforts across the network. Under an upward shift ($1 < \beta < 1/\lambda$), by contrast, peer pressure disciplines

each player i in simultaneous play and raises $x_i^N(\beta)$ above α. Moving first then allows a leader $i \in L$ to cut Stage 1 effort: followers respond downward in Stage 2, which lowers $\beta\bar{x}_i$, eases peer pressure to remain above α, and reduces equilibrium efforts throughout the network. In Stage 2, each follower $j \in F$ places positive weight $1-\lambda$ on individual productivity $\alpha > 0$ and therefore maintains strictly positive effort $x_j^S(L;\beta) > 0$, whereas a leader $i \in L$ whose shifted-reference response $\partial(\beta\bar{x}_i)/\partial x_i$ exceeds unity (which can occur when $\lambda\beta^2 > 1$) finds that cutting x_i lowers $\beta\bar{x}_i$ more than one for one and may hit the zero-effort lower bound $x_i^S(L;\beta) = 0$.

Single-leader selection. Under homogeneous productivities, for a single leader $L = \{i\}$, Stage 2 follower best responses $x_j = (1-\lambda)\alpha + \lambda\beta\bar{x}_j$ $(j \neq i)$ yield the unique strictly positive follower equilibrium $\mathbf{x}_{-i}(x_i;\beta) = (\mathbf{I}_{-i} - \lambda\beta\widehat{\mathbf{G}}_{-i})^{-1}\Big[(1-\lambda)\alpha\mathbf{1}_{-i} + \lambda\beta\widehat{\mathbf{G}}_{-i,i}x_i\Big]$. Because g has no self-loops, leader i's shifted reference $\beta\bar{x}_i = \beta\widehat{\mathbf{G}}_{i,-i}\mathbf{x}_{-i}(x_i;\beta)$ responds to x_i at the constant rate

$$c_i(\beta) := \frac{\partial(\beta\bar{x}_i)}{\partial x_i} = \lambda\beta^2\widehat{\mathbf{G}}_{i,-i}(\mathbf{I}_{-i} - \lambda\beta\widehat{\mathbf{G}}_{-i})^{-1}\widehat{\mathbf{G}}_{-i,i}.$$

Define the simultaneous-move influence matrix in the shifted model as $\mathbf{M}^N(\beta) := (1-\lambda)(\mathbf{I} - \lambda\beta\widehat{\mathbf{G}})^{-1} = [m_{jk}^N(\beta)]$, and let $\ell_i(\beta) := \ell(\{i\};\beta)$ denote player i's leading index.

Corollary B.2 (Single-leader equilibrium)**.** *Suppose that individual productivities are homogeneous ($\alpha_j = \alpha$ for all $j \in N$). If player $i \in N$ is the single leader, the unique subgame-perfect equilibrium satisfies*

$$x_i^S(\{i\};\beta) = \max\left\{0,\, x_i^N(\beta) + \frac{c_i(\beta)}{1+\gamma\Big(1-c_i(\beta)\Big)^2}\Big(\alpha - x_i^N(\beta)\Big)\right\}, \tag{B.33}$$

$$\ell_i(\beta) = \frac{\sum_{j=1}^n m_{ji}^N(\beta)}{m_{ii}^N(\beta)}\Big[x_i^S(\{i\};\beta) - x_i^N(\beta)\Big]. \tag{B.34}$$

Whenever $x_i^S(\{i\};\beta) > 0$, the leading index $\ell_i(\beta)$ equals the product of the curvature-adjusted reference response $c_i(\beta)/\Big[1+\gamma(1-c_i(\beta))^2\Big]$, the propagation multiplier $\Big(\sum_{j=1}^n m_{ji}^N(\beta)\Big)/m_{ii}^N(\beta)$, and the gap $\alpha - x_i^N(\beta)$.

Proof: See Appendix B.5.15.

By (B.32), the difference between individual productivity and Nash effort,

$$\alpha - x_i^N(\beta) = \frac{\lambda(1-\beta)}{1-\lambda\beta}\,\alpha,$$

is identical across all players $i \in N$. The shift parameter β alone therefore determines whether leadership raises or lowers aggregate effort, whereas network position governs relative gains across players through $c_i(\beta)$ and $\left(\sum_{j=1}^{n} m_{ji}^{N}(\beta)\right)/m_{ii}^{N}(\beta)$.

Strong conformity with a downward shift. We next examine equilibrium behavior under a downward shift when the taste for conformity becomes arbitrarily large ($\lambda \to 1$, or equivalently $\gamma \to \infty$).

Corollary B.3 (Strong conformity under a downward shift)**.** *Suppose that individual productivities are homogeneous ($\alpha_i = \alpha$ for all $i \in N$) and $0 < \beta < 1$. For any leader group L,*

$$\lim_{\lambda \to 1} \mathbf{x}^{N}(\beta) = \lim_{\lambda \to 1} \mathbf{x}^{S}(L;\beta) = \mathbf{0},$$

and hence $\lim_{\lambda \to 1} \ell(L;\beta) = 0$.

Proof: See Appendix B.5.16.

Unlike the baseline model ($\beta = 1$), where a single leader's norm response approaches one as $\lambda \to 1$ and anchors effort across the network at α, a downward shift ($0 < \beta < 1$) causes followers to match only the fraction β of neighbors' efforts, which caps each leader's shifted-reference response strictly below $\beta^2 < 1$. Every player therefore remains exposed to peer pressure as $\gamma \to \infty$. Moving first raises aggregate effort at every finite conformity strength ($\ell(L;\beta) > 0$) yet cannot prevent strong peer pressure toward $\beta\bar{x}_i$ from driving all equilibrium efforts and the leadership gain to zero as $\lambda \to 1$.

B.5 Omitted Proofs

B.5.1 Proof of Theorem B.1

Proof. We establish existence and uniqueness of $\boldsymbol{\kappa}^*$ in Step 1 and verify sequential rationality and belief consistency in Step 2.

Step 1: Existence and uniqueness of $\boldsymbol{\kappa}^*$. Row-stochasticity of $\widehat{\mathbf{G}}$ gives $(\mathbf{I}_F - \widehat{\mathbf{G}}_{FF})\mathbf{1}_F = \widehat{\mathbf{G}}_{FL}\mathbf{1}_L$, so $(\mathbf{I}_L - \mathbf{C}_L)\mathbf{1}_L = (1-\lambda)\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\mathbf{1}_F \geq \mathbf{0}$. Because g is connected and $0 < |L| < n$, at least one leader has a neighbor in F; since $\mathbf{U}_F \geq \mathbf{I}_F$ and $\boldsymbol{\alpha}_F > \mathbf{0}$, the corresponding entry of $(\mathbf{I}_L - \mathbf{C}_L)\mathbf{1}_L$ and of $\mathbf{s}_L$ is strictly positive. When $|L| = 1$, $1 - c_i > 0$ is a positive scalar. When $|L| \geq 2$, connectedness of g makes $\mathbf{C}_L \geq \mathbf{0}$ irreducible, because any path in g between two distinct leaders decomposes into segments between consecutive

leaders and each segment contributes positive weight through $\widehat{\mathbf{G}}_{LL}$ or $\lambda\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\widehat{\mathbf{G}}_{FL}$; since $\mathbf{C}_L\mathbf{1}_L \le \mathbf{1}_L$ with strict inequality in at least one row, $\rho(\mathbf{C}_L) < 1$. Hence $\mathbf{I}_L - \mathbf{C}_L$ is invertible, with $(\mathbf{I}_L - \mathbf{C}_L)^{-1} = \sum_{t=0}^{\infty} \mathbf{C}_L^t \ge \mathbf{I}_L$ and $(\mathbf{I}_L - \mathbf{C}_L)^{-1} > \mathbf{0}$ when $|L| \ge 2$, so

$$\underline{\boldsymbol{\kappa}} := (1-\lambda)(\mathbf{I}_L - \mathbf{C}_L)^{-1}\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F = (\mathbf{I}_L - \mathbf{C}_L)^{-1}\mathbf{s}_L > \mathbf{0}.$$

We next show that any vector $\boldsymbol{\kappa} = (\kappa_i)_{i\in L}$ satisfying $\boldsymbol{\kappa} \ge \underline{\boldsymbol{\kappa}}$ and (B.7) lies in $(\mathbf{0}, \boldsymbol{\alpha}_L]$ and satisfies $(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} \ge \mathbf{s}_L$. For $i \in L$, let $e_i(\boldsymbol{\kappa}) := \kappa_i - s_i - \sum_{j\in L} c_{ij}\kappa_j = [(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} - \mathbf{s}_L]_i$ denote leader i's norm deviation per unit of θ. Since $(\mathbf{I}_L - \mathbf{C}_L)\underline{\boldsymbol{\kappa}} = \mathbf{s}_L$ and $\underline{\boldsymbol{\kappa}} > \mathbf{0}$, each leader $i \in L$ satisfies $(1-c_i)\kappa_i \ge (1-c_i)\underline{\kappa}_i = s_i + \sum_{j\in L\setminus\{i\}} c_{ij}\underline{\kappa}_j \ge s_i$, so $\kappa_i > 0$ and $1 - c_i - s_i/\kappa_i \ge 0$. If $\max_{k\in L}(\kappa_k - \alpha_k) > 0$ is attained at $i \in L$, equation (B.7) requires $\gamma(1 - c_i - s_i/\kappa_i)e_i(\boldsymbol{\kappa}) = \alpha_i - \kappa_i < 0$ and hence $e_i(\boldsymbol{\kappa}) < 0$; however, condition (B.5) and $\sum_{j\in L} c_{ij} \le 1$ give

$$e_i(\boldsymbol{\kappa}) \ge (\kappa_i - \alpha_i) - \sum_{j\in L} c_{ij}(\kappa_j - \alpha_j) \ge \Big(1 - \sum_{j\in L} c_{ij}\Big)(\kappa_i - \alpha_i) \ge 0,$$

a contradiction. Thus $\boldsymbol{\kappa} \le \boldsymbol{\alpha}_L$, and (B.7) yields $\gamma(1 - c_i - s_i/\kappa_i)e_i(\boldsymbol{\kappa}) = \alpha_i - \kappa_i \ge 0$ for all $i \in L$. If $e_i(\boldsymbol{\kappa}) < 0$ for some $i \in L$, then $1 - c_i - s_i/\kappa_i = 0$, so $s_i = (1-c_i)\kappa_i \ge (1-c_i)\underline{\kappa}_i = s_i + \sum_{j\in L\setminus\{i\}} c_{ij}\underline{\kappa}_j$; since $\underline{\boldsymbol{\kappa}} > \mathbf{0}$, we obtain $c_{ij} = 0$ for all $j \in L \setminus \{i\}$ and $e_i(\boldsymbol{\kappa}) = (1-c_i)\kappa_i - s_i = 0$, a contradiction. Therefore $(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} \ge \mathbf{s}_L$, and conversely $(\mathbf{I}_L - \mathbf{C}_L)^{-1} \ge \mathbf{I}_L$ implies $\boldsymbol{\kappa} \ge \underline{\boldsymbol{\kappa}}$ whenever $(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} \ge \mathbf{s}_L$.

To construct $\boldsymbol{\kappa}^*$ as a contraction fixed point on $[\mathbf{0}, \boldsymbol{\alpha}_L]$, for any $\mathbf{v} \in [\mathbf{0}, \boldsymbol{\alpha}_L]$ and $i \in L$, define $K_i(\mathbf{v}) := \sum_{j\in L\setminus\{i\}} c_{ij}v_j \ge 0$ and

$$\Phi_i(t; K) := \alpha_i - t - \gamma\Big[(1-c_i)t - s_i - K\Big]\left(1 - c_i - \frac{s_i}{t}\right), \qquad t > 0.$$

Since $\mathbf{v} \le \boldsymbol{\alpha}_L$, $\mathbf{C}_L \ge \mathbf{0}$, and $1 - c_i \ge 1 - \lambda > 0$, condition (B.5) implies $\Big[s_i + K_i(\mathbf{v})\Big]/(1 - c_i) \le \alpha_i$. When $s_i + K_i(\mathbf{v}) > 0$, for all $t \ge \Big[s_i + K_i(\mathbf{v})\Big]/(1-c_i) > 0$ we have

$$\frac{\partial\Phi_i\Big(t; K_i(\mathbf{v})\Big)}{\partial t} = -1 - \gamma(1-c_i)^2 + \frac{\gamma s_i\Big[s_i + K_i(\mathbf{v})\Big]}{t^2} \le -1,$$

together with $\Phi_i\Big([s_i + K_i(\mathbf{v})]/(1-c_i); K_i(\mathbf{v})\Big) = \alpha_i - [s_i + K_i(\mathbf{v})]/(1-c_i) \ge 0$ and $\Phi_i\Big(\alpha_i; K_i(\mathbf{v})\Big) \le 0$, so $\Phi_i\Big(\cdot; K_i(\mathbf{v})\Big)$ admits a unique root $\mathcal{T}_i(\mathbf{v}) \in \Big[[s_i + K_i(\mathbf{v})]/(1-c_i), \alpha_i\Big]$. When $s_i + K_i(\mathbf{v}) = 0$, $\Phi_i(t; 0) = \alpha_i - [1 + \gamma(1-c_i)^2]t$ has the unique positive root $\mathcal{T}_i(\mathbf{v}) = \alpha_i/[1 + \gamma(1-c_i)^2] \in (0, \alpha_i]$.

In either case, $t = \mathcal{T}_i(\mathbf{v}) \geq \alpha_i/[1+\gamma(1-c_i)^2] > 0$ satisfies $-\partial\Phi_i/\partial t \geq 1$ and

$$\begin{aligned}&\gamma(1-c_i)\left(-\frac{\partial\Phi_i}{\partial t}\right) - \Big[1+\gamma(1-c_i)^2\Big]\gamma\left(1-c_i-\frac{s_i}{t}\right)\\ &\quad= \frac{\gamma s_i}{t^2}\Big\{t+\gamma(1-c_i)\Big[(1-c_i)t - s_i - K_i(\mathbf{v})\Big]\Big\} \geq 0.\end{aligned}$$

Implicit differentiation of $\Phi_i\Big(\mathcal{T}_i(\mathbf{v}); K_i(\mathbf{v})\Big) = 0$ therefore gives

$$0 \leq \frac{\partial\mathcal{T}_i}{\partial K_i} = \frac{\gamma(1-c_i-s_i/t)}{-\partial\Phi_i/\partial t} \leq \frac{\gamma(1-c_i)}{1+\gamma(1-c_i)^2}.$$

Since $\sum_{j\in L\setminus\{i\}} c_{ij} \leq 1-c_i \leq 1$, each row sum of the Jacobian of $\mathcal{T} = (\mathcal{T}_i)_{i\in L}$ is at most $\gamma(1-c_i)^2/[1+\gamma(1-c_i)^2] \leq \lambda < 1$. Hence $\mathcal{T}$ is monotone increasing and is a $\|\cdot\|_\infty$-contraction on $[\mathbf{0}, \boldsymbol{\alpha}_L]$. The unique fixed point $\boldsymbol{\kappa}^* > \mathbf{0}$ of $\mathcal{T}$ satisfies $(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa}^* \geq \mathbf{s}_L$, $\boldsymbol{\kappa}^* \geq \underline{\boldsymbol{\kappa}}$, and (B.7). Because every solution to (B.7) with $\boldsymbol{\kappa} \geq \underline{\boldsymbol{\kappa}}$ lies in $[\mathbf{0}, \boldsymbol{\alpha}_L]$, satisfies $(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} \geq \mathbf{s}_L$, and is a fixed point of $\mathcal{T}$, $\boldsymbol{\kappa}^*$ is unique.

Step 2: Sequential rationality and belief consistency. After every Stage 1 history $\mathbf{x}_L \geq \mathbf{0}$, the point belief at $\mu_{\boldsymbol{\kappa}^*}(\mathbf{x}_L) \in [0,\bar\theta]$ is supported on $[0,\bar\theta]$, and (B.2) evaluated at $\mu_{\boldsymbol{\kappa}^*}(\mathbf{x}_L)$ is the unique Stage 2 equilibrium. On the equilibrium path $\mathbf{x}_L^I(\theta) = \boldsymbol{\kappa}^*\theta$, (B.4) gives $\mu_{\boldsymbol{\kappa}^*}(\boldsymbol{\kappa}^*\theta) = \theta$ for all $\theta \in [0,\bar\theta]$, which yields (B.6) and satisfies Bayes' rule.

To verify global Stage 1 optimality, fix $\theta \in [0,\bar\theta]$ and leader $i \in L$, with $x_j = \kappa_j^*\theta$ for all $j \in L\setminus\{i\}$, and set $K_i := \sum_{j\in L\setminus\{i\}} c_{ij}\kappa_j^*$. By Step 1, $1-c_i-s_i/\kappa_i^* \geq 0$ and $e_i(\boldsymbol{\kappa}^*) = (1-c_i)\kappa_i^* - s_i - K_i \geq 0$. Define the strictly concave quadratic function

$$Q_i(x_i;\theta) := \alpha_i\theta x_i - \frac{1}{2}x_i^2 - \frac{\gamma}{2}\left[\left(1-c_i-\frac{s_i}{\kappa_i^*}\right)x_i - K_i\theta\right]^2, \qquad x_i \geq 0.$$

By (B.7), $\partial Q_i(x_i;\theta)/\partial x_i = 0$ at $x_i = \kappa_i^*\theta$, so $x_i = \kappa_i^*\theta$ uniquely maximizes $Q_i(\cdot;\theta)$ on $[0,\infty)$. For $0 \leq x_i \leq \kappa_i^*\theta$, (B.4) gives $\mu_{\boldsymbol{\kappa}^*}(\mathbf{x}_L) = x_i/\kappa_i^*$, so leader i's Stage 1 payoff coincides with $Q_i(x_i;\theta)$. For $x_i > \kappa_i^*\theta$, (B.4) implies $\mu_{\boldsymbol{\kappa}^*}(\mathbf{x}_L) \leq x_i/\kappa_i^*$, and therefore

$$(1-c_i)x_i - s_i\mu_{\boldsymbol{\kappa}^*}(\mathbf{x}_L) - K_i\theta \geq \left(1-c_i-\frac{s_i}{\kappa_i^*}\right)x_i - K_i\theta \geq e_i(\boldsymbol{\kappa}^*)\theta \geq 0.$$

Thus leader i's conformity cost on $(\kappa_i^*\theta,\infty)$ is at least as large as under $Q_i(x_i;\theta)$, so leader i's Stage 1 payoff never exceeds $Q_i(\kappa_i^*\theta;\theta) = u_i(\mathbf{x}^I(L;\theta);\theta)$. No deviation in $[0,\infty)$ is profitable in any state $\theta \in [0,\bar\theta]$. □

B.5.2 Proof of Proposition B.1

Proof. For any leader group L and any $\boldsymbol{\kappa} = (\kappa_i)_{i\in L} \in \mathcal{K}(L)$, let $e_i(\boldsymbol{\kappa}) := [(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} - \mathbf{s}_L]_i$. At $\theta = \bar{\theta} > 0$, an upward deviation $x_i > \kappa_i\bar{\theta}$ by leader i leaves $\mu_{\boldsymbol{\kappa}}(\mathbf{x}_L) = \bar{\theta}$, whereas a downward deviation $x_i \in [0, \kappa_i\bar{\theta}]$ yields $\mu_{\boldsymbol{\kappa}}(\mathbf{x}_L) = x_i/\kappa_i$. Dividing the left and right derivatives of leader i's reduced Stage 1 payoff at $x_i = \kappa_i\bar{\theta}$ by $\bar{\theta}$ therefore gives

$$\begin{aligned} D_i^- &:= \alpha_i - \kappa_i - \gamma e_i(\boldsymbol{\kappa})\left(1 - c_i - \frac{s_i}{\kappa_i}\right) \geq 0, \\ D_i^+ &:= \alpha_i - \kappa_i - \gamma(1-c_i)e_i(\boldsymbol{\kappa}) \leq 0. \end{aligned} \tag{B.35}$$

Under condition (B.5), if $s_i > 0$, then $D_i^- - D_i^+ = \gamma(s_i/\kappa_i)e_i(\boldsymbol{\kappa}) \geq 0$ gives $e_i(\boldsymbol{\kappa}) \geq 0$ and hence $1 - c_i - s_i/\kappa_i \geq 0$, so $D_i^- \geq 0$ implies $\kappa_i \leq \alpha_i$. If $s_i = 0$, (B.35) reduces to $\alpha_i - \kappa_i = \gamma(1-c_i)e_i(\boldsymbol{\kappa})$. Suppose, to the contrary, that $\max_{k\in L}(\kappa_k - \alpha_k) > 0$ is attained at $i \in L$. Then $s_i = 0$, whereas (B.5) and $\sum_{j\in L} c_{ij} \leq 1$ give $e_i(\boldsymbol{\kappa}) \geq (\kappa_i - \alpha_i) - \sum_{j\in L} c_{ij}(\kappa_j - \alpha_j) \geq 0$, which contradicts $\gamma(1-c_i)e_i(\boldsymbol{\kappa}) = \alpha_i - \kappa_i < 0$. Hence $\boldsymbol{\kappa} \leq \boldsymbol{\alpha}_L$, and $\alpha_i - \kappa_i = \gamma(1-c_i)e_i(\boldsymbol{\kappa}) \geq 0$ implies $e_i(\boldsymbol{\kappa}) \geq 0$ when $s_i = 0$ as well. Each $\kappa_i > 0$ therefore lies in $\big[[s_i + K_i(\boldsymbol{\kappa})]/(1-c_i), \alpha_i\big]$, where $\Phi_i\big(\cdot; K_i(\boldsymbol{\kappa})\big)$ from Step 1 of the proof of Theorem B.1 is strictly decreasing. Since $\Phi_i\big(\kappa_i; K_i(\boldsymbol{\kappa})\big) = D_i^- \geq 0 = \Phi_i\big(\mathcal{T}_i(\boldsymbol{\kappa}); K_i(\boldsymbol{\kappa})\big)$, we obtain $\boldsymbol{\kappa} \leq \mathcal{T}(\boldsymbol{\kappa})$. Monotonicity and contraction of $\mathcal{T}$ yield $\boldsymbol{\kappa} \leq \mathcal{T}^m(\boldsymbol{\kappa}) \to \boldsymbol{\kappa}^*$ as $m \to \infty$; together with $\lambda \mathbf{U}_F \widehat{\mathbf{G}}_{FL} \geq \mathbf{0}$, this inequality implies $\mathbf{x}^I(L, \boldsymbol{\kappa}^*; \theta) \geq \mathbf{x}^I(L, \boldsymbol{\kappa}; \theta)$ for all $\theta \in [0, \bar{\theta}]$. □

B.5.3 Proof of Proposition B.2

Proof. Fix $L = \{i\}$ and $F = N \setminus \{i\}$. We first characterize when $\mathcal{K}(\{i\})$ is nonempty and show that its element is unique. In a connected network, $(1-\lambda)\mathbf{U}_F\boldsymbol{\alpha}_F > \mathbf{0}$ and $\widehat{\mathbf{G}}_{i,-i} \neq \mathbf{0}$ imply $s_i > 0$. At any interior state $\theta \in (0, \bar{\theta})$, local deviations in either direction satisfy $\mu_{\kappa_i}(x_i) = x_i/\kappa_i$, so any $\kappa_i \in \mathcal{K}(\{i\})$ satisfies $D_i^- = 0$ in (B.35). Together with $D_i^+ \leq 0$ at $\theta = \bar{\theta}$, the condition $D_i^- = 0$ yields $e_i = (1-c_i)\kappa_i - s_i \geq 0$ and

$$\alpha_i = \kappa_i + \gamma e_i\left(1 - c_i - \frac{s_i}{\kappa_i}\right) = \kappa_i + \frac{\gamma e_i^2}{\kappa_i} \geq \kappa_i \geq \frac{s_i}{1-c_i},$$

which proves necessity of $\alpha_i \geq s_i/(1-c_i)$. Sufficiency follows from Theorem B.1, since $\alpha_i \geq s_i/(1-c_i)$ is condition (B.5) for $L = \{i\}$. Rewriting $D_i^- = 0$ gives the quadratic equation

$$\big[1 + \gamma(1-c_i)^2\big]\kappa_i^2 - \big[\alpha_i + 2\gamma(1-c_i)s_i\big]\kappa_i + \gamma s_i^2 = 0.$$

When $\alpha_i > s_i/(1-c_i)$, the left-hand side takes the negative value $[s_i/(1-c_i)]\big[s_i/(1-c_i)-\alpha_i\big] < 0$ at $\kappa_i = s_i/(1-c_i)$; when $\alpha_i = s_i/(1-c_i)$, the two roots are $s_i/(1-c_i)$ and $\frac{\gamma(1-c_i)^2}{1+\gamma(1-c_i)^2}\frac{s_i}{1-c_i} < \frac{s_i}{1-c_i}$. In either case, only the larger root satisfies $\kappa_i \geq s_i/(1-c_i)$, which establishes uniqueness in $\mathcal{K}(\{i\})$ and gives $x_i^I(\theta) = \kappa_i^*\theta$ with

$$\kappa_i^* = \frac{\alpha_i + 2\gamma(1-c_i)s_i + \sqrt{\alpha_i^2 + 4\gamma s_i\big[(1-c_i)\alpha_i - s_i\big]}}{2\big[1+\gamma(1-c_i)^2\big]}. \tag{B.36}$$

We next compare equilibrium efforts with the complete-information and simultaneous-move benchmarks. In simultaneous play, $x_i^N = (1-\lambda)\alpha_i + \lambda(s_i + c_i x_i^N)$ gives $x_i^N = (\alpha_i + \gamma s_i)/[1+\gamma(1-c_i)]$, so $\alpha_i \geq s_i/(1-c_i)$ is equivalent to $\alpha_i \geq x_i^N$. Under complete information, $x_i^S = \big[\alpha_i + \gamma(1-c_i)s_i\big]/\big[1+\gamma(1-c_i)^2\big]$; subtracting $x_i^S\theta$ from $\kappa_i^*\theta$ in (B.36) yields (B.8). By (B.2) at $\mu(\mathbf{x}_L) = \theta$ and (B.6), $\mathbf{x}_{-i}^I(\theta) - \mathbf{x}_{-i}^S(\theta) = \lambda \mathbf{U}_F \widehat{\mathbf{G}}_{-i,i}(\kappa_i^* - x_i^S)\theta$. Meanwhile, restricting column i of $(\mathbf{I} - \lambda\widehat{\mathbf{G}})\mathbf{M}^N = (1-\lambda)\mathbf{I}$ to the follower rows $N\setminus\{i\}$ gives $(\mathbf{I}_{-i} - \lambda\widehat{\mathbf{G}}_{-i})\mathbf{M}_{-i,i}^N = \lambda\widehat{\mathbf{G}}_{-i,i} m_{ii}^N$, so $\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{-i,i} = \mathbf{M}_{-i,i}^N/m_{ii}^N$ and $1 + \lambda\sum_{j\neq i}(\mathbf{U}_F\widehat{\mathbf{G}}_{-i,i})_j = \sum_{j=1}^n m_{ji}^N/m_{ii}^N$, which proves (B.9). Connectedness and $\lambda > 0$ imply $c_i > 0$, and

$$x_i^S - x_i^N = \frac{\gamma c_i\big[(1-c_i)\alpha_i - s_i\big]}{\big[1+\gamma(1-c_i)^2\big]\big[1+\gamma(1-c_i)\big]},$$

so $\widehat{\ell}_i > 0$ when $\alpha_i > x_i^N$ and $\widehat{\ell}_i = 0$ when $\alpha_i = x_i^N$. □

B.5.4 Proof of Proposition B.3

Proof. For a leader group L and any $\boldsymbol{\kappa} \in \mathcal{K}(L)$, let $e_i(\boldsymbol{\kappa}) := [(\mathbf{I}_L - \mathbf{C}_L)\boldsymbol{\kappa} - \mathbf{s}_L]_i$, and define

$$\mathbf{H}_L := \mathbf{I}_L + \gamma(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{C}_L), \qquad \rho_i := \kappa_i + \gamma(1-c_i)e_i(\boldsymbol{\kappa}) - \alpha_i \quad (i \in L).$$

By (B.35), $0 \leq \rho_i \leq \gamma(s_i/\kappa_i)e_i(\boldsymbol{\kappa})$ for all $i \in L$. Off-diagonal entries of $\mathbf{H}_L$ are nonpositive, and in each row $i \in L$ the diagonal entry exceeds the sum of absolute off-diagonal entries by $1 + \gamma(1-c_i)\big(1 - \sum_{j\in L} c_{ij}\big) \geq 1$. Hence $\mathbf{H}_L$ is a nonsingular M-matrix with $\mathbf{H}_L^{-1} \geq \mathbf{0}$, and $\mathbf{H}_L\mathbf{1}_L \geq \mathbf{1}_L$ implies $\mathbf{H}_L^{-1}\mathbf{1}_L \leq \mathbf{1}_L$, so $\|\mathbf{H}_L^{-1}\|_\infty \leq 1$.

In the complete-information two-stage equilibrium, leaders satisfy $\mathbf{H}_L\mathbf{x}_L^S = \boldsymbol{\alpha}_L + \gamma(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\mathbf{s}_L$, whereas the definition of $\boldsymbol{\rho} = (\rho_i)_{i\in L}$ gives $\mathbf{H}_L\boldsymbol{\kappa} = \boldsymbol{\alpha}_L + \gamma(\mathbf{I}_L - \mathbf{C}_L^{\mathrm{D}})\mathbf{s}_L + \boldsymbol{\rho}$. Subtracting the complete-information system and using $\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL} \geq \mathbf{0}$ yields

$$\begin{aligned} 0 \leq \rho_i &\leq \gamma\frac{s_i}{\kappa_i}e_i(\boldsymbol{\kappa}), \\ \boldsymbol{\kappa} - \mathbf{x}_L^S &= \mathbf{H}_L^{-1}\boldsymbol{\rho} \geq \mathbf{0}, \\ \mathbf{U}_F\Big[(1-\lambda)\boldsymbol{\alpha}_F + \lambda\widehat{\mathbf{G}}_{FL}\boldsymbol{\kappa}\Big] - \mathbf{x}_F^S &= \lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\Big(\boldsymbol{\kappa} - \mathbf{x}_L^S\Big) \geq \mathbf{0}. \end{aligned} \tag{B.37}$$

Multiplying (B.37) by $\theta \geq 0$ gives $\mathbf{x}^I(L,\boldsymbol{\kappa};\theta) \geq \mathbf{x}^S(L;\theta)$ for all $\theta \in [0,\bar{\theta}]$. □

B.5.5 Proof of Proposition B.4

Proof. Since productivities are heterogeneous and $\mathbf{M}^N$ is strictly positive and row-stochastic, any player i with the highest productivity satisfies $\alpha_i > x_i^N$, so $\{i\} \in \mathcal{F}(\lambda)$ and $\widehat{\ell}_i > 0$ by Proposition B.2. Because $\mathbb{E}[\theta] > 0$, any optimal leader group $L^* \in \mathcal{F}(\lambda)$ satisfies $\widehat{\ell}(L^*) > 0$.

Part (i): Leader eligibility. For $L^* \in \mathcal{F}(\lambda)$, let $\mathbf{h} := (\mathbf{I}_{L^*} - \mathbf{C}_{L^*})\boldsymbol{\alpha}_{L^*} - \mathbf{s}_{L^*} \geq \mathbf{0}$. If $\mathbf{h} = \mathbf{0}$, then $\boldsymbol{\kappa}^* = \boldsymbol{\alpha}_{L^*}$ satisfies $(\mathbf{I}_{L^*} - \mathbf{C}_{L^*})\boldsymbol{\kappa}^* = \mathbf{s}_{L^*}$ and (B.7); leaders and followers then play simultaneous-move best responses in every state θ, so $\mathbf{x}^I(L^*;\theta) = \theta\mathbf{x}^N$ and $\widehat{\ell}(L^*) = 0$, contradicting $\widehat{\ell}(L^*) > 0$. Thus $\mathbf{h} \neq \mathbf{0}$. In simultaneous play, $\mathbf{x}_{F^*}^N = \mathbf{U}_{F^*}\Big[(1-\lambda)\boldsymbol{\alpha}_{F^*} + \lambda\widehat{\mathbf{G}}_{F^*L^*}\mathbf{x}_{L^*}^N\Big]$ reduces the equilibrium conditions on L^* to $(\mathbf{I}_{L^*} - \lambda\mathbf{C}_{L^*})\mathbf{x}_{L^*}^N = (1-\lambda)\boldsymbol{\alpha}_{L^*} + \lambda\mathbf{s}_{L^*}$, and hence

$$\boldsymbol{\alpha}_{L^*} - \mathbf{x}_{L^*}^N = \lambda(\mathbf{I}_{L^*} - \lambda\mathbf{C}_{L^*})^{-1}\mathbf{h}.$$

For $|L^*| = 1$, $(\mathbf{I}_{L^*} - \lambda\mathbf{C}_{L^*})^{-1}$ is a positive scalar; for $|L^*| \geq 2$, $\mathbf{C}_{L^*}$ is nonnegative and irreducible with $\rho(\mathbf{C}_{L^*}) < 1$, so $(\mathbf{I}_{L^*} - \lambda\mathbf{C}_{L^*})^{-1} > \mathbf{0}$. Since $\mathbf{h} \geq \mathbf{0}$ is nonzero, $\alpha_i - x_i^N > 0$ for all $i \in L^*$, which is equivalent to $\ell_i > 0$ by Proposition 2 in Section 3.2.

Part (ii): Follower domination. Suppose, to the contrary, that some leader $k \in L^*$ has no neighbor in F^*. Since g is undirected, $\widehat{\mathbf{G}}_{kF^*} = \mathbf{0}$ and $\widehat{\mathbf{G}}_{F^*k} = \mathbf{0}$, so $s_k = c_k = 0$, $c_{kj} = \widehat{g}_{kj}$ for $j \in L^*$, and $c_{ik} = \widehat{g}_{ik}$ for $i \in L^*$. Equation (B.7) for $i = k$ simplifies to

$$\kappa_k^* = (1-\lambda)\alpha_k + \lambda\sum_{j\in L^*\setminus\{k\}}\widehat{g}_{kj}\kappa_j^*,$$

which coincides with the Stage 2 best-response coefficient of an informed follower. Because g is connected and $F^* \neq \varnothing$, leader k has at least one neighbor in $L' := L^* \setminus \{k\}$, so

$0 < |L'| < n$ and $|L^*| \geq 2$. Reassigning k to $F' := F^* \cup \{k\}$ gives $\mathbf{U}_{F'} = \mathrm{diag}(\mathbf{U}_{F^*}, 1)$, and the entries of $\mathbf{s}_{L'}$ and $\mathbf{C}_{L'}$ satisfy

$$s'_i = s_i + (1-\lambda)\widehat{g}_{ik}\alpha_k, \qquad c'_{ij} = c_{ij} + \lambda \widehat{g}_{ik}\widehat{g}_{kj} \qquad (i,j \in L').$$

Consequently, $\alpha_i - s'_i - (\mathbf{C}_{L'}\boldsymbol{\alpha}_{L'})_i = h_i + \lambda \widehat{g}_{ik} h_k \geq 0$ for all $i \in L'$, so $L' \in \mathcal{F}(\lambda)$.

Let $\mathbf{e} := (\mathbf{I}_{L^*} - \mathbf{C}_{L^*})\boldsymbol{\kappa}^* - \mathbf{s}_{L^*} \geq \mathbf{0}$. By (B.7), $\alpha_i - \kappa_i^* = \gamma(1 - c_i - s_i/\kappa_i^*)e_i \geq 0$ and $\mathbf{h} = \mathbf{e} + (\mathbf{I}_{L^*} - \mathbf{C}_{L^*})(\boldsymbol{\alpha}_{L^*} - \boldsymbol{\kappa}^*)$. Because $|L^*| \geq 2$, $\mathbf{C}_{L^*}$ is irreducible: if $e_i = 0$ for some $i \in L^*$, then $\alpha_i - \kappa_i^* = 0$ and $0 \leq h_i = -\sum_{j \in L^* \setminus \{i\}} c_{ij}(\alpha_j - \kappa_j^*) \leq 0$, which forces $\alpha_j - \kappa_j^* = e_j = 0$ whenever $c_{ij} > 0$ and hence $\mathbf{e} = \mathbf{h} = \mathbf{0}$, a contradiction. Thus $e_i > 0$ for all $i \in L^*$. Evaluate the root map $\mathcal{T}'$ for L' at $\boldsymbol{\kappa}^*_{L'} := (\kappa_i^*)_{i \in L'}$. Because κ_k^* equals player k's Stage 2 response coefficient to $\boldsymbol{\kappa}^*_{L'}$, each $i \in L'$ satisfies $[(\mathbf{I}_{L'} - \mathbf{C}_{L'})\boldsymbol{\kappa}^*_{L'} - \mathbf{s}_{L'}]_i = e_i > 0$, while

$$\left(1 - c'_i - \frac{s'_i}{\kappa_i^*}\right) - \left(1 - c_i - \frac{s_i}{\kappa_i^*}\right) = -\widehat{g}_{ik}\left[\lambda \widehat{g}_{ki} + \frac{(1-\lambda)\alpha_k}{\kappa_i^*}\right] \leq 0,$$

with strict inequality for every neighbor $i \in L'$ of k. Hence $\Phi'_i\big(\kappa_i^*; K'_i(\boldsymbol{\kappa}^*_{L'})\big) \geq 0$ for all $i \in L'$, with strict inequality for neighbors of k, which yields $\boldsymbol{\kappa}^*_{L'} \leq \mathcal{T}'(\boldsymbol{\kappa}^*_{L'})$ and $\boldsymbol{\kappa}^*_{L'} \neq \mathcal{T}'(\boldsymbol{\kappa}^*_{L'})$. Monotonicity and contraction of $\mathcal{T}'$ imply that the fixed point $\boldsymbol{\kappa}^{*\prime}$ for L' satisfies $\boldsymbol{\kappa}^{*\prime} \geq \mathcal{T}'(\boldsymbol{\kappa}^*_{L'}) \geq \boldsymbol{\kappa}^*_{L'}$ with at least one strict inequality, and $\lambda \mathbf{U}_{F'}\widehat{\mathbf{G}}_{F'L'} \geq \mathbf{0}$ then gives $\mathbf{x}^I(L';\theta) \geq \mathbf{x}^I(L^*;\theta)$ for all $\theta \in [0,\bar{\theta}]$, with strict inequality on L' for $\theta > 0$. Thus $\widehat{\ell}(L') > \widehat{\ell}(L^*)$, contradicting the optimality of L^*. □

B.5.6 Proof of Proposition B.5

We first establish uniform effort bounds across all separating equilibria in $\mathcal{K}(L)$. Let $\underline{\alpha} := \min_{i \in N} \alpha_i$ and $\overline{\alpha} := \max_{i \in N} \alpha_i$.

Lemma B.1. *For all $\lambda \in (0,1)$, every leader group L, every $\boldsymbol{\kappa} \in \mathcal{K}(L)$, and all $\theta \in [0,\bar{\theta}]$, $\underline{\alpha}\theta\mathbf{1} \leq \mathbf{x}^I(L,\boldsymbol{\kappa};\theta) \leq \overline{\alpha}\theta\mathbf{1}$; in particular, $\underline{\alpha}\mathbf{1}_L \leq \boldsymbol{\kappa} \leq \overline{\alpha}\mathbf{1}_L$.*

Proof. By Proposition B.3, $\mathbf{x}^I(L,\boldsymbol{\kappa};\theta) \geq \mathbf{x}^S(L;\theta) = \theta \mathbf{M}^S(L)\boldsymbol{\alpha} \geq \underline{\alpha}\theta\mathbf{1}$ because $\mathbf{M}^S(L)$ in Section 3.1 is nonnegative and row-stochastic; evaluating the L-block at $\theta = \bar{\theta} > 0$ gives $\boldsymbol{\kappa} \geq \underline{\alpha}\mathbf{1}_L$. For the upper bound, if $i \in L$ has $s_i > 0$, (B.37) gives $e_i(\boldsymbol{\kappa}) \geq 0$ and hence $1 - c_i - s_i/\kappa_i = \big[e_i(\boldsymbol{\kappa}) + \sum_{j \in L \setminus \{i\}} c_{ij}\kappa_j\big]/\kappa_i \geq 0$, so $D_i^- \geq 0$ in (B.35) implies $\kappa_i \leq \alpha_i \leq \overline{\alpha}$. If $\max_{k \in L} \kappa_k > \overline{\alpha}$ were attained at some $i \in L$, then $s_i = 0$, so leader i would have no neighbor in F and row i of $\mathbf{C}_L$ would sum to one; thus $e_i(\boldsymbol{\kappa}) = \kappa_i - \sum_{j \in L} c_{ij}\kappa_j \geq 0$,

and $D_i^+ = D_i^- = 0$ would yield $\kappa_i = \alpha_i - \gamma(1-c_i)e_i(\boldsymbol{\kappa}) \le \overline{\alpha}$, a contradiction. Finally, $(1-\lambda)\mathbf{U}_F\mathbf{1}_F + \lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\mathbf{1}_L = \mathbf{1}_F$ expresses $\mathbf{x}_F^I(L,\boldsymbol{\kappa};\theta)$ as convex combinations of $\theta\boldsymbol{\alpha}_F$ and $\theta\boldsymbol{\kappa}$, which gives $\mathbf{x}_F^I(L,\boldsymbol{\kappa};\theta) \le \overline{\alpha}\theta\mathbf{1}_F$. □

With Lemma B.1 established, we prove parts (i) and (ii) of Proposition B.5.

Proof. Part (i): Weak conformity. For any $L \subseteq N$, let $A := L \cap N_+$, $B := L \setminus N_+$, and $F := N \setminus L$. Since $\alpha_i - \bar{\alpha}_i > 0$ for $i \in A$ and $\alpha_i - \bar{\alpha}_i \le 0$ for $i \in B$,

$$\widehat{\Lambda}(A) - \widehat{\Lambda}(L) = \sum_{i\in A}\sum_{j\in B}\widehat{g}_{ij}\frac{\alpha_j}{\alpha_i}(\alpha_i - \bar{\alpha}_i) - \sum_{i\in B}\sum_{j\in F}\widehat{g}_{ij}\frac{\alpha_j}{\alpha_i}(\alpha_i - \bar{\alpha}_i) \ge 0.$$

Because g is connected and productivities are heterogeneous, some player $i \in \arg\max_{k\in N}\alpha_k$ has a neighbor with strictly lower productivity, so $i \in N_+$ and $\widehat{\Lambda}(\{i\}) = (\bar{\alpha}_i/\alpha_i)(\alpha_i - \bar{\alpha}_i) > 0$, whereas $\arg\min_{k\in N}\alpha_k \subseteq N \setminus N_+$. Hence $\max_{0<|L|<n}\widehat{\Lambda}(L) = \max_{L\subseteq N}\widehat{\Lambda}(L) > 0$, and at least one maximizer L^* satisfies $\varnothing \neq L^* \subseteq N_+$.

As $\lambda \to 0$, $\mathbf{s}_L = \widehat{\mathbf{G}}_{LF}\boldsymbol{\alpha}_F + \mathcal{O}(\lambda)$ and $\mathbf{C}_L = \widehat{\mathbf{G}}_{LL} + \mathcal{O}(\lambda)$. Every nonempty $L \subseteq N_+$ satisfies $\alpha_i - [\mathbf{s}_L + \mathbf{C}_L\boldsymbol{\alpha}_L]_i = (\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda) > 0$ for all $i \in L$ and all sufficiently small $\lambda > 0$, so $L \in \mathcal{F}(\lambda)$. At $\lambda = 0$, (B.7) has solution $\boldsymbol{\kappa}^* = \boldsymbol{\alpha}_L$ with Jacobian $\mathbf{I}_L$; by the implicit function theorem and $\gamma = \lambda + \mathcal{O}(\lambda^2)$,

$$\kappa_i^* = \alpha_i - \lambda(\alpha_i - \bar{\alpha}_i)\left(1 - \frac{\sum_{j\in F}\widehat{g}_{ij}\alpha_j}{\alpha_i}\right) + \mathcal{O}(\lambda^2), \qquad i \in L.$$

Since $x_i^N = \alpha_i - \lambda(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$ and $\mathbf{x}_F^I(L;\theta) - \theta\mathbf{x}_F^N = \lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}(\boldsymbol{\kappa}^* - \mathbf{x}_L^N)\theta = \mathcal{O}(\lambda^2)\theta$, summing over N gives $\widehat{\ell}(L) = \lambda\widehat{\Lambda}(L) + \mathcal{O}(\lambda^2)$ for every nonempty $L \subseteq N_+$.

Now take any leader group L and any $\boldsymbol{\kappa} \in \mathcal{K}(L)$. Lemma B.1 and (B.35) yield $\boldsymbol{\kappa} = \boldsymbol{\alpha}_L + \mathcal{O}(\lambda)$ and $e_i(\boldsymbol{\kappa}) = (\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda)$ uniformly across L and $\boldsymbol{\kappa} \in \mathcal{K}(L)$. By (B.37), $0 \le \rho_i \le \gamma(s_i/\kappa_i)e_i(\boldsymbol{\kappa})$ for each $i \in L$. For sufficiently small $\lambda > 0$, if $\alpha_i > \bar{\alpha}_i$, then $\rho_i \le \lambda\sum_{j\in F}\widehat{g}_{ij}(\alpha_j/\alpha_i)(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\lambda^2)$; if $\alpha_i = \bar{\alpha}_i$, then $e_i(\boldsymbol{\kappa}) = \mathcal{O}(\lambda)$ gives $0 \le \rho_i \le \mathcal{O}(\lambda^2)$; if $\alpha_i < \bar{\alpha}_i$, then $e_i(\boldsymbol{\kappa}) < 0$ forces $s_i = 0$, $\widehat{g}_{ij} = 0$ for all $j \in F$, and $\rho_i = 0$. Since $\mathbf{H}_L^{-1} = \mathbf{I}_L + \mathcal{O}(\lambda)$, $\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL} = \mathcal{O}(\lambda)$, and $\sum_{i=1}^n x_i^S(L) - X^N = \mathcal{O}(\lambda^2)$, equation (B.37) yields

$$\sum_{i=1}^n x_i^I(L,\boldsymbol{\kappa};\theta) - \theta X^N \le \left[\lambda\widehat{\Lambda}(L) + \mathcal{O}(\lambda^2)\right]\theta$$

uniformly across L, $\boldsymbol{\kappa} \in \mathcal{K}(L)$, and $\theta \in [0,\bar{\theta}]$. Because 2^N is finite and $\mathbb{E}[\theta] > 0$, for all sufficiently small $\lambda > 0$ the maximizer $L^* \subseteq N_+$ belongs to $\mathcal{F}(\lambda)$ and achieves strictly

higher expected aggregate effort than every group outside $\arg\max_{0<|L|<n}\widehat{\Lambda}(L)$ under any $\boldsymbol{\kappa}\in\mathcal{K}(L)$. Hence an optimal leader group maximizes $\widehat{\Lambda}(L)$.

Part (ii): Strong conformity. We first show that for each leader group L, as $\lambda\to 1$,

$$\mathbf{0}\le \mathbf{x}^I(L,\boldsymbol{\kappa};\theta)-\mathbf{x}^S(L;\theta),\qquad \sup_{\boldsymbol{\kappa}\in\mathcal{K}(L)}\|\mathbf{x}^I(L,\boldsymbol{\kappa};\theta)-\mathbf{x}^S(L;\theta)\|_\infty=\mathcal{O}(1-\lambda)\theta \quad \text{(B.38)}$$

for all $\theta\in[0,\bar{\theta}]$. Connectedness of g and $L\neq\varnothing$ imply that $(\mathbf{I}_F-\widehat{\mathbf{G}}_{FF})^{-1}$ exists, so $\mathbf{s}_L=\mathcal{O}(1-\lambda)$, $\gamma\mathbf{s}_L=\lambda\widehat{\mathbf{G}}_{LF}\mathbf{U}_F\boldsymbol{\alpha}_F=\mathcal{O}(1)$, and $\|\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\|_\infty\le 1$. When $|L|\ge 2$, $\mathbf{C}_L$ converges as $\lambda\to 1$ to $\widehat{\mathbf{G}}_{LL}+\widehat{\mathbf{G}}_{LF}(\mathbf{I}_F-\widehat{\mathbf{G}}_{FF})^{-1}\widehat{\mathbf{G}}_{FL}$, which is irreducible and row-stochastic, so $c_i(1):=\lim_{\lambda\to1}c_i<1$ for all $i\in L$. Since $\kappa_i\ge\underline{\alpha}$ by Lemma B.1, we have $1-c_i-s_i/\kappa_i\ge[1-c_i(1)]/2>0$ for λ close to one. If $s_i=0$, (B.37) gives $\rho_i=0$; if $s_i>0$, $D_i^-\ge 0$ and Lemma B.1 give $0\le e_i(\boldsymbol{\kappa})\le(\overline{\alpha}-\underline{\alpha})/\big[\gamma(1-c_i-s_i/\kappa_i)\big]=\mathcal{O}(1-\lambda)$ and hence $0\le\rho_i\le(\gamma s_i/\kappa_i)e_i(\boldsymbol{\kappa})=\mathcal{O}(1-\lambda)$ uniformly over $\boldsymbol{\kappa}\in\mathcal{K}(L)$. Together with $\|\mathbf{H}_L^{-1}\|_\infty\le 1$ and $\|\lambda\mathbf{U}_F\widehat{\mathbf{G}}_{FL}\|_\infty\le 1$, (B.37) proves (B.38) for $|L|\ge 2$. When $L=\{i\}$ (with $F=N\setminus\{i\}$), $(\mathbf{I}_{-i}-\widehat{\mathbf{G}}_{-i})\mathbf{1}_{-i}=\widehat{\mathbf{G}}_{-i,i}$ gives $1-c_i=(1-\lambda)\widehat{\mathbf{G}}_{i,-i}\mathbf{U}_F\mathbf{1}_{-i}=\mathcal{O}(1-\lambda)$, so $\gamma(1-c_i)=\mathcal{O}(1)$. By (B.37) and Lemma B.1, $x_i^S(\{i\})\le\kappa_i\le\alpha_i$, while $|\alpha_i-x_i^S(\{i\})|=\gamma(1-c_i)\big|(1-c_i)x_i^S(\{i\})-s_i\big|=\mathcal{O}(1-\lambda)$, which establishes (B.38) for $L=\{i\}$.

Let $\pi_i:=d_i/\sum_{j=1}^n d_j$, so $\sum_{i=1}^n\pi_i\widehat{g}_{ij}=\pi_j$ for all $j\in N$ and $\sum_{i=1}^n\pi_i=1$. As $\lambda\to1$, $\mathbf{x}^N\to(\sum_{j=1}^n\pi_j\alpha_j)\mathbf{1}$ and $\mathbf{x}^S(\{i\})\to\alpha_i\mathbf{1}$. For $2\le|L|<n$, boundedness of $\mathbf{x}^S(L)$ and $1-c_i(1)>0$ give $(\mathbf{I}-\widehat{\mathbf{G}})\mathbf{x}^S(L)\to\mathbf{0}$, so every subsequential limit of $\mathbf{x}^S(L)$ is proportional to $\mathbf{1}$. Weighting leader i's first-order condition $\alpha_i-x_i^S=\gamma(1-c_i)(x_i^S-\bar{x}_i^S)$ by $\pi_i/(1-c_i)$ and follower j's condition $\alpha_j-x_j^S=\gamma(x_j^S-\bar{x}_j^S)$ by π_j and summing over N yields $\lim_{\lambda\to1}\mathbf{x}^S(L)=z_L\mathbf{1}$, where

$$z_L=\frac{\displaystyle\sum_{i\in L}\frac{\pi_i}{1-c_i(1)}\alpha_i+\sum_{j\in F}\pi_j\alpha_j}{\displaystyle\sum_{i\in L}\frac{\pi_i}{1-c_i(1)}+\sum_{j\in F}\pi_j}<\overline{\alpha} \quad \text{(B.39)}$$

under heterogeneous productivities. Any singleton $\{i^*\}$ with $\alpha_{i^*}=\overline{\alpha}$ belongs to $\mathcal{F}(\lambda)$ for all $\lambda\in(0,1)$, admits a unique equilibrium in $\mathcal{K}(\{i^*\})$, and satisfies $\lim_{\lambda\to1}\mathbf{x}^I(\{i^*\};\theta)=\overline{\alpha}\theta\mathbf{1}$. By (B.38), (B.39), and $\mathbb{E}[\theta]>0$, every multi-leader group, singleton with $\alpha_i<\overline{\alpha}$, and simultaneous-move benchmark yields strictly smaller limiting expected aggregate effort uniformly across $\mathcal{K}(L)$. Finiteness of 2^N preserves these strict inequalities on a common interval $(\bar{\lambda},1)$, which proves part (ii). □

B.5.7 Proof of Corollary B.1

Proof. Part (i). In K_n, $\widehat{g}_{ij} = 1/(n-1)$ for all $j \neq i$ and $\widehat{g}_{ii} = 0$. Symmetry and row-stochasticity of $\mathbf{M}^N = (1-\lambda)(\mathbf{I} - \lambda\widehat{\mathbf{G}})^{-1}$ imply $X^N = \sum_{j=1}^n \alpha_j = n\bar{\alpha}$, so $\bar{x}_i^N = (n\bar{\alpha} - x_i^N)/(n-1)$ and $x_i^N = (1-\lambda)\alpha_i + \lambda\bar{x}_i^N$ give

$$x_i^N = \frac{(1-\lambda)(n-1)}{n-1+\lambda}\alpha_i + \frac{\lambda n}{n-1+\lambda}\bar{\alpha}, \qquad \alpha_i - x_i^N = \frac{\lambda n}{n-1+\lambda}(\alpha_i - \bar{\alpha}).$$

Thus $\alpha_i - x_i^N$ and $\alpha_i - \bar{\alpha}$ have the same sign for all $\lambda \in (0,1)$. By Proposition B.2, a single leader i admits no linear separating equilibrium when $\alpha_i < \bar{\alpha}$, satisfies $\widehat{\ell}_i = 0$ when $\alpha_i = \bar{\alpha}$, and achieves $\widehat{\ell}_i > 0$ when $\alpha_i > \bar{\alpha}$. By Proposition B.4(i) and Proposition 2 in Section 3.2, any optimal leader group L^* over $\mathcal{F}(\lambda)$ satisfies $\alpha_i > x_i^N$, and hence $\alpha_i > \bar{\alpha}$, for all $i \in L^*$.

Part (ii). Since $\bar{\alpha}_i = (n\bar{\alpha} - \alpha_i)/(n-1)$, we have $\alpha_i - \bar{\alpha}_i = \frac{n}{n-1}(\alpha_i - \bar{\alpha})$ and $N_+ = \{i \in N : \alpha_i > \bar{\alpha}\}$. For any $L \subseteq N$ with $B := L \setminus N_+ \neq \varnothing$, either $A := L \cap N_+ = \varnothing$, so $\widehat{\Lambda}(L) \leq 0 < \max_{S \subseteq N} \widehat{\Lambda}(S)$, or $A \neq \varnothing$, in which case $\widehat{g}_{ij} = 1/(n-1) > 0$ gives $\widehat{\Lambda}(A) - \widehat{\Lambda}(L) \geq \sum_{i \in A}\sum_{j \in B} \widehat{g}_{ij}(\alpha_j/\alpha_i)(\alpha_i - \bar{\alpha}_i) > 0$. Every maximizer of $\widehat{\Lambda}$ is therefore a nonempty subset of N_+, on which substituting $\widehat{g}_{ij} = 1/(n-1)$ and $\alpha_i - \bar{\alpha}_i = \frac{n}{n-1}(\alpha_i - \bar{\alpha})$ into (B.10) yields (B.11); Proposition B.5(i) then establishes the group characterization for sufficiently small conformity.

For singletons $\{i\}, \{k\} \subseteq N_+$ with $\alpha_i > \alpha_k > \bar{\alpha}$, equation (B.11) gives $\widehat{\Lambda}(\{i\}) = \frac{n}{(n-1)^2}(n\bar{\alpha} - \alpha_i)(1 - \bar{\alpha}/\alpha_i)$, and

$$\widehat{\Lambda}(\{k\}) - \widehat{\Lambda}(\{i\}) = \frac{n}{(n-1)^2}\frac{\alpha_i - \alpha_k}{\alpha_i \alpha_k}\big(\alpha_i\alpha_k - n\bar{\alpha}^2\big).$$

By part (i), no player $i \notin N_+$ achieves $\widehat{\ell}_i > 0$, whereas every $i \in N_+$ belongs to $\mathcal{F}(\lambda)$ for all $\lambda \in (0,1)$ and satisfies $\widehat{\ell}_i = \lambda\widehat{\Lambda}(\{i\}) + \mathcal{O}(\lambda^2)$. When $n \leq 4$, either $N_+ = \{1\}$, so player 1 is the unique candidate with $\widehat{\ell}_i > 0$, or $|N_+| \geq 2$ with $n \in \{3,4\}$; in the latter case, $\alpha_1 + \alpha_k < n\bar{\alpha}$ for every $k \in N_+ \setminus \{1\}$, so $\alpha_1\alpha_k \leq (\alpha_1 + \alpha_k)^2/4 < (n^2/4)\bar{\alpha}^2 \leq n\bar{\alpha}^2$, $\widehat{\Lambda}(\{1\}) > \widehat{\Lambda}(\{k\})$, and hence $\widehat{\ell}_1 > \widehat{\ell}_k$ for all sufficiently small $\lambda > 0$. When $n \geq 5$ and $\alpha_1\alpha_2 > n\bar{\alpha}^2$, positivity of $\alpha_3, \ldots, \alpha_n$ implies $\alpha_1 < n\bar{\alpha}$ and $\alpha_2 > n\bar{\alpha}^2/\alpha_1 > \bar{\alpha}$, so $\{1,2\} \subseteq N_+$, $\widehat{\Lambda}(\{2\}) > \widehat{\Lambda}(\{1\})$, and $\widehat{\ell}_2 > \widehat{\ell}_1$ for all sufficiently small $\lambda > 0$. □

B.5.8 Proof of Proposition B.6

Proof. Step 1: Simultaneous-move equilibrium, followers' responses, and norm-response bounds. In simultaneous play, player i's payoff $u_i(\mathbf{x})$ has first derivative $\alpha_i - x_i - \gamma_i(x_i - \bar{x}_i)$

and second derivative $-(1+\gamma_i) < 0$ with respect to x_i; at $x_i = 0$ with $\mathbf{x}_{-i} \geq \mathbf{0}$, the first derivative is $\alpha_i + \gamma_i \bar{x}_i > 0$, so every best response is interior and satisfies $x_i = (1-\lambda_i)\alpha_i + \lambda_i \bar{x}_i$. Since $\mathbf{W}$ is row-stochastic and $\max_{i \in N} \lambda_i < 1$, both $\mathbf{\Lambda W}$ and $\mathbf{\Lambda}_F \mathbf{W}_{FF}$ have row sums strictly below one, so $\rho(\mathbf{\Lambda W}) < 1$ and $\rho(\mathbf{\Lambda}_F \mathbf{W}_{FF}) < 1$. Irreducibility of $\mathbf{W}$ and $\lambda_i > 0$ for all $i \in N$ imply $(\mathbf{I} - \mathbf{\Lambda W})^{-1} = \sum_{t=0}^{\infty} (\mathbf{\Lambda W})^t > \mathbf{0}$, so $(\mathbf{I} - \mathbf{\Lambda W})\mathbf{x}^N = (\mathbf{I} - \mathbf{\Lambda})\boldsymbol{\alpha}$ has the unique solution $\mathbf{x}^N = \widehat{\mathbf{M}}^N \boldsymbol{\alpha} > \mathbf{0}$ with $\widehat{\mathbf{M}}^N > \mathbf{0}$ in (B.15), and $(\mathbf{I} - \mathbf{\Lambda W})\mathbf{1} = (\mathbf{I} - \mathbf{\Lambda})\mathbf{1}$ gives $\widehat{\mathbf{M}}^N \mathbf{1} = \mathbf{1}$. Likewise, $\widehat{\mathbf{U}}_F = \sum_{t=0}^{\infty} (\mathbf{\Lambda}_F \mathbf{W}_{FF})^t \geq \mathbf{I}_F$ yields the unique Stage 2 equilibrium (B.13) for any $\mathbf{x}_L \geq \mathbf{0}$ and hence (B.14). Since $\mathbf{W}_{FL}\mathbf{1}_L = (\mathbf{I}_F - \mathbf{W}_{FF})\mathbf{1}_F$, we have $\widehat{\mathbf{U}}_F \mathbf{\Lambda}_F \mathbf{W}_{FL} \mathbf{1}_L = \mathbf{1}_F - \widehat{\mathbf{U}}_F (\mathbf{I}_F - \mathbf{\Lambda}_F)\mathbf{1}_F < \mathbf{1}_F$, and the identity $\widehat{\mathbf{U}}_F \mathbf{\Lambda}_F \mathbf{W}_{FL} \mathbf{1}_L = \mathbf{\Lambda}_F \Big(\mathbf{W}_{FL}\mathbf{1}_L + \mathbf{W}_{FF} \widehat{\mathbf{U}}_F \mathbf{\Lambda}_F \mathbf{W}_{FL} \mathbf{1}_L \Big)$ implies $\widehat{\mathbf{U}}_F \mathbf{\Lambda}_F \mathbf{W}_{FL} \mathbf{1}_L \leq \mathbf{\Lambda}_F \mathbf{1}_F$. Because $\widehat{\mathbf{C}}_L = \mathbf{W}_{LL} + \mathbf{W}_{LF} \widehat{\mathbf{U}}_F \mathbf{\Lambda}_F \mathbf{W}_{FL}$ and $w_{ii} = 0$, we obtain $\widehat{\mathbf{C}}_L \mathbf{1}_L \leq \mathbf{1}_L$ and $0 \leq \widehat{c}_i \leq \max_{j \in F} \lambda_j < 1$ for all $i \in L$.

Step 2: Two-stage equilibrium and block influence matrix. Leader i's reduced Stage 1 payoff $u_i\Big(\mathbf{x}_L, \mathbf{x}_F(\mathbf{x}_L)\Big)$ has first derivative $\alpha_i - x_i - \gamma_i(1-\widehat{c}_i)\Big(x_i - \bar{x}_i(\mathbf{x}_L)\Big)$ and second derivative $-\Big[1 + \gamma_i(1-\widehat{c}_i)^2\Big] < 0$ with respect to x_i. At $x_i = 0$, we have $\bar{x}_i(\mathbf{x}_L) \geq 0$ for all $\mathbf{x}_{L\setminus\{i\}} \geq \mathbf{0}$, so the first derivative is at least $\alpha_i > 0$; every Stage 1 best response is therefore interior and satisfies $x_i + \gamma_i(1-\widehat{c}_i)\Big(x_i - \bar{x}_i(\mathbf{x}_L)\Big) = \alpha_i$. Define the $(n \times n)$ diagonal matrix $\mathbf{E}_L$ by $(\mathbf{E}_L)_{ii} := 1 - \widehat{c}_i$ for $i \in L$ and $(\mathbf{E}_L)_{ii} := 1$ for $i \in F$, and set $\mathbf{Q}_L := \mathbf{I} + \mathbf{\Gamma} \mathbf{E}_L (\mathbf{I} - \mathbf{W})$. Together with the Stage 2 system $\mathbf{x}_F + \mathbf{\Gamma}_F(\mathbf{x}_F - \bar{\mathbf{x}}_F) = \boldsymbol{\alpha}_F$, an effort profile $\mathbf{x} \geq \mathbf{0}$ forms a subgame-perfect equilibrium if and only if $\mathbf{Q}_L \mathbf{x} = \boldsymbol{\alpha}$. Factor $\mathbf{Q}_L$ as $\mathbf{Q}_L = \operatorname{diag}\Big(1 + \gamma_i (\mathbf{E}_L)_{ii}\Big)(\mathbf{I} - \mathbf{TW})$, where $\mathbf{T} := \operatorname{diag}\Big(\gamma_i (\mathbf{E}_L)_{ii} / [1 + \gamma_i (\mathbf{E}_L)_{ii}]\Big)_{i \in N}$ has diagonal entries in $(0,1)$. Irreducibility and row-stochasticity of $\mathbf{W}$ imply $(\mathbf{I} - \mathbf{TW})^{-1} = \sum_{t=0}^{\infty} (\mathbf{TW})^t > \mathbf{0}$, so $\mathbf{Q}_L^{-1} = [\mathbf{I} + \mathbf{\Gamma} \mathbf{E}_L (\mathbf{I} - \mathbf{W})]^{-1} > \mathbf{0}$, and $(\mathbf{I} - \mathbf{W})\mathbf{1} = \mathbf{0}$ yields $\mathbf{Q}_L^{-1} \mathbf{1} = \mathbf{1}$. Moreover,

$$\mathbf{I}_L - \mathbf{\Lambda}_L + \mathbf{\Lambda}_L (\mathbf{I}_L - \widehat{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{I}_L - \widehat{\mathbf{C}}_L) = (\mathbf{I}_L - \mathbf{\Lambda}_L \widehat{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{I}_L - \mathbf{H}_L),$$

where $\mathbf{H}_L := (\mathbf{I}_L - \mathbf{\Lambda}_L \widehat{\mathbf{C}}_L^{\mathrm{D}})^{-1} \mathbf{\Lambda}_L (\mathbf{I}_L - \widehat{\mathbf{C}}_L^{\mathrm{D}}) \widehat{\mathbf{C}}_L \geq \mathbf{0}$ has row sums at most $\lambda_i(1-\widehat{c}_i)/(1 - \lambda_i \widehat{c}_i) < 1$, so $\mathbf{I}_L - \mathbf{H}_L$ is invertible and $\widehat{\mathbf{V}}_L$ is well defined. With leaders ordered first, for any $\boldsymbol{\alpha} \in \mathbb{R}^n$, the F-block of $\mathbf{Q}_L \mathbf{x} = \boldsymbol{\alpha}$ gives (B.13), and substituting (B.14) into the L-block $(\mathbf{I}_L - \mathbf{\Lambda}_L)\mathbf{x}_L + \mathbf{\Lambda}_L (\mathbf{I}_L - \widehat{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{x}_L - \bar{\mathbf{x}}_L) = (\mathbf{I}_L - \mathbf{\Lambda}_L)\boldsymbol{\alpha}_L$ and back into (B.13) yields $\mathbf{x} = \widehat{\mathbf{M}}^S(L)\boldsymbol{\alpha}$ with $\widehat{\mathbf{M}}^S(L)$ in (B.16). Hence $\widehat{\mathbf{M}}^S(L) = \mathbf{Q}_L^{-1} > \mathbf{0}$ is row-stochastic, and $\mathbf{x}^S(L) = \widehat{\mathbf{M}}^S(L)\boldsymbol{\alpha} > \mathbf{0}$ is the unique subgame-perfect Nash equilibrium. □

B.5.9 Proof of Proposition B.7

Proof. Step 1: Leader eligibility. For any leader group L, let $\widehat{c}_i(L)$ denote leader i's norm response under L, and let $\mathbf{E}_L$ and $\mathbf{Q}_L = \mathbf{I} + \boldsymbol{\Gamma}\mathbf{E}_L(\mathbf{I} - \mathbf{W})$ be defined as in Step 2 of the proof of Proposition B.6. Since $0 \leq \widehat{c}_i(L) < 1$ for all $i \in L$, $\mathbf{E}_L$ is invertible, and $\boldsymbol{\Gamma}\mathbf{E}_L(\mathbf{I} - \mathbf{W})\mathbf{x}^S(L) = \boldsymbol{\alpha} - \mathbf{x}^S(L)$ gives $\boldsymbol{\Gamma}(\mathbf{I} - \mathbf{W})\mathbf{x}^S(L) = \mathbf{E}_L^{-1}\big(\boldsymbol{\alpha} - \mathbf{x}^S(L)\big)$. Subtracting $\boldsymbol{\Gamma}(\mathbf{I} - \mathbf{W})\mathbf{x}^N = \boldsymbol{\alpha} - \mathbf{x}^N$ and using $\widehat{\mathbf{M}}^N = [\mathbf{I} + \boldsymbol{\Gamma}(\mathbf{I} - \mathbf{W})]^{-1}$ yields

$$\mathbf{x}^S(L) - \mathbf{x}^N = \widehat{\mathbf{M}}^N\big(\mathbf{E}_L^{-1} - \mathbf{I}\big)\big(\boldsymbol{\alpha} - \mathbf{x}^S(L)\big) = \sum_{i \in L} \frac{\widehat{c}_i(L)}{1 - \widehat{c}_i(L)} \widehat{\mathbf{M}}^N_{\cdot i}\big(\alpha_i - x_i^S(L)\big). \quad \text{(B.40)}$$

Because $\widehat{\mathbf{M}}^N > \mathbf{0}$ is row-stochastic and productivities are heterogeneous, any player i with the highest productivity satisfies $\alpha_i > x_i^N$, so $\ell_i > 0$ by (B.18), and every optimal leader group L^* satisfies $\ell(L^*) > 0$.

For an optimal leader group L^* of minimum cardinality, set $B := \{i \in L^* : \alpha_i \leq x_i^S(L^*)\}$ and $A := L^* \setminus B$. Since $\ell(L^*) > 0$, $\widehat{c}_i(L^*) \in [0, 1)$, and $\widehat{\mathbf{M}}^N > \mathbf{0}$, identity (B.40) implies $0 < |A| \leq |L^*| < n$. Reassigning B to Stage 2 preserves every closed directed walk from $i \in A$ to i through $N \setminus L^*$ and adds only nonnegative walk weights in (B.19), so $\widehat{c}_i(A) \geq \widehat{c}_i(L^*)$ for all $i \in A$. Using $\mathbf{Q}_{L^*}\mathbf{x}^S(L^*) = \boldsymbol{\alpha}$, we obtain for each $i \in N$,

$$\big(\mathbf{Q}_A\mathbf{x}^S(L^*) - \boldsymbol{\alpha}\big)_i = \gamma_i\big[(\mathbf{E}_A)_{ii} - (\mathbf{E}_{L^*})_{ii}\big]\big(x_i^S(L^*) - \bar{x}_i^S(L^*)\big) \leq 0 :$$

for $i \in B$, $(\mathbf{E}_A)_{ii} - (\mathbf{E}_{L^*})_{ii} = \widehat{c}_i(L^*) \geq 0$ and $\gamma_i\big(1 - \widehat{c}_i(L^*)\big)\big(x_i^S(L^*) - \bar{x}_i^S(L^*)\big) = \alpha_i - x_i^S(L^*) \leq 0$; for $i \in A$, $(\mathbf{E}_A)_{ii} - (\mathbf{E}_{L^*})_{ii} = \widehat{c}_i(L^*) - \widehat{c}_i(A) \leq 0$ and $x_i^S(L^*) - \bar{x}_i^S(L^*) > 0$; for $i \in F^*$, $(\mathbf{E}_A)_{ii} = (\mathbf{E}_{L^*})_{ii} = 1$. Since $\mathbf{Q}_A^{-1} = \widehat{\mathbf{M}}^S(A) > \mathbf{0}$, we obtain $\mathbf{x}^S(A) \geq \mathbf{x}^S(L^*)$ and $\ell(A) \geq \ell(L^*)$. Minimality of $|L^*|$ gives $B = \varnothing$, so $\alpha_i > x_i^S(L^*)$ for all $i \in L^*$.

Step 2: Directed cycles and effort comparisons. Suppose, to the contrary, that $\widehat{c}_k(L^*) = 0$ for some $k \in L^*$. Since $\ell(L^*) > 0$, identity (B.40) implies $|L^*| \geq 2$, so $L' := L^* \setminus \{k\}$ satisfies $0 < |L'| < n$. Because $(\mathbf{E}_{L'})_{kk} = 1 = (\mathbf{E}_{L^*})_{kk}$ and $\widehat{c}_i(L') \geq \widehat{c}_i(L^*)$ for all $i \in L'$, the residual inequality above gives $\mathbf{Q}_{L'}\mathbf{x}^S(L^*) - \boldsymbol{\alpha} \leq \mathbf{0}$ and hence $\mathbf{x}^S(L') \geq \mathbf{x}^S(L^*)$, contradicting the minimum cardinality of L^*. Thus $\widehat{c}_i(L^*) > 0$ for all $i \in L^*$; by (B.19), a shortest positive-weight closed directed walk from i to i with intermediate players in F^* has distinct intermediate players and forms a directed cycle in $\mathbf{W}$ whose other players lie in F^*. Finally, $0 < \widehat{c}_i(L^*) < 1$, $\alpha_i > x_i^S(L^*)$ for all $i \in L^*$, and $\widehat{\mathbf{M}}^N > \mathbf{0}$ in (B.40) yield $\mathbf{x}^S(L^*) > \mathbf{x}^N$, which establishes (B.20) and $\ell_i > 0$ via (B.18). □

B.5.10 Proof of Proposition B.8

Proof. Part (i): Weak conformity. As $\gamma \to 0$, $\lambda_i = \gamma\nu_i/(1+\gamma\nu_i) = \gamma\nu_i + \mathcal{O}(\gamma^2)$, so equations (B.15) and (B.19) yield

$$\alpha_i - x_i^N = \gamma\nu_i(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\gamma^2), \qquad \hat{c}_i = \gamma \sum_{j\in F} \nu_j w_{ij} w_{ji} + \mathcal{O}(\gamma^2) \quad (i \in L),$$

while $\widehat{\mathbf{M}}^S(L) = \mathbf{I} + \mathcal{O}(\gamma)$. Summing (B.17) over $j = 1, \ldots, n$ establishes (B.22), $\ell(L) = \gamma^2\Phi(L) + \mathcal{O}(\gamma^3)$, uniformly across all leader groups L. Because $\mathbf{W}$ is irreducible and row-stochastic and productivities are heterogeneous, some player $i \in \arg\max_{k\in N} \alpha_k$ places positive weight on a lower-productivity player and belongs to N_+, whereas $\arg\min_{k\in N} \alpha_k \subseteq N \setminus N_+$; hence $0 < |N_+| < n$ and $\max_{0<|L|<n} \Phi(L) \geq \Phi(\{i\}) \geq 0 = \Phi(\varnothing) = \Phi(N)$. Pick $L_1 \in \arg\max_{0<|L|<n} \Phi(L)$. Since there are finitely many leader groups, every group L with $\Phi(L) < \Phi(L_1)$ satisfies $\ell(L) < \ell(L_1)$ for all sufficiently small $\gamma > 0$, so any optimal leader group maximizes $\Phi(L)$. Moreover, for any $L \subseteq N$ with $A := L \cap N_+$ and $B := L \setminus N_+$, we have $\Phi(A) - \Phi(L) = \sum_{i\in A}\sum_{j\in B} \nu_i\nu_j w_{ij}w_{ji}(\alpha_i - \bar{\alpha}_i) - \sum_{i\in B}\sum_{j\in F} \nu_i\nu_j w_{ij}w_{ji}(\alpha_i - \bar{\alpha}_i) \geq 0$, and for $S \subseteq N_+$ and distinct $i, j \in N_+ \setminus S$, $\Phi(S\cup\{i,j\}) - \Phi(S\cup\{i\}) - \Phi(S\cup\{j\}) + \Phi(S) = -\nu_i\nu_j w_{ij}w_{ji}\big[(\alpha_i - \bar{\alpha}_i) + (\alpha_j - \bar{\alpha}_j)\big] \leq 0$, so Φ is submodular on 2^{N_+}.

Part (ii): Strong conformity. Let $\underline{\alpha} := \min_{k\in N} \alpha_k$ and $\overline{\alpha} := \max_{k\in N} \alpha_k$. Row-stochasticity of $\widehat{\mathbf{M}}^N$ and $\widehat{\mathbf{M}}^S(L)$ gives $\underline{\alpha}\mathbf{1} \leq \mathbf{x}^N, \mathbf{x}^S(L) \leq \overline{\alpha}\mathbf{1}$ for all $\gamma > 0$. As $\gamma \to \infty$, the simultaneous-move system $\gamma\,\mathrm{diag}(\nu_k)(\mathbf{I} - \mathbf{W})\mathbf{x}^N = \boldsymbol{\alpha} - \mathbf{x}^N$ implies $(\mathbf{I} - \mathbf{W})\mathbf{x}^N = \mathcal{O}(\gamma^{-1})$, so irreducibility of $\mathbf{W}$ ensures that every subsequential limit of $\mathbf{x}^N$ is proportional to $\mathbf{1}$; premultiplying by $\boldsymbol{\pi}^\top \mathrm{diag}(\nu_k^{-1})$ gives $\sum_{j=1}^n (\pi_j/\nu_j)x_j^N = \sum_{j=1}^n (\pi_j/\nu_j)\alpha_j$ for all $\gamma > 0$, which proves the simultaneous-move limit in (B.23).

For a singleton $L = \{i\}$ with $F = N \setminus \{i\}$, irreducibility of $\mathbf{W}$ implies that $(\mathbf{I}_{-i} - \mathbf{W}_{-i})^{-1}$ exists and $\widehat{\mathbf{U}}_F \to (\mathbf{I}_{-i} - \mathbf{W}_{-i})^{-1}$ as $\gamma \to \infty$. Since $\mathbf{I}_{-i} - \boldsymbol{\Lambda}_{-i} = \mathcal{O}(\gamma^{-1})$, the intercept $a_i := \mathbf{W}_{i,-i}\widehat{\mathbf{U}}_F(\mathbf{I}_{-i} - \boldsymbol{\Lambda}_{-i})\boldsymbol{\alpha}_{-i}$ in (B.14) satisfies $a_i = \mathcal{O}(\gamma^{-1})$, while $\widehat{\mathbf{U}}_F\boldsymbol{\Lambda}_{-i}\mathbf{W}_{-i,i} = \mathbf{1}_{-i} - \widehat{\mathbf{U}}_F(\mathbf{I}_{-i} - \boldsymbol{\Lambda}_{-i})\mathbf{1}_{-i} = \mathbf{1}_{-i} + \mathcal{O}(\gamma^{-1})$ and $\mathbf{W}_{i,-i}\mathbf{1}_{-i} = 1$ give $1 - \hat{c}_i = \mathcal{O}(\gamma^{-1})$. Leader i's first-order condition $\big[1 + \gamma\nu_i(1-\hat{c}_i)^2\big]x_i^S(\{i\}) = \alpha_i + \gamma\nu_i(1-\hat{c}_i)a_i$ therefore yields $x_i^S(\{i\}) = \alpha_i + \mathcal{O}(\gamma^{-1})$, and (B.13) implies $\lim_{\gamma\to\infty} \mathbf{x}^S(\{i\}) = \alpha_i\mathbf{1}$ in (B.23).

For $|L| \geq 2$, as $\gamma \to \infty$, $\widehat{\mathbf{C}}_L$ converges to $\mathbf{W}_{LL} + \mathbf{W}_{LF}(\mathbf{I}_F - \mathbf{W}_{FF})^{-1}\mathbf{W}_{FL}$, whose rows sum to one. By irreducibility of $\mathbf{W}$, a shortest directed path from each $i \in L$ to $L \setminus \{i\}$ has all intermediate players in F and places positive limiting weight on some $k \in L\setminus\{i\}$; hence $p_i := \lim_{\gamma\to\infty} \hat{c}_i \in [0, 1)$, and every diagonal entry $(\mathbf{E}_L)_{kk}$ from the proof

of Proposition B.6 remains bounded away from zero for large γ. From $\gamma \operatorname{diag}(\nu_k)\mathbf{E}_L(\mathbf{I}-\mathbf{W})\mathbf{x}^S(L) = \boldsymbol{\alpha} - \mathbf{x}^S(L)$, we obtain $(\mathbf{I}-\mathbf{W})\mathbf{x}^S(L) = \mathcal{O}(\gamma^{-1})$, and premultiplying by $\boldsymbol{\pi}^\top \operatorname{diag}\Big[(\nu_k(\mathbf{E}_L)_{kk})^{-1}\Big]$ gives $\sum_{k=1}^n \Big[\pi_k/\Big(\nu_k(\mathbf{E}_L)_{kk}\Big)\Big] x_k^S(L) = \sum_{k=1}^n \Big[\pi_k/\Big(\nu_k(\mathbf{E}_L)_{kk}\Big)\Big]\alpha_k$, so $\lim_{\gamma\to\infty} \mathbf{x}^S(L) = z_L\mathbf{1}$ with

$$z_L = \frac{\displaystyle\sum_{i\in L} \frac{\pi_i\alpha_i}{\nu_i(1-p_i)} + \sum_{j\in F} \frac{\pi_j\alpha_j}{\nu_j}}{\displaystyle\sum_{i\in L} \frac{\pi_i}{\nu_i(1-p_i)} + \sum_{j\in F} \frac{\pi_j}{\nu_j}}. \tag{B.41}$$

Under heterogeneous productivities, (B.41) places strictly positive weight on every player's productivity and satisfies $z_L < \overline{\alpha}$. Thus $\lim_{\gamma\to\infty} X^S(L) = nz_L < n\overline{\alpha}$ for every multi-leader group L and $\lim_{\gamma\to\infty} X^S(\{i\}) = n\alpha_i < n\overline{\alpha}$ whenever $\alpha_i < \overline{\alpha}$, whereas any singleton $\{i^*\}$ with $\alpha_{i^*} = \overline{\alpha}$ satisfies $\lim_{\gamma\to\infty} X^S(\{i^*\}) = n\overline{\alpha}$. Since there are finitely many leader groups, for all sufficiently large γ an optimal leader group is a singleton consisting of a player with the highest productivity. □

B.5.11 Proof of Proposition B.9

Proof. Let $\underline{\alpha} := \min_{i\in N}\alpha_i > 0$ and $\overline{\alpha} := \max_{i\in N}\alpha_i > 0$, and consider $\chi \in (0,1)$, so $\xi = \chi/(1+\chi) \in (0,\chi)$ and $\xi(1+\chi) = \chi$. For any leader group L, Steps 1 and 2 hold for all $\chi \in (0,1)$: Step 1 establishes unique Stage 2 play after any Stage 1 effort profile, and Step 2 bounds Stage 1 best responses globally on $[0,\infty)$. Step 3 then shows that for all sufficiently small $\chi > 0$, the two-stage and simultaneous-move games admit the unique interior equilibria in (B.26) and (B.27).

Step 1: Follower subgames after any Stage 1 effort profile. Since $0 < \xi < 1$, payoff (B.25) is strictly concave in x_k with unique best response $\max\{0, (1+\chi)\alpha_k - \chi\bar{x}_k\}$ on $[0,\infty)$. For any $\mathbf{x}_L \in [0,\infty)^{|L|}$, the Stage 2 best-response map $T_F(\cdot;\mathbf{x}_L)$ on $[0,\infty)^{|F|}$, given for each $j \in F$ by

$$T_{F,j}(\mathbf{x}_F;\mathbf{x}_L) := \max\Big\{0, (1+\chi)\alpha_j - \chi(\widehat{\mathbf{G}}_{FL}\mathbf{x}_L)_j - \chi(\widehat{\mathbf{G}}_{FF}\mathbf{x}_F)_j\Big\},$$

maps $[0,\infty)^{|F|}$ into $[0,(1+\chi)\overline{\alpha}]^{|F|}$ and satisfies $\|T_F(\mathbf{x}_F;\mathbf{x}_L) - T_F(\mathbf{x}'_F;\mathbf{x}_L)\|_\infty \le \chi\|\mathbf{x}_F - \mathbf{x}'_F\|_\infty$ with $\chi < 1$. Every follower subgame therefore admits a unique Nash equilibrium $\mathbf{x}_F^*(\mathbf{x}_L) \in [0,(1+\chi)\overline{\alpha}]^{|F|}$. For any $\mathbf{x}_L, \mathbf{x}'_L \in [0,\infty)^{|L|}$, row-stochasticity of $\widehat{\mathbf{G}}$ gives

$$\|\mathbf{x}_F^*(\mathbf{x}_L) - \mathbf{x}_F^*(\mathbf{x}'_L)\|_\infty \le \chi \max\Big\{\|\mathbf{x}_L - \mathbf{x}'_L\|_\infty,\ \|\mathbf{x}_F^*(\mathbf{x}_L) - \mathbf{x}_F^*(\mathbf{x}'_L)\|_\infty\Big\}.$$

Since $\chi < 1$, this inequality yields $\|\mathbf{x}_F^*(\mathbf{x}_L) - \mathbf{x}_F^*(\mathbf{x}'_L)\|_\infty \le \chi \|\mathbf{x}_L - \mathbf{x}'_L\|_\infty$. Because $\widehat{g}_{ii} = 0$, leader i's induced social norm $\bar{x}_i^*(\mathbf{x}_L) := \sum_{h \in L\setminus\{i\}} \widehat{g}_{ih} x_h + \sum_{j \in F} \widehat{g}_{ij} x_j^*(\mathbf{x}_L)$ is Lipschitz in x_i with constant at most χ.

Step 2: Global upper bound on leaders' best responses. Fix $i \in L$ and $\mathbf{x}_{L\setminus\{i\}} \in [0,\infty)^{|L|-1}$, and set $\tilde{u}_i(x_i) := u_i\Big(x_i, \mathbf{x}_{L\setminus\{i\}}, \mathbf{x}_F^*(x_i, \mathbf{x}_{L\setminus\{i\}})\Big)$, $q(x_i) := x_i - \bar{x}_i^*(x_i, \mathbf{x}_{L\setminus\{i\}})$, and $R := \overline{\alpha}/(1-\chi) > (1+\chi)\overline{\alpha}$. For any $x_i > R$, Step 1 implies $0 < (1-\chi)(x_i - R) \le q(x_i) - q(R) \le (1+\chi)(x_i - R)$, while $\bar{x}_i^* \ge 0$ implies $q(x_i) + q(R) \le x_i + R$. Hence

$$\begin{aligned} q(x_i)^2 - q(R)^2 &= \Big[q(x_i) - q(R)\Big]\Big[q(x_i) + q(R)\Big] \\ &\le \Big[q(x_i) - q(R)\Big] \max\{0, q(x_i) + q(R)\} \le (1+\chi)(x_i^2 - R^2). \end{aligned}$$

Using $\xi(1+\chi) = \chi$ and $(x_i + R)/2 > R$ for $x_i > R$, we obtain

$$\begin{aligned} \tilde{u}_i(x_i) - \tilde{u}_i(R) &= \alpha_i(x_i - R) - \frac{x_i^2 - R^2}{2} + \frac{\xi}{2}\Big[q(x_i)^2 - q(R)^2\Big] \\ &\le \overline{\alpha}(x_i - R) - \frac{1-\chi}{2}(x_i^2 - R^2) < \Big[\overline{\alpha} - (1-\chi)R\Big](x_i - R) = 0. \end{aligned}$$

Since $\tilde{u}_i$ is continuous on $[0,\infty)$, $\tilde{u}_i$ attains a global maximum on $[0,\infty)$ and every global best response of leader i lies in $[0, R]$.

Step 3: Unique interior equilibrium and influence matrices. For any $\mathbf{x}_L \in [0,R]^{|L|}$, Step 1 gives $\mathbf{x}_F^*(\mathbf{x}_L) \in [0, (1+\chi)\overline{\alpha}]^{|F|} \subset [0,R]^{|F|}$, so every player's social norm lies in $[0,R]$ and $(1+\chi)\alpha_j - \chi\bar{x}_j \ge (1+\chi)\underline{\alpha} - \chi R$ for all $j \in F$. Because $R \to \overline{\alpha}$ as $\chi \to 0$, $(1+\chi)\underline{\alpha} - \chi R > 0$ holds for all sufficiently small $\chi > 0$. Throughout $[0,R]^{|L|}$, followers' equilibrium efforts and leaders' induced social norms therefore satisfy

$$\begin{aligned} \mathbf{x}_F^*(\mathbf{x}_L) = \mathbf{x}_F(\mathbf{x}_L) &= \widetilde{\mathbf{U}}_F\Big[(1+\chi)\boldsymbol{\alpha}_F - \chi\widehat{\mathbf{G}}_{FL}\mathbf{x}_L\Big] > \mathbf{0}, \\ \bar{\mathbf{x}}_L(\mathbf{x}_L) &= (1+\chi)\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\boldsymbol{\alpha}_F + \widetilde{\mathbf{C}}_L\mathbf{x}_L. \end{aligned}$$

By Step 1, $|\tilde{c}_i| \le \chi$, while $\widetilde{\mathbf{U}}_F = \sum_{t=0}^{\infty}(-\chi)^t\widehat{\mathbf{G}}_{FF}^t$ satisfies $\|\widetilde{\mathbf{U}}_F\|_\infty \le (1-\chi)^{-1}$ and gives $\|\widetilde{\mathbf{C}}_L\|_\infty \le 1 + \chi(1-\chi)^{-1} = (1-\chi)^{-1}$. For each $i \in L$ and $\mathbf{x}_{L\setminus\{i\}} \in [0,R]^{|L|-1}$, $0 \le R - \bar{x}_i(R, \mathbf{x}_{L\setminus\{i\}}) \le R$ and $0 < 1 - \tilde{c}_i \le 1 + \chi$ imply that $\tilde{u}_i$ on $[0,R]$ is quadratic with

$$\begin{aligned} &\tilde{u}_i''(x_i) = -\Big[1 - \xi(1-\tilde{c}_i)^2\Big] \le -\Big[1 - \chi(1+\chi)\Big], \\ &\tilde{u}_i'(0) \ge \underline{\alpha} - \xi(1+\chi)R = \underline{\alpha} - \chi R, \qquad \tilde{u}_i'(R) \le \overline{\alpha} - (1-\chi)R = 0. \end{aligned}$$

Since $1-\chi(1+\chi)\to 1$ and $\underline{\alpha}-\chi R\to\underline{\alpha}>0$ as $\chi\to 0$, for all sufficiently small $\chi>0$ we have $\tilde{u}_i''(x_i)<0$ on $[0,R]$ and $\tilde{u}_i'(0)>0\geq\tilde{u}_i'(R)$. Together with Step 2, leader i's unique global best response $x_i^*(\mathbf{x}_{L\setminus\{i\}})$ on $[0,\infty)$ lies in $(0,R]$, satisfies $\tilde{u}_i'(x_i^*)=0$, and solves

$$\Big[1-\xi(1-\widetilde{c}_i)^2\Big]x_i^*=\alpha_i-\xi(1-\widetilde{c}_i)\Big[(1+\chi)(\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\boldsymbol{\alpha}_F)_i+\sum_{h\in L\setminus\{i\}}\widetilde{c}_{ih}x_h\Big].$$

Since $\sum_{h\in L\setminus\{i\}}|\widetilde{c}_{ih}|\leq\|\widetilde{\mathbf{C}}_L\|_\infty\leq(1-\chi)^{-1}$, each row sum of absolute partial derivatives of $\mathbf{x}_L^*$ is at most $\chi/\big\{(1-\chi)[1-\chi(1+\chi)]\big\}$, which converges to 0 as $\chi\to 0$. For all sufficiently small $\chi>0$, $\mathbf{x}_L^*$ is therefore a $\|\cdot\|_\infty$-contraction mapping $[0,R]^{|L|}$ into $(0,R]^{|L|}$ with a unique fixed point $\mathbf{x}_L^S>\mathbf{0}$, and Step 2 rules out equilibria or profitable deviations outside $[0,R]^{|L|}$. Multiplying the first-order system by $1+\chi=(1-\xi)^{-1}$ gives

$$\begin{aligned}&\Big[(1+\chi)\mathbf{I}_L-\chi(\mathbf{I}_L-\widetilde{\mathbf{C}}_L^{\mathrm{D}})(\mathbf{I}_L-\widetilde{\mathbf{C}}_L)\Big]\mathbf{x}_L^S\\&\qquad=(1+\chi)\Big[\boldsymbol{\alpha}_L-\chi(\mathbf{I}_L-\widetilde{\mathbf{C}}_L^{\mathrm{D}})\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\boldsymbol{\alpha}_F\Big],\end{aligned}$$

where strict row diagonal dominance for sufficiently small $\chi>0$ makes the left-hand matrix invertible with inverse $\widetilde{\mathbf{V}}_L$; substituting $\mathbf{x}_L^S$ into $\mathbf{x}_F^S=\mathbf{x}_F(\mathbf{x}_L^S)>\mathbf{0}$ yields (B.27).

Finally, in simultaneous play on $[0,\infty)^n$, the best-response map $\mathbf{x}\mapsto\max\{\mathbf{0},(1+\chi)\boldsymbol{\alpha}-\chi\widehat{\mathbf{G}}\mathbf{x}\}$ maps $[0,\infty)^n$ into $[0,(1+\chi)\overline{\alpha}]^n\subset[0,R]^n$ with $\|\cdot\|_\infty$-Lipschitz constant at most $\chi<1$, and $(1+\chi)\underline{\alpha}-\chi R>0$ ensures for all sufficiently small $\chi>0$ that the unique fixed point is interior and equals $\mathbf{x}^N=\widetilde{\mathbf{M}}^N\boldsymbol{\alpha}>\mathbf{0}$ with $\widetilde{\mathbf{M}}^N$ in (B.26). Row-stochasticity $\widehat{\mathbf{G}}\mathbf{1}=\mathbf{1}$ gives $(\mathbf{I}+\chi\widehat{\mathbf{G}})\mathbf{1}=(1+\chi)\mathbf{1}$ and $\widetilde{\mathbf{M}}^N\mathbf{1}=\mathbf{1}$, while $(1+\chi)\widetilde{\mathbf{U}}_F\mathbf{1}_F-\chi\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL}\mathbf{1}_L=\mathbf{1}_F$ implies $(1+\chi)\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\mathbf{1}_F=(\mathbf{I}_L-\widetilde{\mathbf{C}}_L)\mathbf{1}_L$ and $\widetilde{\mathbf{M}}^S(L)\mathbf{1}=\mathbf{1}$. □

B.5.12 Proof of Proposition B.10

Proof. Consider a single leader $L=\{i\}$ with $F=N\setminus\{i\}$ for sufficiently small $\chi>0$.

Step 1: Norm response and propagation multiplier. We first show that $\widetilde{c}_i<0$. Let $\mathbf{D}:=\mathrm{diag}(d_1,\ldots,d_n)$ denote the degree matrix of g, and set $\mathbf{S}:=\mathbf{D}^{1/2}\widehat{\mathbf{G}}\mathbf{D}^{-1/2}$. Because g is undirected, $\mathbf{S}$ is symmetric with eigenvalues in $[-1,1]$, so $\mathbf{I}_{-i}+\chi\mathbf{S}_{-i}$ is positive definite for $0<\chi<1$. With $\mathbf{v}:=\mathbf{S}_{-i,i}\neq\mathbf{0}$ by connectedness of g, similarity between $\widehat{\mathbf{G}}$ and $\mathbf{S}$ yields

$$-\frac{\widetilde{c}_i}{\chi}=\widehat{\mathbf{G}}_{i,-i}(\mathbf{I}_{-i}+\chi\widehat{\mathbf{G}}_{-i})^{-1}\widehat{\mathbf{G}}_{-i,i}=\mathbf{v}^\top(\mathbf{I}_{-i}+\chi\mathbf{S}_{-i})^{-1}\mathbf{v}>0,$$

so $\widetilde{c}_i < 0$. By Step 1 of the proof of Proposition B.9, $|\widetilde{c}_i| \leq \chi < 1$, so $1 + \chi\widetilde{c}_i > 0$. In column i of $(\mathbf{I} + \chi\widehat{\mathbf{G}})\widetilde{\mathbf{M}}^N = (1+\chi)\mathbf{I}$, the follower rows and row i give

$$\widetilde{\mathbf{M}}^N_{-i,i} = -\chi\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{-i,i}\widetilde{m}^N_{ii}, \qquad \widetilde{m}^N_{ii} = \frac{1+\chi}{1+\chi\widetilde{c}_i} > 1 + \chi > 1,$$

which establishes (B.28) and $\partial\mathbf{x}_{-i}(x_i)/\partial x_i = -\chi\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{-i,i} = \widetilde{\mathbf{M}}^N_{-i,i}/\widetilde{m}^N_{ii}$. Since $\widetilde{\mathbf{U}}_F = (\mathbf{I}_{-i} + \chi\widehat{\mathbf{G}}_{-i})^{-1} = \mathbf{I}_{-i} + \mathcal{O}(\chi)$ as $\chi \to 0$ and $\sum_{j\neq i}\widehat{g}_{ji} > 0$ by connectedness of g,

$$\begin{aligned}\frac{\sum_{j=1}^n \widetilde{m}^N_{ji}}{\widetilde{m}^N_{ii}} &= 1 + \sum_{j\neq i}\frac{\widetilde{m}^N_{ji}}{\widetilde{m}^N_{ii}} = 1 - \chi\sum_{j\neq i}(\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{-i,i})_j \\ &= 1 - \chi\sum_{j\neq i}\widehat{g}_{ji} + \mathcal{O}(\chi^2) \in (0,1)\end{aligned}$$

for all sufficiently small $\chi > 0$.

Step 2: Single-leader gain and sign equivalences. By Step 3 of the proof of Proposition B.9, both $\mathbf{x}^N$ and $\mathbf{x}^S(\{i\})$ belong to $[0,R]^n$, where $\mathbf{x}_{-i}(x_i)$ is affine and leader i's reduced payoff $\tilde{u}_i(x_i)$ is quadratic with second derivative $-\left[1-\xi(1-\widetilde{c}_i)^2\right] < 0$ and $\tilde{u}_i'\left(x_i^S(\{i\})\right) = 0$. At $x_i = x_i^N$, $\mathbf{x}_{-i}(x_i^N) = \mathbf{x}^N_{-i}$ and $\alpha_i - x_i^N + \xi(x_i^N - \bar{x}_i^N) = 0$ give $\tilde{u}_i'(x_i^N) = \alpha_i - x_i^N + \xi(1-\widetilde{c}_i)(x_i^N - \bar{x}_i^N) = \widetilde{c}_i(\alpha_i - x_i^N)$. Since $(1+\chi)\left[1-\xi(1-\widetilde{c}_i)^2\right] = 1 + \chi\widetilde{c}_i(2-\widetilde{c}_i)$, we obtain

$$x_i^S(\{i\}) - x_i^N = \frac{\widetilde{c}_i(\alpha_i - x_i^N)}{1-\xi(1-\widetilde{c}_i)^2} = \frac{(1+\chi)\widetilde{c}_i}{1+\chi\widetilde{c}_i(2-\widetilde{c}_i)}(\alpha_i - x_i^N).$$

Summing $x_j^S(\{i\}) - x_j^N = (\widetilde{m}^N_{ji}/\widetilde{m}^N_{ii})\left[x_i^S(\{i\}) - x_i^N\right]$ across $j \in N$ establishes (B.29). Because $\widetilde{c}_i < 0$, $1-\xi(1-\widetilde{c}_i)^2 > 0$, and $\left(\sum_{j=1}^n \widetilde{m}^N_{ji}\right)/\widetilde{m}^N_{ii} > 0$, equation (B.29) implies $\ell_i > 0$ if and only if $x_i^N > \alpha_i$, which is equivalent to $\alpha_i > \bar{x}_i^N$ since $x_i^N - \alpha_i = \chi(\alpha_i - \bar{x}_i^N)$. □

B.5.13 Proof of Proposition B.11

Proof. For sufficiently small $\chi > 0$ and any leader group L, define the $(n \times n)$ diagonal matrix $\mathbf{E}_L$ by $(\mathbf{E}_L)_{ii} := 1 - \widetilde{c}_i$ for $i \in L$ and $(\mathbf{E}_L)_{ii} := 1$ for $i \in F$, and set $\mathbf{Q}_L := \mathbf{I} - \xi\mathbf{E}_L(\mathbf{I} - \widehat{\mathbf{G}})$. By Proposition B.9, $\mathbf{Q}_L\mathbf{x}^S(L) = \boldsymbol{\alpha}$ and $-\xi(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{x}^N = \boldsymbol{\alpha} - \mathbf{x}^N$, so subtracting these two systems gives

$$\mathbf{Q}_L\left(\mathbf{x}^S(L) - \mathbf{x}^N\right) = (\mathbf{I} - \mathbf{E}_L)(\boldsymbol{\alpha} - \mathbf{x}^N).$$

As $\chi \to 0$, $\xi = \chi + \mathcal{O}(\chi^2)$, $\mathbf{Q}_L^{-1} = \mathbf{I} + \mathcal{O}(\chi)$, $\widetilde{\mathbf{M}}^N = \mathbf{I} + \chi(\mathbf{I} - \widehat{\mathbf{G}}) + \mathcal{O}(\chi^2)$, and $\widetilde{\mathbf{U}}_F = \mathbf{I}_F + \mathcal{O}(\chi)$. Hence $\alpha_i - x_i^N = -\chi(\alpha_i - \bar{\alpha}_i) + \mathcal{O}(\chi^2)$ for all $i \in N$ and $\widetilde{c}_i =$

$-\chi(\widehat{\mathbf{G}}_{LF}\widetilde{\mathbf{U}}_F\widehat{\mathbf{G}}_{FL})_{ii} = -\chi\sum_{j\in F} r_{ij} + \mathcal{O}(\chi^2)$ for all $i \in L$, which yields $\mathbf{x}^S(L) - \mathbf{x}^N = (\mathbf{I} - \mathbf{E}_L)(\boldsymbol{\alpha} - \mathbf{x}^N) + \mathcal{O}(\chi^3)$ and

$$\begin{aligned}\ell(L) &= \sum_{i=1}^{n}\Big[x_i^S(L) - x_i^N\Big] = \sum_{i\in L}\widetilde{c}_i(\alpha_i - x_i^N) + \mathcal{O}(\chi^3)\\ &= \chi^2\Lambda(L) + \mathcal{O}(\chi^3) = \xi^2\Lambda(L) + \mathcal{O}(\xi^3),\end{aligned}$$

with $\Lambda(L)$ in (B.30).

Because g is connected and undirected and productivities are heterogeneous, some player $i \in \arg\max_{k\in N}\alpha_k$ has a lower-productivity neighbor, so $r_{ij} > 0$ for every neighbor j of i gives $\Lambda(\{i\}) = \sum_{j\neq i} r_{ij}(\alpha_i - \bar{\alpha}_i) > 0 = \Lambda(\varnothing) = \Lambda(N)$ and $\max_{0<|L|<n}\Lambda(L) = \max_{L\subseteq N}\Lambda(L) > 0$. Pick $L_1 \in \arg\max_{0<|L|<n}\Lambda(L)$. Since there are finitely many leader groups, the $\mathcal{O}(\chi^3)$ remainder above is uniform across $0 < |L| < n$, so every group L with $\Lambda(L) < \Lambda(L_1)$ satisfies $\ell(L) < \ell(L_1)$ for all sufficiently small $\chi > 0$; hence an optimal leader group maximizes $\Lambda(L)$. □

B.5.14 Proof of Proposition B.12

Proof. We first characterize the simultaneous-move equilibrium, Stage 2 follower responses, and Stage 1 best responses under $\alpha_i = \alpha$ for all $i \in N$, and then prove parts (i)–(iii).

Simultaneous-move equilibrium and two-stage reduction. Payoff (B.31) with $\alpha_i = \alpha$ is strictly concave in x_i with marginal payoff $\alpha-(1+\gamma)x_i+\gamma\beta\bar{x}_i$. Since $\alpha > 0$ and $\bar{x}_i \geq 0$ on $[0,\infty)^n$, player i's unique simultaneous-move best response is $x_i = (1-\lambda)\alpha + \lambda\beta\bar{x}_i > 0$. Because $\widehat{\mathbf{G}}\mathbf{1} = \mathbf{1}$ and $\lambda\beta < 1$, the best-response map $\mathbf{x} \mapsto (1-\lambda)\alpha\mathbf{1} + \lambda\beta\widehat{\mathbf{G}}\mathbf{x}$ is a $\|\cdot\|_\infty$-contraction on $[0,\infty)^n$ with modulus $\lambda\beta < 1$ and unique fixed point $\mathbf{x}^N(\beta)$ in (B.32).

For any leader group L, $\mathbf{U}_F(\beta) := (\mathbf{I}_F - \lambda\beta\widehat{\mathbf{G}}_{FF})^{-1} = \sum_{t=0}^{\infty}(\lambda\beta)^t\widehat{\mathbf{G}}_{FF}^{\,t} \geq \mathbf{I}_F$ is well defined because $\lambda\beta < 1$. After any Stage 1 effort profile $\mathbf{x}_L \in [0,\infty)^{|L|}$, Stage 2 best responses form a $\|\cdot\|_\infty$-contraction on $[0,\infty)^{|F|}$ with modulus at most $\lambda\beta < 1$ and unique equilibrium

$$\mathbf{x}_F(\mathbf{x}_L;\beta) = (1-\lambda)\alpha\mathbf{U}_F(\beta)\mathbf{1}_F + \lambda\beta\mathbf{U}_F(\beta)\widehat{\mathbf{G}}_{FL}\mathbf{x}_L > \mathbf{0}. \tag{B.42}$$

By (B.42), the leaders' shifted references satisfy $\beta\bar{\mathbf{x}}_L(\mathbf{x}_L;\beta) = \mathbf{a}_L(\beta) + \mathbf{C}_L(\beta)\mathbf{x}_L$, where

$$\begin{aligned}\mathbf{a}_L(\beta) &:= (1-\lambda)\alpha\beta\widehat{\mathbf{G}}_{LF}\mathbf{U}_F(\beta)\mathbf{1}_F \geq \mathbf{0},\\ \mathbf{C}_L(\beta) &:= \beta\widehat{\mathbf{G}}_{LL} + \lambda\beta^2\widehat{\mathbf{G}}_{LF}\mathbf{U}_F(\beta)\widehat{\mathbf{G}}_{FL} = \Big[c_{ih}(\beta)\Big]_{i,h\in L} \geq \mathbf{0}.\end{aligned}$$

Each leader $i \in L$ has diagonal entry $c_i(\beta) := \big(\mathbf{C}_L(\beta)\big)_{ii} = \lambda\beta^2 \widehat{\mathbf{G}}_{iF}\mathbf{U}_F(\beta)\widehat{\mathbf{G}}_{Fi} = \partial\big(\beta\bar{x}_i(\mathbf{x}_L;\beta)\big)/\partial x_i$. Since g is undirected and $\mathbf{U}_F(\beta) \geq \mathbf{I}_F$, $c_i(\beta) > 0$ holds if and only if leader i has a neighbor in F; connectedness of g and $0 < |L| < n$ ensure that $c_i(\beta) > 0$ for at least one $i \in L$.

To bound $c_i(\beta)$, differentiate the Stage 2 system $x_j = (1-\lambda)\alpha + \lambda\beta\big(\sum_{h\in L}\widehat{g}_{jh}x_h + \sum_{s\in F}\widehat{g}_{js}x_s\big)$ $(j \in F)$ with respect to x_i. Since $\partial x_j(\mathbf{x}_L;\beta)/\partial x_i \geq 0$ by (B.42) and each row of $\widehat{\mathbf{G}}$ sums to one,

$$\max_{j\in F}\frac{\partial x_j(\mathbf{x}_L;\beta)}{\partial x_i} \leq \lambda\beta\max\left\{1, \max_{j\in F}\frac{\partial x_j(\mathbf{x}_L;\beta)}{\partial x_i}\right\},$$

so $\lambda\beta < 1$ implies $0 \leq \partial x_j(\mathbf{x}_L;\beta)/\partial x_i \leq \lambda\beta$ and hence

$$0 \leq c_i(\beta) = \beta\sum_{j\in F}\widehat{g}_{ij}\frac{\partial x_j(\mathbf{x}_L;\beta)}{\partial x_i} \leq \lambda\beta^2 < \beta. \tag{B.43}$$

For $\mathbf{x}_{L\setminus\{i\}} \in [0,\infty)^{|L|-1}$, let $R_i(\mathbf{x}_{L\setminus\{i\}};\beta) := a_i(\beta) + \sum_{h\in L\setminus\{i\}} c_{ih}(\beta)x_h \geq 0$, where $a_i(\beta)$ is the ith entry of $\mathbf{a}_L(\beta)$. Leader i's reduced Stage 1 payoff $u_i\big(x_i, \mathbf{x}_{L\setminus\{i\}}, \mathbf{x}_F(\mathbf{x}_L;\beta)\big)$ is quadratic on $[0,\infty)$ with second derivative $-\big[1+\gamma\big(1-c_i(\beta)\big)^2\big] < 0$ and unique global best response

$$\mathrm{BR}_i(\mathbf{x}_{L\setminus\{i\}}) = \max\left\{0, \frac{\alpha + \gamma\big(1-c_i(\beta)\big)R_i(\mathbf{x}_{L\setminus\{i\}};\beta)}{1+\gamma\big(1-c_i(\beta)\big)^2}\right\}. \tag{B.44}$$

Whenever $x_i = \mathrm{BR}_i(\mathbf{x}_{L\setminus\{i\}}) > 0$, optimality is equivalent to the first-order condition

$$\alpha = x_i + \gamma\big(1-c_i(\beta)\big)\big(x_i - \beta\bar{x}_i(\mathbf{x}_L;\beta)\big). \tag{B.45}$$

Parts (i) and (ii): Downward or zero shift ($0 < \beta \leq 1$). When $0 < \beta \leq 1$, bound (B.43) gives $1 - c_i(\beta) > 0$ for all $i \in L$, so $\mathrm{BR}_i(\mathbf{x}_{L\setminus\{i\}}) > 0$ in (B.44) for all $\mathbf{x}_{L\setminus\{i\}} \geq \mathbf{0}$. Define $\mathbf{E}_L(\beta) := \mathrm{diag}\big(e_k(\beta)\big)_{k\in N}$, with $e_k(\beta) := 1 - c_k(\beta) \in (0,1]$ for $k \in L$ and $e_k(\beta) := 1$ for $k \in F$. Combining (B.42) and (B.45) shows that $\mathbf{x}^S(L;\beta) > \mathbf{0}$ is a subgame-perfect equilibrium if and only if

$$\mathbf{Q}_L(\beta)\mathbf{x}^S(L;\beta) = \alpha\mathbf{1}, \qquad \mathbf{Q}_L(\beta) := \mathbf{I} + \gamma\mathbf{E}_L(\beta)(\mathbf{I} - \beta\widehat{\mathbf{G}}). \tag{B.46}$$

Factor $\mathbf{Q}_L(\beta) = \mathrm{diag}\big(1+\gamma e_k(\beta)\big)\big[\mathbf{I} - \mathbf{T}_L(\beta)\widehat{\mathbf{G}}\big]$, where $\mathbf{T}_L(\beta) := \mathrm{diag}\big(\gamma\beta e_k(\beta)/[1+\gamma e_k(\beta)]\big)_{k\in N}$ has diagonal entries in $(0,\lambda\beta] \subset (0,1)$. Connectedness of g and row-stochasticity of $\widehat{\mathbf{G}}$ imply $\big[\mathbf{I} - \mathbf{T}_L(\beta)\widehat{\mathbf{G}}\big]^{-1} = \sum_{t=0}^{\infty}\big(\mathbf{T}_L(\beta)\widehat{\mathbf{G}}\big)^t > \mathbf{0}$, so $\mathbf{Q}_L(\beta)^{-1} > \mathbf{0}$,

and $\mathbf{x}^S(L;\beta) = \alpha\mathbf{Q}_L(\beta)^{-1}\mathbf{1} > \mathbf{0}$ is the unique subgame-perfect equilibrium. At $\beta = 1$, $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{1} = \mathbf{0}$ gives $\mathbf{Q}_L(1)\mathbf{1} = \mathbf{1}$ and $x_i^S(L;1) = x_i^N(1) = \alpha$ for all $i \in N$, proving part (ii).

When $0 < \beta < 1$, $\mathbf{Q}_L(\beta)(\alpha\mathbf{1}) = \alpha\mathbf{1} + \gamma\alpha(1-\beta)\mathbf{E}_L(\beta)\mathbf{1} > \alpha\mathbf{1}$ and $\mathbf{Q}_L(\beta)^{-1} > \mathbf{0}$ imply $\mathbf{x}^S(L;\beta) < \alpha\mathbf{1}$. Subtracting $[\mathbf{I} + \gamma(\mathbf{I} - \beta\widehat{\mathbf{G}})]\mathbf{x}^N(\beta) = \alpha\mathbf{1}$ from (B.46) yields

$$\mathbf{x}^S(L;\beta) - \mathbf{x}^N(\beta) = \mathbf{Q}_L(\beta)^{-1}\Big[\mathbf{I} - \mathbf{E}_L(\beta)\Big]\Big[\alpha\mathbf{1} - \mathbf{x}^N(\beta)\Big].$$

Since $\alpha\mathbf{1} - \mathbf{x}^N(\beta) = \frac{\lambda(1-\beta)}{1-\lambda\beta}\alpha\mathbf{1} > \mathbf{0}$, $\mathbf{I} - \mathbf{E}_L(\beta) \geq \mathbf{0}$ has at least one positive diagonal entry $c_i(\beta) > 0$, and $\mathbf{Q}_L(\beta)^{-1} > \mathbf{0}$, we obtain $x_i^N(\beta) < x_i^S(L;\beta) < \alpha$ for all $i \in N$, proving part (i).

Part (iii): Upward shift ($1 < \beta < 1/\lambda$). Let $x^N(\beta) := \frac{1-\lambda}{1-\lambda\beta}\alpha > \alpha$ denote the common Nash effort in (B.32). On $[0, x^N(\beta)]^{|L|}$, if $0 \leq c_i(\beta) \leq 1$, then $1 - c_i(\beta) \geq 0$ and $c_{ih}(\beta) \geq 0$ make $\mathrm{BR}_i(\mathbf{x}_{L\setminus\{i\}})$ in (B.44) nondecreasing in $\mathbf{x}_{L\setminus\{i\}}$. At $\mathbf{x}_L = x^N(\beta)\mathbf{1}_L$, Stage 2 uniqueness gives $\mathbf{x}_F\Big(x^N(\beta)\mathbf{1}_L;\beta\Big) = x^N(\beta)\mathbf{1}_F$, and $\alpha - x^N(\beta) = \gamma(1-\beta)x^N(\beta)$ implies that leader i's reduced Stage 1 marginal payoff equals $c_i(\beta)\Big(\alpha - x^N(\beta)\Big) \leq 0$, so the unconstrained maximizer at $\mathbf{x}_{L\setminus\{i\}} = x^N(\beta)\mathbf{1}_{L\setminus\{i\}}$ is

$$x^N(\beta) + \frac{c_i(\beta)}{1 + \gamma\Big(1 - c_i(\beta)\Big)^2}\Big(\alpha - x^N(\beta)\Big) \in \Big[\alpha,\, x^N(\beta)\Big].$$

If instead $c_i(\beta) > 1$, then $1 - c_i(\beta) < 0$ and $R_i(\mathbf{x}_{L\setminus\{i\}};\beta) \geq 0$ in (B.44) give

$$0 \leq \mathrm{BR}_i(\mathbf{x}_{L\setminus\{i\}}) \leq \frac{\alpha}{1 + \gamma\Big(1 - c_i(\beta)\Big)^2} < \alpha < x^N(\beta). \tag{B.47}$$

Hence $\mathrm{BR}_L = (\mathrm{BR}_i)_{i\in L}$ maps $[0, x^N(\beta)]^{|L|}$ continuously into $[0, x^N(\beta)]^{|L|}$ and, by Brouwer's fixed-point theorem, admits a fixed point $\mathbf{x}_L^S$, which together with (B.42) yields a pure-strategy subgame-perfect equilibrium with $\mathbf{x}_F^S > \mathbf{0}$.

Now take any pure-strategy subgame-perfect equilibrium $\mathbf{x}^S(L;\beta)$ and set $z := \max_{k\in N} x_k^S(L;\beta) > 0$. Suppose, to the contrary, that $z > x^N(\beta)$. No follower $j \in F$ can attain z, since $\bar{x}_j^S(L;\beta) \leq z$, $\lambda\beta < 1$, and $(1-\lambda)\alpha = (1-\lambda\beta)x^N(\beta)$ give $x_j^S(L;\beta) \leq (1-\lambda)\alpha + \lambda\beta z < z$; nor can a leader with $c_i(\beta) > 1$ attain z by (B.47). Thus some leader $i \in L$ with $0 \leq c_i(\beta) \leq 1$ satisfies $x_i^S(L;\beta) = z > 0$. Since $\bar{x}_i^S(L;\beta) \leq z$, $1 - c_i(\beta) \in [0,1]$, $\beta - 1 > 0$, and $1 - \gamma(\beta-1) = (1-\lambda\beta)/(1-\lambda) > 0$, equation (B.45) gives

$$\begin{aligned}\alpha &= z + \gamma\Big(1 - c_i(\beta)\Big)\Big(z - \beta\bar{x}_i^S(L;\beta)\Big) \geq \Big[1 - \gamma\Big(1 - c_i(\beta)\Big)(\beta - 1)\Big]z \\ &\geq \Big[1 - \gamma(\beta-1)\Big]z > \Big[1 - \gamma(\beta-1)\Big]x^N(\beta) = \alpha,\end{aligned}$$

a contradiction. Hence $x_k^S(L;\beta) \leq x^N(\beta)$ for all $k \in N$.

It remains to show that $x_k^S(L;\beta) < x^N(\beta)$ holds strictly for all $k \in N$. Leaders with $c_i(\beta) > 1$ satisfy $x_i^S(L;\beta) < x^N(\beta)$ by (B.47), and if a leader $i \in L$ with $0 < c_i(\beta) \le 1$ satisfied $x_i^S(L;\beta) = x^N(\beta)$, then $\bar{x}_i^S(L;\beta) \le x^N(\beta)$ and (B.45) would give

$$\alpha \ge \Big[1 - \gamma\Big(1 - c_i(\beta)\Big)(\beta - 1)\Big] x^N(\beta) = \alpha + \gamma c_i(\beta)(\beta - 1) x^N(\beta) > \alpha,$$

a contradiction; thus $x_i^S(L;\beta) < x^N(\beta)$ whenever $c_i(\beta) > 0$. Every follower $k \in F$ and every leader $k \in L$ with $c_k(\beta) = 0$ satisfies $x_k^S(L;\beta) = (1-\lambda)\alpha + \lambda\beta\bar{x}_k^S(L;\beta)$ and can attain $x^N(\beta)$ only if every neighbor of k in g also attains $x^N(\beta)$. Connectedness of g would then propagate equality to a leader with $c_i(\beta) > 0$, a contradiction. Together with $x_i^S(L;\beta) \ge 0$ for $i \in L$ and $x_j^S(L;\beta) > 0$ for $j \in F$ from (B.42), we obtain $0 \le x_i^S(L;\beta) < x_i^N(\beta)$ for all $i \in L$ and $0 < x_j^S(L;\beta) < x_j^N(\beta)$ for all $j \in F$, proving part (iii). □

B.5.15 Proof of Corollary B.2

Proof. For $L = \{i\}$ and $F = N \setminus \{i\}$, equation (B.42) gives the affine Stage 2 equilibrium $\mathbf{x}_{-i}(x_i;\beta) = (1-\lambda)\alpha \mathbf{U}_F(\beta)\mathbf{1}_{-i} + \lambda\beta \mathbf{U}_F(\beta)\widehat{\mathbf{G}}_{-i,i}x_i > \mathbf{0}$ on $[0,\infty)$. Leader i's reduced Stage 1 payoff $\tilde{u}_i(x_i) := u_i\Big(x_i, \mathbf{x}_{-i}(x_i;\beta)\Big)$ is a strictly concave quadratic on $[0,\infty)$ with constant second derivative $-\Big[1+\gamma\Big(1-c_i(\beta)\Big)^2\Big] < 0$ for all $\beta \in (0, 1/\lambda)$, so the single-leader equilibrium is unique. At $x_i = x_i^N(\beta)$, Stage 2 uniqueness implies $\mathbf{x}_{-i}\Big(x_i^N(\beta);\beta\Big) = \mathbf{x}_{-i}^N(\beta)$, and $\alpha - x_i^N(\beta) = \gamma\Big(x_i^N(\beta) - \beta\bar{x}_i^N(\beta)\Big)$ yields

$$\tilde{u}_i'\Big(x_i^N(\beta)\Big) = \alpha - x_i^N(\beta) - \gamma\Big(1 - c_i(\beta)\Big)\Big(x_i^N(\beta) - \beta\bar{x}_i^N(\beta)\Big) = c_i(\beta)\Big(\alpha - x_i^N(\beta)\Big).$$

Hence the unconstrained maximizer of $\tilde{u}_i$ is $x_i^N(\beta) + \frac{c_i(\beta)}{1+\gamma(1-c_i(\beta))^2}\Big(\alpha - x_i^N(\beta)\Big)$, and truncation at zero proves (B.33).

In column i of $(\mathbf{I} - \lambda\beta\widehat{\mathbf{G}})\mathbf{M}^N(\beta) = (1-\lambda)\mathbf{I}$, the follower rows give $(\mathbf{I}_{-i} - \lambda\beta\widehat{\mathbf{G}}_{-i})\mathbf{M}_{-i,i}^N(\beta) = \lambda\beta\widehat{\mathbf{G}}_{-i,i}m_{ii}^N(\beta)$, with $m_{ii}^N(\beta) \ge 1 - \lambda > 0$. Comparison with (B.42) gives $\partial x_j(x_i;\beta)/\partial x_i = \Big[\lambda\beta\mathbf{U}_F(\beta)\widehat{\mathbf{G}}_{-i,i}\Big]_j = m_{ji}^N(\beta)/m_{ii}^N(\beta)$ for all $j \ne i$. Because $\mathbf{x}_{-i}(x_i;\beta)$ is affine on $[0,\infty)$ and $\mathbf{x}_{-i}\Big(x_i^N(\beta);\beta\Big) = \mathbf{x}_{-i}^N(\beta)$, both interior and zero-effort solutions in (B.33) satisfy $x_j^S(\{i\};\beta) - x_j^N(\beta) = \Big[m_{ji}^N(\beta)/m_{ii}^N(\beta)\Big]\Big[x_i^S(\{i\};\beta) - x_i^N(\beta)\Big]$ for all $j \ne i$. Summing over $j \in N$ establishes (B.34), and (B.33) yields the three-factor product whenever $x_i^S(\{i\};\beta) > 0$. □

B.5.16 Proof of Corollary B.3

Proof. Under $\alpha_i = \alpha$ for all $i \in N$ and $0 < \beta < 1$, equation (B.32) gives $0 < x_i^N(\beta) = \frac{1-\lambda}{1-\lambda\beta}\alpha \le \frac{1-\lambda}{1-\beta}\alpha \to 0$ as $\lambda \to 1$, so $\lim_{\lambda\to 1} \mathbf{x}^N(\beta) = \mathbf{0}$.

For any leader group L, bound (B.43) gives $0 \le c_i(\beta) \le \lambda\beta^2 < \beta^2 < 1$ for all $i \in L$ and $\lambda \in (0,1)$. By Proposition B.12(i), $\mathbf{x}^S(L;\beta) > \mathbf{0}$, and (B.45) together with the Stage 2 first-order conditions implies that each player $j \in N$ satisfies

$$\alpha - x_j^S(L;\beta) = \gamma e_j(\beta)\Big(x_j^S(L;\beta) - \beta\bar{x}_j^S(L;\beta)\Big),$$

where $e_j(\beta) := 1 - c_j(\beta) > 1 - \beta^2 > 0$ for $j \in L$ and $e_j(\beta) := 1 > 1 - \beta^2$ for $j \in F$. Choose $k \in \arg\max_{j\in N} x_j^S(L;\beta)$. Row-stochasticity of $\widehat{\mathbf{G}}$ gives $\bar{x}_k^S(L;\beta) \le x_k^S(L;\beta)$ and hence $x_k^S(L;\beta) - \beta\bar{x}_k^S(L;\beta) \ge (1-\beta)x_k^S(L;\beta) > 0$. Since $e_k(\beta) > 1 - \beta^2$ and $\gamma = \lambda/(1-\lambda) \to \infty$ as $\lambda \to 1$, player k's first-order condition yields

$$0 < \max_{j\in N} x_j^S(L;\beta) \le \frac{\alpha}{1 + \gamma(1-\beta^2)(1-\beta)} \longrightarrow 0 \qquad \text{as } \lambda \to 1.$$

Hence $\lim_{\lambda\to 1} \mathbf{x}^S(L;\beta) = \mathbf{0}$, and subtracting $\lim_{\lambda\to 1} X^N(\beta) = 0$ gives $\lim_{\lambda\to 1} \ell(L;\beta) = 0$. □

References

Ghuge, Rohan, Joseph Kwon, Viswanath Nagarajan, and Adetee Sharma (2022). "Constrained Assortment Optimization Under the Paired Combinatorial Logit Model". In: *Operations Research* 70.2, pp. 786–804.

Hermalin, Benjamin E. (1998). "Toward an Economic Theory of Leadership: Leading by Example". In: *American Economic Review* 88.5, pp. 1188–206.

Zhou, Junjie and Ying-Ju Chen (2016). "Targeted Information Release in Social Networks". In: *Operations Research* 64.3, pp. 721–35.